

Provable Subexponential Algorithms for NIST Third-Round Lattice Families

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Abstract

We give provable classical subexponential algorithms for secret recovery in growing parameter families associated with NIST third-round lattice candidates. For the Kyber/ML-KEM, FrodoKEM, SABER, NTRU LPRime, and Dilithium/ML-DSA families studied here, polynomial moduli and polylogarithmic coefficient scales yield recovery of the short secret component in expected time and space $2^{(1/2+o(1))n/\ln \ln n}$.

For noisy or rounded linear relations, we exploit an exact gap in the squared Euclidean norm of a comparison vector defined by each coordinate guess. One Gaussian list suffices to identify every secret coordinate by binary search, without enumerating the others. We establish the required sampling guarantees through new geometric bounds for structured public operators over prime and power of two moduli. The construction builds on the Wagner-style Gaussian sampling framework of Ducas, Engelberts, and Loyer (CRYPTO 2025) and the low-error decision-LWE algorithm of Han, Gao, and Hu (2026).

For quotient relations of NTRU type, we develop an affine slice search: fixing coordinates restricts candidate pairs to slices of the public lattice, and their estimated Gaussian masses guide the choice of each next coordinate. In the stated modulus window, Falcon’s key generation quality condition supplies the required mass bound. The algorithm then recovers an equivalent signing key with high probability in time and space $2^{O(n/\ln \ln n)}$. The same search recovers the short key core for cyclic NTRU-HPS/HRSS.

Together, these results give subexponential algorithms for problem families associated with all seven NIST third-round lattice candidates. Despite the subexponential complexity, our results do not establish a reduction in the concrete security of the currently specified parameter sets.

AI disclosure. Building on our earlier work [HGH26], the authors used large language model tools through Codex (GPT-5.6-sol and GPT-6-astra) to explore extensions to growing parameter families associated with Kyber. Further extensions to other NIST third-round lattice candidates were explored with assistance from GPT-5.6-sol. The research proceeded through iterative exchanges: the authors contributed mathematical ideas and approaches, while the tools also proposed ideas and generated candidate mathematical derivations. The authors then organized these approaches into the two frameworks presented in this paper, with further assistance from GPT-6-astra in organizing and drafting the manuscript. The final manuscript was substantially verified and revised by the authors, who remain responsible for all mathematical claims, proofs, references, and conclusions.

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1 Introduction

Small secrets and errors are central to efficient cryptography based on lattices. They simplify arithmetic, reduce key and ciphertext sizes, and make decryption reliable. These advantages explain their use in many lattice candidates considered in NIST’s standardization process [Kyber21, Frodo20, SaberSpec, NTRU20b, FIPS204, Falcon20, NTRU20a]. The usual estimates of concrete security account for coefficient size through heuristic models of lattice reduction, sieving, and hybrid attacks [APS15]. In these models, the cost of the lattice reduction step is typically modeled as exponential in its block size. This raises an asymptotic question: can narrow coefficient distributions lead to provable recovery in time subexponential in the dimension?

For LWE, small errors have long been known to permit subexponential algorithms. A central question is how many samples these algorithms need. Kirchner and Fouque [KF15] obtained subexponential running times with a refined BKW algorithm for sufficiently small secrets and errors. Their claim that a linear number of original samples suffices relied on sample amplification. Herold, Kirshanova, and May [HKM18] identified a gap in that argument and proved the required sample amplification guarantee from $\Theta(n \ln n)$ original samples. Algebraic methods provide another route. Arora and Ge [AG11] gave a provable subexponential algorithm for sufficiently small Gaussian errors, using a subexponential number of samples. Subsequent work reduced the sample requirements for binary errors [ACFP14, STA20, Ste24], including heuristic estimates with linear samples and a provable result with linear samples for a biased binary error distribution [STA20].

Ducas, Engelberts, and Loyer [DEL25] avoided sample amplification by developing a discrete Gaussian sampler based on Wagner’s algorithm [Wag02]. Their method gave a provable subexponential algorithm for SIS^∞ with only slightly more variables than equations. Applying this sampler to LWE required further control of its accuracy and the dependencies among its outputs. In our earlier work [HGH26], we used random auxiliary lattices and a direct analysis of the dependent Gaussian samples to obtain a subexponential algorithm for decision LWE. For polynomial prime moduli and sufficiently small discrete Gaussian errors, the algorithm allows an arbitrary secret distribution and uses $n + \Theta(n / \ln \ln n)$ original samples, with expected time and space $2^{O(n / \ln \ln n)}$. However, this result still uses more than n samples and does not cover structured LWE, such as the module LWE underlying ML-KEM.

We build on these sampling methods to develop a general framework for recovering small secrets from noisy or rounded linear relations. The framework uses the small size of both the secret and the error to turn each coordinate guess into a measurable difference in squared norm. A single Gaussian list then identifies every secret coordinate. We separate this recovery procedure from the geometric properties of the public operator and prove the bounds needed for structured instances. This gives recovery from exactly n coefficient observations in the linear families studied here, including rounded products and compressed public keys.

We also develop a framework for finding valid short pairs from public quotients of NTRU type. Together, the two frameworks give provable subexponential algorithms for growing parameter families associated with all seven lattice candidates in the third round of NIST’s standardization process. Under the conditions stated below, the algorithms are classical and randomized, and succeed with high probability in expected time and space $2^{O(n / \ln \ln n)}$, where n is the relevant coefficient dimension.

1.1 Contributions

We develop two algorithmic frameworks and the geometric estimates needed to apply them. The contributions have four parts.

Generic recovery from low energy linear relations. A guess for one secret coordinate changes the associated error vector’s squared norm by an exact integer gap. All other secret coordinates contribute a common offset, so comparing the candidate scores identifies the coordinate without enumerating the remaining secret. A single Gaussian list estimates these scores simultaneously for every coordinate and every admissible public label. The resulting recovery theorem separates the coordinate tests and list construction from the algebraic structure of the public operator.

Prefix geometry for structured public operators. We prove the smoothing estimates required by the Gaussian list for scalar matrices over prime fields, finite field multipliers, prime cyclotomic modules, and operators modulo powers of two. These estimates connect the generic theorem to the five linear families. Uniform images, ideal decompositions, and additive order layers provide the appropriate mass bounds in each algebraic setting, including coefficient prefixes that end within a polynomial block.

Short pair search from public quotients. For a public relation $hf = g \pmod{q}$, Gaussian averages estimate the mass in successive affine slices of the public lattice. The algorithm uses these masses to choose coordinates until it reaches a short pair passing the public algebraic checks. A bound on the total Gaussian mass controls the probability of following a valid pair’s path. This framework gives the Falcon and cyclic NTRU results, with the required geometry supplied by key quality or by spectral and saturation estimates.

Applications to seven growing families. We instantiate the two frameworks for seven families associated with NIST’s third-round lattice candidates. Kyber and Dilithium were subsequently standardized as ML-KEM and ML-DSA, respectively [FIPS203, FIPS204]. For the five linear families, polynomial moduli and polylogarithmic coefficient scales give subexponential recovery. Joint secret and error energy controls the linear recovery cost, while the Gaussian mass of the public lattice controls the short pair search probability.

Constructing the Gaussian lists is the expensive step in both frameworks. Their construction combines the Gaussian sampling tools of [DEL25, HGH26, ADS15] with the geometric estimates proved here. We first define the seven growing families and describe the related SIS-infinity result of DEL, then state the general recovery results and their seven corollaries.

1.2 The NIST Third-Round Lattice Families

Table 1 lists the seven NIST third-round lattice candidates [NIST22] and their public relations, alongside the SIS-infinity problem studied by DEL. We define the growing families through their public operators, coefficient distributions, and parameter growth. Their public relations and coefficient profiles follow the cited specifications. Our NTRU Prime application concerns its NTRU LPrime branch. We use boldface for matrices and module or coefficient vectors, and ordinary italic symbols for individual ring elements and coefficients. The map `coeff` converts ring objects to coefficient vectors. We specify integer representatives where needed. Throughout, lengths are Euclidean lengths of coefficient vectors. A polylogarithmic quantity is bounded by a fixed power of $\ln n$. Parameters are local to each family. For NTRU LPrime, p denotes the ring degree. For ML-DSA, d denotes the number of dropped public key bits, as in FIPS 204.

For the five linear families, public matrices or multipliers are sampled uniformly in the algebraic spaces specified below. Their relations have the form $\mathbf{y} = \mathbf{M}\mathbf{s} + \mathbf{e} \pmod{q}$, where n counts the coefficients of the integer secret. Module ranks are fixed. For Kyber/ML-KEM and SABER, the

Table 1: Results and prior work associated with the seven NIST third-round lattice candidates. Unrounded linear and quotient relations are interpreted modulo q . Rounded public outputs use the conventions of the corresponding specifications. For SABER, the output is modulo p . Module and quotient relations use the schemes’ ring notation.

Family	Relation	Structure	Modulus type
<i>This work</i>			
FrodoKEM	$\mathbf{B} = \mathbf{A}\mathbf{s} + \mathbf{E}$	Unstructured matrix	Power of two
Kyber/ML-KEM	$\mathbf{t} = \mathbf{A}\mathbf{s} + \mathbf{e}$	Cyclotomic module	Prime
SABER	$\mathbf{b} = \text{Round}_{q \rightarrow p}(\mathbf{A}^T \mathbf{s})$	Cyclotomic module	Power of two
NTRU LPRime	$A = \text{Round}(aG)$	Extension field	Prime
Dilithium/ML-DSA	$\mathbf{A}\mathbf{s}_1 + \mathbf{s}_2 = 2^d \mathbf{t}_1 + \mathbf{t}_0$	Cyclotomic module	Prime
Falcon	$h = gf^{-1}$	Cyclotomic ring	Prime
NTRU-HPS/HRSS	$hf = 3g$	Cyclic ring	Power of two
<i>Prior work</i> [DEL25, Theorem 4.1] ¹			
Dilithium/ML-DSA	Scalar SIS [∞]	Unstructured matrix	Prime

¹DEL25 gives a heuristic cost analysis for Dilithium in Appendix A.1.

degree d grows through powers of two and $n = kd$. For ML-DSA, the degree m grows through powers of two and $n = \ell m$ counts the coefficients of $\mathbf{s}_1$. For quotient relations, m is the number of coefficients in each ring element: $m = n$ for Falcon and $m = N$ for cyclic NTRU. We choose all public parameters and energy bounds before drawing the public operator. The coefficient support bounds below may depend on n , and constant bounds are included.

FrodoKEM. For the FrodoKEM relation [Frodo20], let $\mathbf{A}$ be uniform in $(\mathbb{Z}/q\mathbb{Z})^{n \times n}$, and take

$$q = 2^{\lceil \kappa \log_2 n \rceil}, \quad \kappa > 1/2.$$

The public matrix $\mathbf{B} = \mathbf{A}\mathbf{s} + \mathbf{E}$ contains polynomially many prescribed columns. Fix a polylogarithmic V such that, with probability $1 - o(1)$, all prescribed columns simultaneously have joint secret and error energy at most nV .

If the secret and error coefficients are bounded in absolute value by public positive integers $H_s, H_e = (\ln n)^{O(1)}$, respectively, take

$$B = H_s, \quad V = \max\{1, H_s^2 + H_e^2\}.$$

These bounds hold deterministically for every prescribed column. The finite sampling tables in FrodoKEM provide such support bounds. All prescribed columns share the same public matrix.

Kyber/ML-KEM. For Kyber/ML-KEM [Kyber21, FIPS203], let

$$R = \mathbb{Z}[X]/(X^d + 1), \quad R_q = R/qR, \quad \mathbf{A} \leftarrow \text{Unif}(R_q^{k \times k}), \quad \mathbf{t} = \mathbf{A}\mathbf{s}_R + \mathbf{e}_R \pmod{q}.$$

Here $\mathbf{s}_R, \mathbf{e}_R \in R^k$ are integer module vectors and $\mathbf{t} \in R_q^k$. With $\mathbf{M}$ the coefficient matrix of $\mathbf{A}$, set $\mathbf{s} = \text{coeff}(\mathbf{s}_R)$, $\mathbf{e} = \text{coeff}(\mathbf{e}_R)$, and $\mathbf{y} = \text{lift}_q^+(\text{coeff}(\mathbf{t}))$. Take prime $q = n^{\kappa+o(1)}$ for fixed $\kappa \geq 1$, with constant splitting degree $\text{ord}_{2d}(q)$. Assume that a public polylogarithmic $V \geq 1$ gives a joint secret

and error energy bound nV with probability $1 - o(1)$. For centered binomial coefficients with public positive integer parameters $\eta_s, \eta_e = (\ln n)^{O(1)}$, the respective supports are $[-\eta_s, \eta_s]$ and $[-\eta_e, \eta_e]$. One may take

$$B = \eta_s, \quad V = \max\{1, \eta_s^2 + \eta_e^2\},$$

giving a deterministic joint energy bound nV . ML-KEM key generation uses $\eta_s = \eta_e = \eta_1$. The parameter η_2 is used for encryption errors. The k ring equations give exactly n coefficient observations.

SABER. For SABER [SaberSpec], let $\mathbf{A}$ be uniform in $R_q^{k \times k}$, where $R_q = (\mathbb{Z}/q\mathbb{Z})[X]/(X^d + 1)$. Choose moduli $p \mid q$ that are powers of two, with

$$q = n^{\kappa + o(1)}, \quad \kappa > 1/2, \quad \Delta = q/p \geq 2.$$

The secret coefficient bound η and the rounding step Δ are polylogarithmic. Let $\mathbf{s}_R \in (\mathbb{Z}[X]/(X^d + 1))^k$, $\mathbf{s} = \text{coeff}(\mathbf{s}_R)$, and let $\mathbf{M}$ be the coefficient matrix of $\mathbf{A}^T$. The public vector $\mathbf{b} \in R_p^k$ uses the biased rounding map defined coefficientwise in (9.1). Let $\mathbf{b}^+ = \text{lift}_p^+(\text{coeff}(\mathbf{b}))$ be the nonnegative integer coefficient lift. The public lift $\mathbf{y} = \Delta \mathbf{b}^+$ satisfies

$$\mathbf{y} = \mathbf{M}\mathbf{s} + \mathbf{e}_{\text{rnd}} \pmod{q}, \quad \|\mathbf{e}_{\text{rnd}}\|^2 \leq n\Delta^2/4.$$

Thus $B = \eta$ and $V = \eta^2 + \Delta^2/4$ are deterministic bounds.

NTRU LPRime. Take supplied parameters (p, q, w) satisfying the arithmetic conditions of NTRU LPRime [NTRU20b], with primes p, q and $X^p - X - 1$ irreducible over $\mathbb{F}_q$. Our growing family assumes $q = \Theta(p)$ and uses recovery dimension $n = p$. Define

$$O = \mathbb{Z}[X]/(X^p - X - 1), \quad R_q = O/qO = \mathbb{F}_q[X]/(X^p - X - 1).$$

The public multiplier G is uniform in R_q , a degree- p extension of the prime field $\mathbb{F}_q$. Let $a \in O$ be a ternary secret of weight w , with $0 \leq w \leq p$. The public product uses centered representatives and is rounded coefficientwise to the nearest multiples of three. Writing $A := \text{Round}(aG) \in O$ for the rounded ring element and $\mathbf{y} := \text{coeff}(A) \in \mathbb{Z}^n$ for its integer coefficient vector gives

$$\mathbf{y} = \mathbf{M}_G \mathbf{a} + \mathbf{e}_{\text{rnd}} \pmod{q}, \quad \|\mathbf{a}\|^2 + \|\mathbf{e}_{\text{rnd}}\|^2 \leq w + n.$$

Here $\mathbf{a} = \text{coeff}(a)$, $B = 1$, and $V = 1 + w/n$.

Dilithium/ML-DSA. For ML-DSA [FIPS204, Algorithms 6 and 35], fix integers $k \geq \ell \geq 1$, let $m \rightarrow \infty$ through powers of two, and put $n = \ell m$, $R = \mathbb{Z}[X]/(X^m + 1)$, and $R_q = R/qR$. Take prime $q \equiv 1 \pmod{2m}$ with $q \leq n^{C_q}$ for fixed C_q . Choose public integers $\eta \geq 1$, $d \geq 1$, with $\eta, \Delta = 2^d = (\ln n)^{O(1)}$, and draw $\mathbf{A} \leftarrow \text{Unif}(R_q^{k \times \ell})$. Let $\mathbf{s}_1 \in R^\ell$, $\mathbf{s}_2 \in R^k$ have coefficient absolute values at most η . Set $\mathbf{t} = \mathbf{A}\mathbf{s}_1 + \mathbf{s}_2 \pmod{q}$ and apply Power2Round_d coefficientwise to obtain $\mathbf{t}_1, \mathbf{t}_0 \in R^k$. The public key retains $\mathbf{t}_1$ and discards $\mathbf{t}_0$, giving

$$\Delta \mathbf{t}_1 = \mathbf{A}\mathbf{s}_1 + \mathbf{e}_R \pmod{q}, \quad \mathbf{e}_R = \mathbf{s}_2 - \mathbf{t}_0, \quad \|\mathbf{e}_R\|_\infty \leq \eta + \Delta/2.$$

The coefficient bounds give the parameters

$$B = \eta, \quad V = \eta^2 + (\eta + \Delta/2)^2.$$

For both quotient families, the ring degree m tends to infinity. Their common modulus window is

$$2m + 1 \leq q \leq m^{A_q}, \quad \ln(q/m) \leq (\ln m)^{1-c}, \quad A_q > 0, \quad 0 < c < 1, \quad (1.1)$$

with fixed constants.

Falcon. For Falcon [Falcon20], let $n \rightarrow \infty$ through powers of two, put $R = \mathbb{Z}[x]/(x^n + 1)$ and $R_q = R/qR$, and take prime $q \equiv 1 \pmod{2n}$ in the window (1.1) with $m = n$. Fix a rational constant $C \geq 1$ independent of n . Falcon uses $C = 117/100$. The family consists of public keys $h \in R_q$ admitting $f, g, F_0, G_0 \in R$ with

$$\begin{aligned} f \bmod q &\in R_q^\times, & h &= gf^{-1} \pmod{q}, \\ fG_0 - gF_0 &= q, & \gamma_q(f, g)^2 &\leq C^2 q. \end{aligned} \tag{1.2}$$

Here γ_q is the key generation quality measure defined in (B.27). We call these public keys *quality-qualified*.

NTRU-HPS/HRSS. For cyclic NTRU-HPS/HRSS [NTRU20a, Sections 1.3.2–1.3.3], let $N \rightarrow \infty$ along a supplied sequence of primes satisfying

$$\text{ord}_N(2) = \text{ord}_N(3) = N - 1.$$

Work in $R_N = \mathbb{Z}[x]/(x^N - 1)$ and $S_N = \mathbb{Z}[x]/(\Phi_N)$. Both profiles use a power of two modulus $q = \Theta(N)$ with $q \geq 2N + 1$. Their input blocks are independent. The coefficient laws and optional HPS sorting variant are defined in Definition B.22.

For HPS, put $w = q/8 - 2 \leq 2N/3$. Sample f with independent ternary coefficients. Sample g uniformly from polynomials with exactly $w/2$ coefficients of each sign and all remaining coefficients zero. Both polynomials have degree at most $N - 2$. Thus $w = \Theta(N)$ while the coefficient sizes remain bounded.

For HRSS, take $q > 8\sqrt{2}(N - 1)$, for example the specification's choice $q = 2^{\lceil 7/2 + \log_2 N \rceil}$. Sample f, g_0 independently with the TernaryPlus sampler, using blocks of length $N - 1$, and set $g = (x - 1)g_0$.

Condition each raw distribution on the image of f in S_N being invertible modulo 2 and 3, and that of g being invertible modulo 2. The public representative satisfies $h(1) = 0 \pmod{q}$ and $hf = 3g \pmod{q}$ in R_N . Setting $\hat{h} = 3^{-1}h \bmod q$ gives $\hat{h}f = g \pmod{q}$.

Prior work: scalar SIS-infinity. Ducas, Engelberts, and Loyer [DEL25, Theorem 4.1] study SIS-infinity for a uniformly random matrix over $\mathbb{Z}_q$:

$$\mathbf{x} \in \mathbb{Z}^m \setminus \{\mathbf{0}\}, \quad \mathbf{A}\mathbf{x} = \mathbf{0} \pmod{q}, \quad \|\mathbf{x}\|_\infty \leq \beta, \quad \mathbf{A} \leftarrow \text{Unif}(\mathbb{Z}_q^{n \times m}).$$

It uses a polynomial prime modulus, $m = n + \omega(n/\ln \ln n)$, and $q^{1-n/m} \geq 6$, with

$$\beta = \frac{q}{f} \sqrt{\ln m}, \quad f > 1, \quad 0 < \varepsilon \leq \frac{1}{mq^4}, \quad \frac{q}{f} \geq \sqrt{\ln(1/\varepsilon)}.$$

For $m \leq \text{poly}(n)$, $q = n^{\Theta(1)}$, and admissible polylogarithmic f with ε inverse polynomial in n , the expected running time is $2^{O(n/\ln \ln n)}$ and success probability is $1 - n^{-\Omega(1)}$. They also develop the heuristic cost model applied to Dilithium in Appendix A.1.

1.3 Low Energy Linear Recovery

The linear recovery framework takes as input

$$\mathbf{y} = \mathbf{M}\mathbf{s} + \mathbf{e} \pmod{q}, \quad \mathbf{M} \in (\mathbb{Z}/q\mathbb{Z})^{h \times n}, \quad 1 \leq h \leq n, \tag{1.3}$$

for an integer $q \geq 2$, with public bounds

$$\|\mathbf{s}\|^2 + \|\mathbf{e}\|^2 \leq nV, \quad \|\mathbf{s}\|_\infty \leq B.$$

Let p_E bound the probability that these inequalities fail. The rational $V \geq 1$ and integer $1 \leq B \leq \lfloor \sqrt{nV} \rfloor$ have polynomial bit length and are fixed before the public draw. If an energy bound alone is given, take $B = \lfloor \sqrt{nV} \rfloor$.

The candidate radius and effective depth are

$$R^2 = nV + 3B^2, \quad U = R/\sqrt{n}, \quad Q = \ln(q/\sqrt{n}) > 0, \quad D = \ln \frac{Q}{1 + \ln U}. \quad (1.4)$$

The radius R bounds the error vector for every coordinate candidate. The normalized radius U measures the secret and error scale, while Q measures the modulus relative to $\sqrt{n}$. At fixed modulus and coordinate bound, smaller joint energy gives a smaller U , a larger depth D , and a lower recovery exponent.

Let $\mathbf{M}_m$ consist of the first m columns of $\mathbf{M}$. The public operator enters through the prefix lattices

$$\Lambda_m(\mathbf{M}) = \{(\mathbf{u}, \mathbf{v}) \in \mathbb{Z}^h \times \mathbb{Z}^m : \mathbf{M}_m^T \mathbf{u} + \mathbf{v} = 0 \pmod{q}\}.$$

The sampling schedule fixes prefixes m_i , widths t_i , and a tolerance ε_0 from the public parameters. The required geometric event is

$$\eta_{\varepsilon_0}(\Lambda_{m_i}(\mathbf{M})) \leq t_i/2 \quad \text{for every scheduled prefix } i.$$

The prefix and auxiliary smoothing bounds keep each Gaussian lifting and pairing step close to its target distribution with controlled error. The following is an informal form of Theorem 5.1.

Theorem 1.1 (Low energy linear recovery). *Assume $D \rightarrow \infty$ and $\ln q \leq (\ln n)^{c_q}$ for a fixed c_q . If the prescribed geometric event fails with probability at most $\omega_n = o(1)$, a randomized algorithm recovers the original integer secret from the given h observations with probability at least*

$$1 - p_E - \omega_n - o(1),$$

in expected time and space $2^{(1/2+o(1))n/D}$. One Gaussian list serves all coordinate tests and all admissible labels for the same public operator.

On the geometric event, recovery holds for all inputs within the energy bound, including secrets and errors that depend on $\mathbf{M}$. Thus the theorem covers both sampled errors and deterministic rounding. For polynomial $q = n^{\kappa+o(1)}$ with fixed $\kappa > 1/2$ and polylogarithmic U , one has $D \sim \ln \ln n$. Write

$$T_{\text{lin}}(n) = 2^{(1/2+o(1))n/\ln \ln n}. \quad (1.5)$$

The five linear families of Section 1.2 satisfy these conditions, and their geometric estimates give the following corollaries.

Corollary 1.2. *For the FrodoKEM family, one Gaussian list recovers all prescribed secret columns from $\mathbf{B} = \mathbf{A}\mathbf{S} + \mathbf{E}$ with probability $1 - o(1)$, in expected time and space $T_{\text{lin}}(n)$. Here n is the dimension of one column.*

Corollary 1.3. *For the Kyber/ML-KEM family, the original coefficient secret is recovered from the k ring equations $\mathbf{t} = \mathbf{A}\mathbf{s}_R + \mathbf{e}_R$ with probability $1 - o(1)$, in expected time and space $T_{\text{lin}}(n)$.*

Corollary 1.4. *For the SABER family, the original secret is recovered from the k rounded ring equations with probability $1 - o(1)$, in expected time and space $T_{\text{lin}}(n)$.*

Corollary 1.5. *For the NTRU LPRime family, the original ternary secret a is recovered from $A = \text{Round}(aG)$ with probability $1 - o(1)$, in expected time and space $T_{\text{lin}}(n)$, where $n = p$.*

Corollary 1.6. *For the Dilithium/ML-DSA family, the original vector $\mathbf{s}_1$ is recovered from the compressed public key with probability $1 - o(1)$, in expected time and space $T_{\text{lin}}(n)$, where $n = \ell m$.*

The corollaries use the following geometric estimates. FrodoKEM requires control over additive orders modulo a power of two. Kyber/ML-KEM requires estimates for proper ideals and prefixes that end inside a polynomial block. SABER combines coefficient prefix analysis with the ideal structure modulo a power of two. NTRU LPRime uses the uniform image of a nonzero element under a random field multiplier. For ML-DSA, fix ℓ rows before drawing $\mathbf{A}$ and restrict the public relation to this square block. The secret and restricted error each have $n = \ell m$ coefficients, giving joint energy at most nV . With the discarded bits included in the error, the same module estimate used for Kyber/ML-KEM applies. Sections 7–11 prove these estimates and give the complete application statements.

The method also applies to square ternary LWE over a uniform matrix modulo a prime. This gives a direct comparison with May’s meet-in-the-middle attacks based on representations [May21]. For ternary secret distributions with a fixed positive entropy rate, May’s heuristic cost estimates are exponential in n , whereas our recovery bound is $T_{\text{lin}}(n) = 2^{o(n)}$. Both use the original n equations in the common square model. Section 6 gives the precise comparison and its parameter conditions.

1.4 Short Pair Search from Public Quotients

A quotient relation supplies a public lattice containing a short pair. For a public $m \times m$ matrix $\mathbf{H}$, define

$$\Lambda_{\mathbf{H}} = \{(\mathbf{u}, \mathbf{v}) \in \mathbb{Z}^{2m} : \mathbf{H}\mathbf{u} = \mathbf{v} \pmod{q}\}.$$

For a ring pair (u, v) in either quotient family, the corresponding coefficient vector is written $\text{coeff}(u, v) = (\mathbf{u}, \mathbf{v}) \in \Lambda_{\mathbf{H}_h}$, where $\mathbf{H}_h$ is the coefficient matrix of multiplication by h . The cyclic NTRU application instead uses $\mathbf{H}_{\hat{h}}$. The target is a short nonzero vector that passes a public algebraic test. Such search problems, specified by a ring, a public distribution, and a length bound, have been studied as NTRU problems [PS21] and through related module problems of rank two [FPS22].

The algorithm chooses coordinates using Gaussian masses. Fixing some coordinates of $\mathbf{u}$ leaves an affine slice of the public lattice. Each value for the next coordinate defines a smaller slice. The algorithm samples that value with weight proportional to the slice’s Gaussian mass, which gives greater weight to shorter completions.

Poisson summation expresses each slice mass through a Gaussian average on the dual of its direction lattice. Reusable dual Gaussian lists estimate these averages throughout the search. Once $\mathbf{u}$ is fixed, the centered representative of $\mathbf{H}\mathbf{u} \bmod q$ completes $\mathbf{v}$. Exact norm and algebraic checks select valid short pairs.

A public bound on the lattice’s total Gaussian mass gives a lower bound on the probability of reaching each fixed valid short pair. Repeating the search then gives high success probability.

Use the modulus window (1.1), and fix the finite schedule with

$$\begin{aligned} D_Q &\sim \ln \ln m, & s_{\text{mass}}^2 &\sim q/m^\beta, \\ s_{\text{search}}^2 &= c_{\text{search}} q D_Q / m, & 1/2 < \beta < 1, & \quad c_{\text{search}} > 0. \end{aligned}$$

Here s_{mass} is the Gaussian mass width and s_{search} is the search width, both in the Gaussian width convention used below. The public input supplies a rational mass bound A_{mass} satisfying

$$\begin{aligned} \rho_{s_{\text{mass}}}(\Lambda_{\mathbf{H}}) &\leq A_{\text{mass}}, & \ln A_{\text{mass}} &= O(m/D_Q), \\ s_{\text{mass}}^2 &\geq 8s_{\text{search}}^2. \end{aligned}$$

Here $\rho_s(L) = \sum_{\mathbf{z} \in L} e^{-\pi \|\mathbf{z}\|^2 / s^2}$. A public radius $1 \leq B_Q = O(\sqrt{q}) < q/2$ specifies the target length. The two families take $B_Q^2 = C_{\text{len}}^2 q$ for a fixed rational constant C_{len} depending on the family. Numerical bounds have encodings of polynomial length. The next statement is an informal form of Theorem 12.1.

Theorem 1.7 (Verifiable short pair search). *Suppose a polynomial-time verifier accepts some nonzero vector in $\Lambda_{\mathbf{H}}$ of norm at most B_Q . Under the public mass bound above, a randomized algorithm finds such a vector with probability at least $1 - m^{-K}$ for every fixed $K > 0$, in time and space $2^{O(m/\ln \ln m)}$.*

The algorithm uses the public matrix, length and mass bounds, and verifier. For Falcon, the key generation quality condition supplies the Gaussian mass bound for every quality-qualified public key. For cyclic NTRU, spectral estimates and a bound on the logarithmic saturation index give the required mass bound with probability $1 - o(1)$ over the accepted HPS and HRSS distributions. These bounds and the algebraic checks give the following results.

Corollary 1.8. *For the Falcon family, the algorithm outputs $u, v, F, G \in R$ such that*

$$\begin{aligned} u \bmod q &\in R_q^\times, & vu^{-1} &= h \bmod q, \\ uG - vF &= q, & \gamma_q(u, v)^2 &\leq C^2 q, \end{aligned}$$

with probability at least $1 - n^{-K}$ for every fixed $K > 0$ and every quality-qualified public key, over its internal randomness. Its time and space are $2^{O(n/\ln \ln n)}$.

The output has $\|(u, v)\|_2 \leq C\sqrt{q}$ and gives an integer NTRU basis with maximum Gram-Schmidt norm at most $C\sqrt{q}$. Theorem 12.2 derives this guarantee directly from the key generation quality condition.

Corollary 1.9. *For the NTRU-HPS/HRSS family, a valid short pair and its decryption inverses are recovered with probability $1 - o(1)$ over the accepted input law and independent algorithmic randomness, in time and space $2^{O(N/\ln \ln N)}$.*

This output is the short key core defined in Lemma B.15. Theorems 12.3 and 12.4, and Corollary 12.5, give the explicit failure bounds and their proofs.

1.5 Consequences and Organization

Small coefficients lower the recovery cost. For linear relations, reducing the joint energy increases D and lowers the leading recovery exponent. Constant and polylogarithmic coefficient scales both lead to $D \sim \ln \ln n$ in the families above with polynomial moduli. For quotient relations, a short valid pair and a bound on the total Gaussian mass make the pair accessible to the slice search.

The recovery exponent also measures the cost of composing a reduction with our algorithm. If a problem with N input variables is reduced to a matching recovery law of low energy and dimension $n(N)$, the relevant scale is $n(N)/D_{n(N)}$. Appendix D gives a short distributional composition argument and, under randomized ETH, excludes matching reductions with $n(N)/D_{n(N)} = o(N)$. It explains how lowering the target energy restricts the allowable dimension growth.

The finite schedules give the route to concrete cost estimates for the parameter sets considered here. Appendix C explains how geometric guarantees transfer to the actual public distributions, including those produced by seed expansion.

Section 2 develops the recovery ideas before the formal proofs. Sections 4 and 5 prove the coordinate test and construct its Gaussian list. Sections 6–11 verify the linear applications. Section 12

gives the quotient search theorem and its applications, with proofs in Appendix B. Appendix A supplies finite arithmetic and sampling procedures. Appendix E tabulates parameter values, joint energy bounds, and formal leading term values for five linear public families. Section 13.1 discusses the common RLWE public key relation in homomorphic encryption.

2 Technique Overview

Our algorithm for noisy or rounded linear relations involves three tasks. First, a coordinate guess must create a detectable difference in the public data. Second, one Gaussian list must estimate all the resulting scores accurately. Third, the public lattice must support construction of that list at subexponential cost. We explain these tasks in order, then describe how Gaussian mass estimation leads to the quotient search.

2.1 A coordinate guess reveals an energy gap

Consider $\mathbf{y} = \mathbf{M}\mathbf{s} + \mathbf{e} \pmod{q}$. A direct guess for part of $\mathbf{s}$ leaves the contribution of the unguessed coordinates in the equation. We instead place the entire secret into a short error vector and arrange for a guess to change just one of its coordinates. This is possible because both $\mathbf{s}$ and $\mathbf{e}$ have small joint energy.

Append the identity matrix and define

$$\mathbf{F} = [\mathbf{M}^T \mid \mathbf{I}_n], \quad \mathbf{v}_{i,c} = (\mathbf{y}, c\mathbf{e}_i), \quad \mathbf{E}_{i,c} = (\mathbf{e}, c\mathbf{e}_i - \mathbf{s}).$$

Here $\mathbf{e}_i$ is the i th standard basis vector. The candidate label $\mathbf{v}_{i,c}$ is public, and

$$\mathbf{v}_{i,c} = \mathbf{F}^T \mathbf{s} + \mathbf{E}_{i,c} \pmod{q}.$$

The hidden vector has the particularly simple energy

$$\|\mathbf{E}_{i,c}\|^2 = \|\mathbf{e}\|^2 + \sum_{j \neq i} s_j^2 + (c - s_i)^2.$$

Everything except the last term is independent of the candidate c . Consequently,

$$\|\mathbf{E}_{i,c}\|^2 - \|\mathbf{E}_{i,s_i}\|^2 = (c - s_i)^2. \tag{2.1}$$

The correct integer candidate uniquely minimizes the energy, and every incorrect candidate increases it by at least one.

For example, if $s_i \in \{-1, 0, 1\}$, the three candidate energies are a common unknown offset plus $(-1 - s_i)^2$, s_i^2 , and $(1 - s_i)^2$. Recovering s_i therefore requires comparing three values of one public statistic. The offset contains every other secret coordinate, but is the same in all three comparisons. The appended identity coordinates are chosen by the algorithm, so this construction uses the original observations.

The energy bound controls all these candidates at once. If $\|\mathbf{s}\|^2 + \|\mathbf{e}\|^2 \leq nV$ and $|s_i|, |c| \leq B$, then $\|\mathbf{E}_{i,c}\|^2 \leq nV + 3B^2 = R^2$. The next step is a public estimate of energy that is accurate throughout this ball.

2.2 From a Gaussian score to a shared list

Set

$$\Lambda = \{\mathbf{x} \in \mathbb{Z}^{h+n} : \mathbf{F}\mathbf{x} = 0 \pmod{q}\}.$$

For $\mathbf{x} \in \Lambda$, the term containing the unknown secret vanishes from the phase:

$$\cos\left(\frac{2\pi}{q}\langle \mathbf{x}, \mathbf{v}_{i,c} \rangle\right) = \cos\left(\frac{2\pi}{q}\langle \mathbf{x}, \mathbf{E}_{i,c} \rangle\right).$$

Thus the public label and the hidden error vector give exactly the same cosine. Averaging over a discrete Gaussian on Λ turns this equality into information about the error norm.

For $\mathbf{X} \sim D_{\Lambda, q/f}$, Poisson summation gives

$$\mathbb{E}\left[\cos\left(\frac{2\pi}{q}\langle \mathbf{X}, \mathbf{v}_{i,c} \rangle\right)\right] = \frac{\rho_f(\mathcal{D} - \mathbf{E}_{i,c})}{\rho_f(\mathcal{D})}, \quad \mathcal{D} = q\Lambda^*.$$

The zero vector in $\mathcal{D}$ contributes $e^{-\pi\|\mathbf{E}_{i,c}\|^2/f^2}$ to the numerator. A bound on the Gaussian mass of the nonzero vectors controls all remaining terms. Theorem 4.2 makes this approximation uniform over $\|\mathbf{E}_{i,c}\| \leq R$. Combining it with (2.1) gives a public score whose maximum occurs at the correct coordinate value.

We estimate the score by averaging over a list of N lattice vectors. The list is built by lifting and combining earlier samples, so its entries are dependent. The useful invariant is *conditional similarity*: even after fixing the other entries, each entry has a distribution close to the target Gaussian. A martingale concentration bound then controls the empirical cosine average. This is the dependence analysis inherited from DEL and HGH [DEL25, HGH26].

A single comparison would require concentration for one bounded function. We instead take a union bound over all public labels and coordinate candidates. Theorem 4.4 shows that one list then answers every admissible coordinate test for the same operator. This simultaneous event has two useful consequences. First, it covers a rounding error that is determined by the operator and the secret. Second, it allows the same list to serve all secret columns of a FrodoKEM-type matrix relation. The costly sampling is performed once, while each additional column adds polynomial scoring work.

For a larger candidate interval, adjacent scores also give a binary search. The identity

$$\|\mathbf{E}_{i,c+1}\|^2 - \|\mathbf{E}_{i,c}\|^2 = 2(c - s_i) + 1$$

determines whether s_i lies to the left or right of c . The difference is a nonzero integer, so the same score accuracy preserves its sign. Each coordinate requires $O(\log(2B + 1))$ comparisons. The simultaneous event already includes these adaptively selected candidates.

Comparison with provable dual attack. Pouly and Shen [PS24] generate a Gaussian list for a chosen block of secret coordinates and reuse it to score all candidates for that block. Their lattice vectors are orthogonal modulo q to the unguessed columns of the public matrix. This removes the contribution of the unguessed secret from the score without requiring it to be small. Moving to another block changes this lattice and requires a new list in their construction. In contrast, we exploit the small size of the entire secret by including it in the error vector. For each coordinate test, the other secret coordinates contribute a common energy term that is independent of the candidate. Our embedding uses one fixed lattice for all coordinates.

The same energy structure also makes the coordinate search efficient. The uniform score estimates preserve the ordering of adjacent candidates, so comparing c and $c + 1$ reveals whether $s_i \leq c$ or $s_i \geq c + 1$, even when both candidates are incorrect. This gives a binary search using $O(\log(2B + 1))$ comparisons per coordinate.

2.3 Building the list via Wagner-Style Gaussian Sampling

The list construction starts in $\mathbb{Z}^h$, where Gaussian sampling is easy, and adds the n secret coordinate constraints in blocks. At each level, a temporary auxiliary lattice makes Gaussian lifting possible. Pairing samples with the same quotient label then enforces the new constraints. This is the Gaussian list construction of DEL [DEL25], with the random auxiliary lattices introduced by HGH [HGH26].

Pairing multiplies the Gaussian width by $\sqrt{2}$. Early levels therefore operate at small widths and handle relatively few new coordinates. Later levels have larger widths and can handle larger blocks. The central allocation question is whether these capacities sum to n before the width reaches the target value q/f .

At a threshold t_i , a list of length N can handle approximately

$$b_i \approx \frac{\ln N}{\ln(q/t_i)}$$

new coordinates. The thresholds grow geometrically, so the denominators decrease in equal steps. Summing their reciprocals produces a logarithm. More precisely, the allocation argument gives

$$\sum_i b_i \sim 2 \log_2 N W, \quad W \sim D = \ln \frac{\ln(q/\sqrt{n})}{1 + \ln U}, \quad R = \sqrt{n} U.$$

Taking $\log_2 N \sim n/(2D)$ therefore provides enough capacity. This accumulated capacity is the source of the depth factor in the exponent. The number of public observations stays fixed throughout.

The choice of f balances score accuracy and sampling cost. The smallest leading score on the candidate ball is $\exp(-\pi R^2/f^2)$, while the list has Gaussian width q/f . The schedule takes f^2 proportional to $U^2 D$ and chooses a list long enough to resolve the coordinate gap. The random auxiliary lattices permit the required accuracy at each lifting step. The shifted sampler of Aggarwal, Dadush, and Stephens-Davidowitz [ADS15] implements those steps, and the block dimensions keep its extra cost below the leading exponent. Section 5 verifies the width and error conditions.

For $q = n^{\kappa+o(1)}$ with fixed $\kappa > 1/2$, $\ln(q/\sqrt{n}) = \Theta(\ln n)$. If U is polylogarithmic, then

$$D = \ln \ln n - O(\ln \ln \ln n) \sim \ln \ln n.$$

The resulting time and space are $2^{(1/2+o(1))n/\ln \ln n}$. A larger error, including one produced by coarser rounding, increases U and reduces the available depth. Appendix A gives finite arithmetic and a direct auxiliary coset implementation as well.

2.4 The geometry of the public operator

The allocation argument determines which prefixes the sampler uses. To carry out each lifting and pairing step, the corresponding prefix lattice must have a sufficiently small smoothing parameter. Proving this condition is where the public distribution matters.

For a uniform scalar matrix over a prime field, a fixed nonzero label has a uniform image. Averaging the Gaussian mass over that image gives a first moment bound for the dual lattice. Multiplication by a uniform field element has the same property, even though all entries of its coefficient matrix are related. This observation gives the NTRU LPrime geometry directly.

For a prime cyclotomic module, a nonzero label may generate a proper ideal. Its image is uniform within that ideal. Large ideals have many elements, which lowers the probability assigned to each possible image. Small ideals have fewer elements, but their nonzero integer coefficient representatives are long. The proof combines these two estimates to bound their Gaussian contributions.

This is the main geometric argument for Kyber/ML-KEM. It also applies to the square row block selected from the ML-DSA public operator. For that family, restricting $\mathbf{y}_R = \Delta \mathbf{t}_1$ and $\mathbf{e}_R = \mathbf{s}_2 - \mathbf{t}_0$ to the predetermined rows and taking coefficients makes the discarded public-key bits enter the same joint energy certificate.

Modulo a power of two, additive order plays the corresponding role for a scalar matrix. Grouping labels by their orders gives an exact theta series identity and the FrodoKEM smoothing bound. For SABER's cyclotomic module, these groups are replaced by the powers of the ramified ideal generated by $X + 1$. The order-two part requires a separate distance estimate, because a nonzero systematic coordinate is at least $1/2$ from an integer.

The common task is to bound the expected nonzero dual mass, which gives the required prefix smoothing estimates. Uniform images provide this bound over fields, while ideals and additive orders describe how labels spread in the structured rings. All of these estimates are made in the original coefficient norm. They also cover prefixes that stop partway through a polynomial. For example, a sampling level may add several complete ring elements and then a proper prefix of the next one. Treating that partial block is essential to using the common allocation without changing the recovery algorithm. The family sections organize their proofs around these specific geometric tasks and then substitute the energy bounds into Theorem 5.1.

2.5 From Gaussian mass to a short pair search

The two remaining NIST third round families considered here, Falcon and NTRU HPS/HRSS, are based on public quotient relations. A quotient $h = gf^{-1} \pmod{q}$ determines a lattice containing the short pair (f, g) . Our aim is to find a short pair satisfying this relation and the corresponding public checks.

The linear method above relies on geometric conditions that make the solution unique within the prescribed energy bound. However, a quotient relation has a different homogeneous structure and admits both the zero pair and a valid nonzero pair. A norm ball centered at the origin and containing the valid pair also contains its negative, and may contain further short solutions. We therefore retain Gaussian mass estimation, but use it to guide a search through affine slices for a pair satisfying the public checks. A bound on the total Gaussian mass accounts for these additional lattice points and controls the probability of reaching each fixed valid pair.

Here each ring element has m coefficients, so the full pair lattice has dimension $2m$. For a norm bound B , a width s_{search} , and a total mass bound $\rho_{s_{\text{search}}}(\Lambda_h) \leq A$, every fixed pair $\mathbf{z}$ of norm at most B has Gaussian probability at least

$$p_B = A^{-1} \exp(-\pi B^2 / s_{\text{search}}^2).$$

In the parameter range of the quotient theorem, $\ln A = O(m/D_Q)$, $B = O(\sqrt{q})$, and $s_{\text{search}}^2 = \Theta(qD_Q/m)$. Hence $p_B = \exp(-O(m/D_Q))$. This gives the scale of the search.

To reach a pair, the algorithm fixes coordinates successively. At each stage, the current candidates form an affine slice of the lattice. Poisson summation expresses its Gaussian mass as a cosine average on the dual of its direction lattice. A list for that direction therefore estimates the weights of the next possible coordinate values. Choosing according to these weights follows the Gaussian mass toward a complete short pair. A positive coordinate anchor excludes zero, and retaining both signs covers either sign of a valid pair.

The sampling difficulty is that the search uses many direction lattices. One persistent random extension serves all of them. We first build lists in this enlarged lattice and then take differences of predetermined disjoint pairs. A difference is retained when it belongs to the original lattice. Theorem B.2 proves that a public source mass bound suffices for these lists. Theorem B.3 then

shows that the resulting search contains each fixed short witness with high probability. The proof combines simultaneous accuracy for all queried phases with a lower bound on the probability of the witness’s path. Scanning the candidate table with a public verifier gives Theorem 12.1.

For Falcon, the key generation quality condition gives a full coefficient basis with Gram–Schmidt lengths at most $C\sqrt{q}$. The identity $q\Lambda_h^* = \mathbf{J}\Lambda_h$, with $\mathbf{J}$ orthogonal, converts its reversed dual basis into a basis of Λ_h with Gram–Schmidt lengths at least $\sqrt{q}/C$. Bounding each one-dimensional Gaussian sum in these coordinates gives $\rho_s(\Lambda_h) \leq \theta(Cs/\sqrt{q})^{2n}$. At the search width this is $1+n^{-\omega(1)}$, so a public budget $A_{\text{mass}} = 2$ works uniformly for every quality-qualified key. The verifier imposes the same exact quality threshold on the output.

Spectral bounds on a short pair provide another way to bound the Gaussian mass of a negacyclic public lattice. They control lengths in the pair’s polynomial span and in the orthogonal projection. Appendix B.3.3 derives the corresponding recovery guarantee.

For cyclic NTRU of degree N , the span and projection estimate also includes a saturation index. We bound this index by $N|f(1)|\text{Norm}(f(\zeta_N), g(\zeta_N))$. A bound on the expected logarithm of this ideal norm gives $\ln \Delta_{\text{sat}} \leq N/D_N$ with probability $1 - o(1)$, where $D_N \sim \ln \ln N$. The spectral estimates and the key generation acceptance probabilities then supply a public mass budget for HPS and HRSS. The slice search and exact verification give their recovery results.

3 Preliminaries

3.1 Notation and coefficient embeddings

Bold symbols denote vectors (including module vectors) or matrices; Greek vector and matrix symbols are bold as well. Ordinary italic symbols denote scalars, coefficients, and individual ring elements. Thus s_i, e_i are scalar coordinates, whereas $\mathbf{e}_i$ is a standard basis vector when so declared.

Let R be an integer ring with a specified coefficient basis of length d , and let $R_q = R/qR$ for an integer $q \geq 2$. The map `coeff` lists coefficients in this basis and concatenates the blocks of a ring vector in the stated order. For example, for $\mathbf{s}_R \in R^k$ and $\mathbf{A} \in R_q^{r \times k}$, let

$$\mathbf{s} = \text{coeff}(\mathbf{s}_R) \in \mathbb{Z}^{kd}, \quad \mathbf{M} = \text{Mat}_{\text{coeff}}(\mathbf{A}) \in \mathbb{Z}_q^{rd \times kd},$$

so that

$$\text{coeff}(\mathbf{A}\mathbf{s}_R) = \mathbf{M}\mathbf{s} \pmod{q}.$$

Here r is the number of ring equations and $h = rd$ is the number of coefficient observations. For a residue-ring element, `coeff` first denotes its residue-class coefficient vector. We write lift_q^+ for the coordinatewise lift to $\{0, \dots, q-1\}$, and $\text{lift}_q^{\text{cen}}$ for the lift to $[-\lfloor q/2 \rfloor, \lfloor q/2 \rfloor - 1]$. An integer lift is specified before using a norm or an integer observation; scheme-specific rounding and tie rules are stated locally. All norms and inner products use the Euclidean coefficient embedding. In particular, norms of integer ring elements and ring vectors refer to their coefficient vectors, not to residue classes without a lift. Polynomial coefficients are indexed by $0, \dots, d-1$, and abstract vector coordinates by $[n] = \{1, \dots, n\}$, unless otherwise stated; thus $\mathbf{e}_0 = \text{coeff}(1)$ is compatible with this rule. For a matrix $\mathbf{M}$ with n columns, $\mathbf{M}_m$ denotes its first m columns, with $\mathbf{M}_0$ the empty column matrix. The transpose of a coefficient matrix is its ordinary matrix transpose. We write $\ln$ for the natural logarithm and $\text{Unif}(S)$ for the uniform distribution on a finite set S , also denoted by $U(S)$ in some cited specifications. The scalar U below is instead a normalized radius.

Rings and local parameters are redeclared in each scheme or theorem. In particular, R may denote a ring or a candidate radius, h a public ring element or the number of coefficient observations, and f a ring element or a scoring width, as explicitly defined locally. The smoothing parameter $\eta_\epsilon(L)$ is distinct from a scheme’s coefficient bound η .

3.2 Lattices and discrete Gaussians

A rank- d lattice in $\mathbb{R}^k$ is a set

$$L = \mathcal{L}(\mathbf{B}) = \{\mathbf{B}\mathbf{z} : \mathbf{z} \in \mathbb{Z}^d\}, \quad \mathbf{B} = (\mathbf{b}_1, \dots, \mathbf{b}_d) \in \mathbb{R}^{k \times d},$$

where the columns of $\mathbf{B}$ are linearly independent. Its span is $V = \text{span}_{\mathbb{R}}(L)$, and its determinant is the d -dimensional volume $\det L = \sqrt{\det(\mathbf{B}^T \mathbf{B})}$ of a fundamental parallelepiped. For a full-rank lattice, this is $|\det \mathbf{B}|$. The dual lattice and first minimum are

$$L^* = \{\mathbf{u} \in V : \langle \mathbf{u}, \mathbf{x} \rangle \in \mathbb{Z} \text{ for every } \mathbf{x} \in L\}, \quad \lambda_1(L) = \min_{0 \neq \mathbf{x} \in L} \|\mathbf{x}\|.$$

Thus $(L^*)^* = L$ and $\det L^* = (\det L)^{-1}$, with the dual always taken inside V . If $L \subseteq L'$ have the same rank, their finite quotient has size $[L' : L] = \det L / \det L'$. A sublattice $S \subseteq L'$ is *primitive* when $S = L' \cap \text{span}_{\mathbb{R}}(S)$. Its orthogonal projection $\pi_{S^\perp}(L')$ is a lattice. Each fiber is a translate of S . These fibers are used in Gaussian lifting below.

For $t > 0$ and a discrete subset T of Euclidean space, put

$$\rho_t(\mathbf{x}) = e^{-\pi\|\mathbf{x}\|^2/t^2}, \quad \rho_t(T) = \sum_{\mathbf{x} \in T} \rho_t(\mathbf{x}), \quad \theta(t) = \rho_t(\mathbb{Z}).$$

The discrete Gaussian $D_{L,t}$ assigns probability $\rho_t(\mathbf{x})/\rho_t(L)$ to $\mathbf{x} \in L$. More generally,

$$\Pr_{\mathbf{X} \sim D_{L,t,\mathbf{c}}} [\mathbf{X} = \mathbf{x}] = \frac{\rho_t(\mathbf{x} - \mathbf{c})}{\rho_t(L - \mathbf{c})} \quad (\mathbf{x} \in L).$$

The parameter t is a Gaussian width. For the corresponding one-dimensional continuous Gaussian the standard deviation is $t/\sqrt{2\pi}$. We also use the centered law $D_{L+\mathbf{c},t}$ supported on a coset. These conventions give $\mathbf{X} - \mathbf{c} \sim D_{L-\mathbf{c},t}$ when $\mathbf{X} \sim D_{L,t,\mathbf{c}}$.

3.3 Smoothing and Poisson summation

For $0 < \varepsilon < 1$, the smoothing parameter is

$$\eta_\varepsilon(L) = \inf\{t > 0 : \rho_{1/t}(L^* \setminus \{0\}) \leq \varepsilon\}. \quad (3.1)$$

It measures the width at which the nonzero dual frequencies have small total Gaussian mass. Scaling gives $\eta_\varepsilon(aL) = a\eta_\varepsilon(L)$ for $a > 0$. For lattices of the same rank and span, $L \subseteq L'$ implies $\eta_\varepsilon(L') \leq \eta_\varepsilon(L)$, because $(L')^* \subseteq L^*$.

We use Poisson summation in the conventions of [MR07]. For a rank- d lattice L and $\mathbf{c} \in V$, it gives

$$\rho_t(L + \mathbf{c}) = \frac{t^d}{\det L} \sum_{\mathbf{u} \in L^*} e^{-\pi t^2 \|\mathbf{u}\|^2} e^{2\pi i \langle \mathbf{u}, \mathbf{c} \rangle}. \quad (3.2)$$

Taking $L = \mathbb{Z}$ and $\mathbf{c} = 0$ yields $\theta(t) = t\theta(1/t)$. The triangle inequality gives $\rho_t(L + \mathbf{c}) \leq \rho_t(L)$, and, if $t \geq \eta_\varepsilon(L)$, separating the zero frequency gives

$$(1 - \varepsilon) \frac{t^d}{\det L} \leq \rho_t(L + \mathbf{c}) \leq (1 + \varepsilon) \frac{t^d}{\det L}. \quad (3.3)$$

This uniformity over the center is the reason smoothing controls Gaussian lifting and convolution.

The same formula, applied to a Gaussian times a character, gives the exact Fourier identity

$$\mathbb{E}_{\mathbf{X} \sim D_{L,t}} \left[e^{2\pi i \langle \mathbf{X}, \mathbf{w} \rangle} \right] = \frac{\rho_{1/t}(L^* - \mathbf{w})}{\rho_{1/t}(L^*)} \quad (\mathbf{w} \in V). \quad (3.4)$$

The right side is real, so it is also the expected cosine. For a modular kernel, substituting $t = q/f$ and $\mathbf{w} = \mathbf{v}/q$ produces the periodic Gaussian score used in Section 4.

3.4 Modular kernels and the LWE relation

For $q \geq 2$, write $\mathbb{Z}_q = \mathbb{Z}/q\mathbb{Z}$. When q is prime this is the field $\mathbb{F}_q$. For a modular matrix $\mathbf{F} \in \mathbb{Z}_q^{a \times b}$, define

$$\ker_q(\mathbf{F}) = \Lambda_q^\perp(\mathbf{F}) = \{\mathbf{x} \in \mathbb{Z}^b : \mathbf{F}\mathbf{x} = 0 \pmod{q}\}.$$

This lattice has rank b , since it contains $q\mathbb{Z}^b$, regardless of the number of modular constraints. If $\mathbf{F} = [\mathbf{C} \mid \mathbf{I}_a]$, then the map $\mathbf{x} \mapsto \mathbf{F}\mathbf{x}$ is surjective, even for composite q , and

$$\det \ker_q(\mathbf{F}) = q^a, \quad \ker_q(\mathbf{F})^* = \mathbb{Z}^b + q^{-1}\mathbf{F}^T\mathbb{Z}^a. \quad (3.5)$$

The determinant is the index of the kernel in $\mathbb{Z}^b$. The displayed dual contains $\mathbb{Z}^b$, and its q^a distinct cosets are distinguished by the identity coordinates. This proves equality, since the dual has determinant q^{-a} .

In the conventional LWE notation [Reg05], the public input is $\mathbf{A} \in \mathbb{Z}_q^{n \times m}$ together with

$$\mathbf{b} = \mathbf{A}^T \mathbf{s} + \mathbf{e} \pmod{q}.$$

Search-LWE asks for the secret $\mathbf{s}$. Decision-LWE distinguishes this input from an independent uniform $\mathbf{b} \in \mathbb{Z}_q^m$. For every $\mathbf{x} \in \Lambda_q^\perp(\mathbf{A})$, $\langle \mathbf{x}, \mathbf{b} \rangle = \langle \mathbf{x}, \mathbf{e} \rangle \pmod{q}$, so a short kernel vector supplies a public statistic of the error. The DEL25 sampler constructs lists of such vectors, and HGH26 uses their Fourier scores for decision-LWE. Our recovery problem instead writes the public relation as $\mathbf{y} = \mathbf{M}\mathbf{s} + \mathbf{e}$ with $\mathbf{M} \in \mathbb{Z}_q^{h \times n}$ and allows $h \leq n$.

3.5 Conditional similarity and concentration

The analysis of dual-sieve attacks has highlighted the need to justify score predictions for the actual sampling process [DP23, DP26]. In our setting, correct one-sample marginals alone would not imply concentration: identical copies of one Gaussian sample have the same marginals but provide no averaging. Conditional similarity supplies the additional control used below.

For discrete laws P, Q on a countable set, their statistical distance is $\text{TV}(P, Q) = \frac{1}{2} \sum_x |P(x) - Q(x)|$. A random variable is γ -similar to a law D if its point probabilities lie between $e^{-\gamma}D(x)$ and $e^\gamma D(x)$. A list $(\mathbf{X}_1, \dots, \mathbf{X}_N)$ is *strongly conditionally γ -similar* to D , or to D^N , when

$$e^{-\gamma}D(\mathbf{x}) \leq \Pr[\mathbf{X}_j = \mathbf{x} \mid \mathbf{X}_{-j} = \mathbf{x}_{-j}] \leq e^\gamma D(\mathbf{x}) \quad (3.6)$$

for every index j and every value of the other entries having positive probability. The notation D^N specifies the reference marginals, not independence of the actual list. This is the conditional similarity notion of DEL25 [DEL25], following the Gaussian list framework of [ALS21], and used in HGH26 [HGH26]. The word “strongly” emphasizes conditioning on all other entries. Averaging proves the same bounds when conditioning only on a prefix.

The following concentration bound will be used for every score.

Lemma 3.1. *If $(\mathbf{X}_1, \dots, \mathbf{X}_N)$ is strongly conditionally γ -similar to D , where $0 \leq \gamma \leq 1$, then for every function H with values in $[-1, 1]$ and every $u > 0$,*

$$\Pr \left[\left| \frac{1}{N} \sum_{j=1}^N H(\mathbf{X}_j) - \mathbb{E}_D[H] \right| > 2\gamma + u \right] \leq 2e^{-Nu^2/2}. \quad (3.7)$$

Proof. Let $a_j = \mathbb{E}[H(\mathbf{X}_j) \mid \mathbf{X}_1, \dots, \mathbf{X}_{j-1}]$. The prefix version of (3.6) gives $|a_j - \mathbb{E}_D[H]| \leq e^\gamma - 1 \leq 2\gamma$. Conditional on the preceding entries, $H(\mathbf{X}_j) - a_j$ has mean zero and lies in an interval of length two. Hoeffding's lemma [Hoe63] therefore gives

$$\mathbb{E}[e^{\lambda(H(\mathbf{X}_j) - a_j)} \mid \mathbf{X}_1, \dots, \mathbf{X}_{j-1}] \leq e^{\lambda^2/2}.$$

Iterating this bound and optimizing at $\lambda = u$ bounds either tail of the sum by $e^{-Nu^2/2}$. Add the conditional mean discrepancy to obtain (3.7). $\square$

Conditional similarity and implementation error play different roles. The former permits concentration for a dependent reference list. If an implemented list is within statistical distance ξ of that reference law, the probability of any event concerning the whole list changes by at most ξ . In particular, after a union bound over many scores under the reference law, the implementation cost in probability is still a single additive ξ .

3.6 Gaussian lifting and pairing

To construct a Gaussian list on the final prefix lattice, we enlarge each prefix by a small auxiliary quotient. Gaussian lifting extends samples from the previous prefix to this enlarged lattice, and pairing samples with the same quotient label places their differences in the desired prefix lattice. The smoothing conditions control the distortion introduced by both operations.

For the systematic matrix $[\mathbf{M}^T \mid \mathbf{I}_n]$, where $\mathbf{M} \in \mathbb{Z}_q^{h \times n}$, and a block partition $n = b_1 + \dots + b_r$, put $m_i = \sum_{j \leq i} b_j$ and

$$L_i = \{(\mathbf{u}, \mathbf{v}) \in \mathbb{Z}^{h+m_i} : \mathbf{M}_{m_i}^T \mathbf{u} + \mathbf{v} = 0 \pmod{q}\}, \quad L_0 = \mathbb{Z}^h.$$

An auxiliary lattice $\Gamma_i \supseteq q\mathbb{Z}^{b_i}$ gives $L'_i = L_i + (0 \oplus \Gamma_i)$ and

$$0 \longrightarrow \Gamma_i \longrightarrow L'_i \longrightarrow L_{i-1} \longrightarrow 0, \quad L'_i/L_i \simeq \Gamma_i/q\mathbb{Z}^{b_i}. \quad (3.8)$$

Gaussian lifting uses DEL [DEL25, Lemma 3.1], with $0 < \varepsilon \leq 1/2$. For a primitive fiber $S \subseteq L'$ and $P_0 = \pi_{S^\perp}(L')$, shifted Gaussian lifting with fresh randomness and $t \geq \eta_\varepsilon(S)$ preserves list length and gives

$$(D_{P_0, t}, \gamma) \longmapsto (D_{L', t}, \gamma + 3\varepsilon). \quad (3.9)$$

Here each pair records the reference Gaussian and its strong conditional similarity parameter.

Write BUCKETANDCOMBINE for the ordered pairing operation in Algorithm 1. We use the cyclic quotient version of HGH [HGH26, Algorithm 2], which specializes DEL [DEL25, Algorithm 3]. The two passes preserve the input order, and every input entry is used at most once. In particular, the pairing indices and their output order depend only on the original coset labels.

Algorithm 1: BUCKETANDCOMBINE: cyclic quotient pairing [HGH26, Algorithm 2]

Input: Full-rank lattices $L \subseteq L' \subset \mathbb{R}^d$ with $L'/L \simeq \mathbb{Z}/P\mathbb{Z}$, and an ordered list $\mathcal{X} = (\mathbf{x}_1, \dots, \mathbf{x}_{M_0})$ in L' , with $M_0 \geq 3P$.

Output: An ordered list of $\lfloor M_0/3 \rfloor$ vectors in L .

```

1 Initialize an empty ordered bucket  $B(c)$  for each coset  $c \in L'/L$ ;
2 for  $a \leftarrow 1$  to  $M_0$  do
3   Record  $c_a \leftarrow \mathbf{x}_a \bmod L$ ;
4   Append  $\mathbf{x}_a$  to  $B(c_a)$ ;
5 Initialize  $\mathcal{X}_{\text{out}} \leftarrow ()$ ;
6 for  $a \leftarrow 1$  to  $M_0$  do
7   if  $|\mathcal{X}_{\text{out}}| < \lfloor M_0/3 \rfloor$  and  $|B(c_a)| \geq 2$  then
8     Remove the first two entries  $\mathbf{x}, \mathbf{x}'$  from  $B(c_a)$ , in that order;
9     Append  $\mathbf{x} - \mathbf{x}'$  to  $\mathcal{X}_{\text{out}}$ ;
10 return  $\mathcal{X}_{\text{out}}$ ;

```

We record the correctness and distribution guarantee of HGH [HGH26, Lemma 3.5] in our notation. Its smoothing condition is on the target sublattice L .

Lemma 3.2 (Correctness and distribution of cyclic pairing). *Let $L \subseteq L'$ be full-rank lattices in $\mathbb{R}^d$ with $L'/L \simeq \mathbb{Z}/P\mathbb{Z}$, and let $M_0 \geq 3P$.*

1. *For every ordered input list of M_0 vectors in L' , Algorithm 1 returns exactly $\lfloor M_0/3 \rfloor$ vectors in L , with no input entry reused.*
2. *If $\gamma \geq 0$, $0 < \varepsilon \leq 1/2$, $t \geq \sqrt{2}\eta_\varepsilon(L)$, and the input list is strongly conditionally γ -similar to $D_{L',t}$, then the output list is strongly conditionally $(4\gamma + 3\varepsilon)$ -similar to $D_{L,\sqrt{2}t}$. Thus ordered pairing gives*

$$(D_{L',t}, \gamma) \longmapsto (D_{L,\sqrt{2}t}, 4\gamma + 3\varepsilon), \quad M_0 \longmapsto \lfloor M_0/3 \rfloor. \quad (3.10)$$

Proof. Two vectors in the same coset modulo L have difference in L . If bucket c initially contains M_c entries, scanning all its original labels makes $\lfloor M_c/2 \rfloor$ disjoint pairs available. Since

$$\sum_c \lfloor M_c/2 \rfloor \geq (M_0 - P)/2 \geq M_0/3,$$

the algorithm reaches the prescribed output length. This is the counting argument of DEL [DEL25, Lemma 3.2].

The distributional claim is exactly HGH [HGH26, Lemma 3.5], with the notation $(\Lambda_i, \Lambda'_i, M, s, \delta)$ replaced by (L, L', M_0, t, γ) . Its cyclic quotient label is the coset label used by Algorithm 1; hence its ordered, disjoint pairing rule is the same operation. $\square$

Starting from $\gamma_0 = 0$, lifting followed by Lemma 3.2 gives $\gamma_i = 4(\gamma_{i-1} + 3\varepsilon) + 3\varepsilon$. Consequently, combining the two operations at each level gives

$$\gamma_r = 5(4^r - 1)\varepsilon, \quad 3^r N \longmapsto N, \quad t_0 \longmapsto 2^{r/2}t_0, \quad (3.11)$$

provided the two smoothing conditions and the quotient-size bound hold at every level.

3.7 Auxiliary lattices and shifted sampling

Random auxiliary lattices. Choose a prime P and a uniform nonzero $\mathbf{z} \in \mathbb{F}_P^b$. Define

$$\Gamma = q\mathbb{Z}^b + (q/P)\mathbf{z}\mathbb{Z}, \quad \Gamma/q\mathbb{Z}^b \simeq \mathbb{Z}/P\mathbb{Z}, \quad \det \Gamma = q^b/P. \quad (3.12)$$

The quotient has only P classes, independent of the block dimension. Its dual is $\Gamma^* = q^{-1}\{\mathbf{u} \in \mathbb{Z}^b : \langle \mathbf{u}, \mathbf{z} \rangle = 0 \pmod{P}\}$. If $P^{1/b} \geq 16$ and $\sqrt{b}P^{1/b} \leq P$, then [HGH26, Lemma 3.2] gives, with probability at least $1 - 2^{1-b}$, simultaneously for $0 < \varepsilon < 1$,

$$\eta_\varepsilon(\Gamma) \leq 16qP^{-1/b} \sqrt{\max\{1, \ln(1/\varepsilon)/b\}}. \quad (3.13)$$

Implementing the shifted sampler. To sample from the random auxiliary lattices, we use the shifted Gaussian algorithm of Aggarwal, Dadush, and Stephens-Davidowitz [ADS15, Theorem 22], which supports the required widths. HGH26 uses the same algorithm in its lifting steps. We record the specialization needed here.

Lemma 3.3 (Shifted Gaussian sampling). *Let $L \subset \mathbb{R}^b$ be a full-rank lattice given by a rational basis, and let $\mathbf{c} \in \mathbb{Q}^b$. There is an absolute constant $c_{\text{DGS}} > 0$ such that, whenever*

$$t > \text{dist}(\mathbf{c}, L)e^{-\sqrt{b}}, \quad (3.14)$$

one can sample within statistical distance $2^{-c_{\text{DGS}}b^2}$ of $D_{L,t,\mathbf{c}}$ in time and space $2^{b+O(\sqrt{b}\log b)} = 2^{b+o(b)}$, up to polynomial input-bit factors.

Apply [ADS15, Theorem 22] with $F(b) = e^{\sqrt{b}}$. It supplies at least one sample from $D_{L-\mathbf{c},t}$. Retain the first and translate by $\mathbf{c}$. The time bound becomes $2^{b+O(\sqrt{b}\log b)}$, with error $2^{-\Omega(b^2)}$.

Since $q\mathbb{Z}^b \subseteq \Gamma$, every center is within $q\sqrt{b}/2$ of Γ , so the width condition can be checked before the adaptive centers are known.

Lemma 3.4. *Suppose call j uses fresh randomness and, for every possible common history, its implemented law is within total variation distance ξ_j of the reference call. The joint output laws of at most J adaptive calls are then within distance $\sum_{j=1}^J \xi_j$.*

This is the sequential composition bound of [HGH26, Lemma 3.3]. Section 5 verifies the shifted sampling conditions and error budget. Section A.1 gives finite sampling tools for both recovery frameworks. Section A.2 specifies the finite parameters for linear recovery.

3.8 Gaussian mass from uniform images

The prefix smoothing proof reduces to bounding the expected nonzero dual mass. The relevant property of a public operator is how widely it distributes the image of each nonzero label. Over a field, uniform images give a direct mass calculation even when the matrix entries are dependent. The structured rings considered later require a finer decomposition by ideals or additive orders.

Let q be prime and let $\mathbf{A}_m \in \mathbb{F}_q^{n \times m}$ be random. Define the systematic prefix

$$\begin{aligned} L(\mathbf{A}_m) &= \{(\mathbf{u}, \mathbf{v}) \in \mathbb{Z}^n \times \mathbb{Z}^m : \mathbf{A}_m^T \mathbf{u} + \mathbf{v} = 0 \pmod{q}\}, \\ L(\mathbf{A}_m)^* &= \mathbb{Z}^{n+m} + q^{-1} \begin{pmatrix} \mathbf{A}_m \\ \mathbf{I}_m \end{pmatrix} \mathbb{Z}^m. \end{aligned} \quad (3.15)$$

The identity block proves both surjectivity and the displayed dual decomposition. Call the family *uniform-image* if, for every fixed $0 \neq \mathbf{z} \in \mathbb{F}_q^m$, the vector $\mathbf{A}_m \mathbf{z}$ is uniform in $\mathbb{F}_q^n$. No joint independence between the images of different labels is required.

Lemma 3.5. *If $\mathbf{A}_m$ has the uniform-image property, then for every $T > 0$,*

$$\mathbb{E}\left[\rho_{1/T}(L(\mathbf{A}_m)^* \setminus \mathbb{Z}^{n+m})\right] = q^{-n}\theta(q/T)^n(\theta(q/T)^m - \theta(1/T)^m). \quad (3.16)$$

Let $I_m(T) = \theta(1/T)^{n+m} - 1$. Whenever $I_m(T) < \varepsilon$, one also has

$$\Pr[\eta_\varepsilon(L(\mathbf{A}_m)) > T] \leq \frac{q^{-n}\theta(q/T)^n(\theta(q/T)^m - \theta(1/T)^m)}{\varepsilon - I_m(T)}. \quad (3.17)$$

Proof. The identity coordinates make the dual cosets indexed by $\mathbf{z} \in \mathbb{F}_q^m$ disjoint. For $\mathbf{z} \neq 0$, their Gaussian masses factor as

$$\rho_{1/T}(\mathbb{Z}^n + \mathbf{A}_m \mathbf{z}/q) \rho_{1/T}(\mathbb{Z}^m + \mathbf{z}/q).$$

The first factor has expectation

$$q^{-n} \sum_{\mathbf{a} \in \mathbb{F}_q^n} \rho_{1/T}(\mathbb{Z}^n + \mathbf{a}/q) = q^{-n}\theta(q/T)^n.$$

The residue cosets partition $q^{-1}\mathbb{Z}^n$, which gives the last equality. For the systematic factor, the same partition in dimension m gives

$$\sum_{\mathbf{z} \neq 0} \rho_{1/T}(\mathbb{Z}^m + \mathbf{z}/q) = \theta(q/T)^m - \theta(1/T)^m.$$

Multiplying these expressions and using linearity of expectation proves (3.16). The argument uses the marginal uniformity of each fixed image.

The nonzero integer part of the dual mass is $I_m(T) = \theta(1/T)^{n+m} - 1$. If $\eta_\varepsilon(L(\mathbf{A}_m)) > T$, the remaining mass exceeds $\varepsilon - I_m(T)$. Markov's inequality gives (3.17). $\square$

Lemma 3.6. *Let $\ln n = o(L_0)$, $\varepsilon_0 = 4ne^{-L_0}$, and let $\Lambda_i \subseteq \mathbb{Z}^{a_i}$ be full-rank random lattices with $a_i \leq 2n$. If the prescribed widths satisfy $T_i \geq 8\sqrt{L_0}$ and*

$$\mathbb{E}[\rho_{1/T_i}(\Lambda_i^* \setminus \mathbb{Z}^{a_i})] \leq e^{-3L_0} \quad (1 \leq i \leq r),$$

then, for all sufficiently large n ,

$$\Pr[\exists i : \eta_{\varepsilon_0}(\Lambda_i) > T_i] \leq \frac{r}{2n} e^{-2L_0}.$$

Proof. Write

$$I_i = \rho_{1/T_i}(\mathbb{Z}^{a_i} \setminus \{0\}), \quad S_i = \rho_{1/T_i}(\Lambda_i^* \setminus \mathbb{Z}^{a_i}).$$

Since $\Lambda_i \subseteq \mathbb{Z}^{a_i}$, the total nonzero dual mass is $I_i + S_i$. The bound $\theta(1/T) - 1 \leq 4e^{-\pi T^2}$ for $T \geq 1$ gives

$$I_i \leq \exp(8ne^{-\pi T_i^2}) - 1 \leq 2ne^{-L_0} = \varepsilon_0/2$$

for all sufficiently large n . Thus $\eta_{\varepsilon_0}(\Lambda_i) > T_i$ implies $S_i > \varepsilon_0/2$. By Markov's inequality,

$$\Pr[\eta_{\varepsilon_0}(\Lambda_i) > T_i] \leq \frac{\mathbb{E}[S_i]}{\varepsilon_0/2} \leq \frac{e^{-3L_0}}{2ne^{-L_0}} = \frac{1}{2n} e^{-2L_0}.$$

A union bound over the prescribed indices proves the claim, including when the lattices depend on the same public draw. $\square$

Corollary 3.7. *Let $L_0 \rightarrow \infty$, $L_0 = o(n)$, and $\ln n = o(L_0)$. Prescribe $(m_i, T_i)_{i=1}^r$ before the public draw, with*

$$1 \leq m_i \leq n, \quad 8\sqrt{L_0} \leq T_i \leq q, \quad \max_i \frac{m_i \ln(q/T_i)}{n \ln T_i} = o(1), \quad (3.18)$$

and assume every $\mathbf{A}_{m_i}$ has the uniform-image property under the same distribution of the public matrix. For $\varepsilon_0 = 4ne^{-L_0}$ and all sufficiently large n ,

$$\Pr[\exists i : \eta_{\varepsilon_0}(L(\mathbf{A}_{m_i})) > T_i] \leq \frac{r}{2n} e^{-2L_0}. \quad (3.19)$$

Proof. Put $c_\theta = \ln \theta(1)$. For $T_i \leq q$, Poisson summation gives $\theta(q/T_i) = (q/T_i)\theta(T_i/q) \leq (q/T_i)\theta(1)$. Lemma 3.5 therefore bounds the expected noninteger dual mass by e^{-H_i} , where

$$H_i = n \ln T_i - m_i \ln(q/T_i) - (n + m_i)c_\theta.$$

The prefix ratio in (3.18) makes the second term $o(n \ln T_i)$ uniformly in i . Also $m_i \leq n$ and $\ln T_i \rightarrow \infty$ uniformly, so the last term is $o(n \ln T_i)$. Consequently $H_i = (1 - o(1))n \ln T_i$ uniformly. Since $T_i \geq 8\sqrt{L_0}$ and $L_0 = o(n)$,

$$\frac{n \ln T_i}{L_0} \geq \frac{n}{L_0} \left(\ln 8 + \frac{1}{2} \ln L_0 \right) \rightarrow \infty.$$

Hence $H_i \geq 3L_0$ for all prescribed indices and all sufficiently large n . Apply Lemma 3.6 to obtain (3.19). $\square$

We will apply this calculation to independent prime-modulus matrices and to multiplication by a random field element.

4 Secret Recovery by Coordinate Scoring

Let $q \geq 2$, let $1 \leq h \leq n$, and let $\mathbf{M} \in (\mathbb{Z}/q\mathbb{Z})^{h \times n}$ be a fixed public matrix. The observed relation is

$$\mathbf{y} = \mathbf{M}\mathbf{s} + \mathbf{e} \pmod{q}, \quad \mathbf{s} \in \mathbb{Z}^n, \quad \mathbf{e} \in \mathbb{Z}^h. \quad (4.1)$$

A rounded observation enters this model through an integer lift, with $\mathbf{e}$ equal to the resulting modular residue. The argument allows $\mathbf{s}$ and $\mathbf{e}$ to depend on $\mathbf{M}$.

4.1 Systematic coordinates and the joint energy bound

To test a coordinate, append its proposed value to the public label. The corresponding lattice therefore has one identity coordinate for each secret coordinate. Let $\mathbf{M}_m \in \mathbb{Z}_q^{h \times m}$ consist of the first m columns of $\mathbf{M}$, and define

$$\begin{aligned} \mathbf{F} &= [\mathbf{M}^T \mid \mathbf{I}_n], \quad \Lambda = \Lambda_n, \\ \Lambda_m &= \{(\mathbf{u}, \mathbf{v}) \in \mathbb{Z}^h \times \mathbb{Z}^m : \mathbf{M}_m^T \mathbf{u} + \mathbf{v} = 0 \pmod{q}\}, \quad 0 \leq m \leq n, \\ \mathcal{D} &= q\Lambda^* = q\mathbb{Z}^{h+n} + \mathbf{F}^T \mathbb{Z}^n. \end{aligned} \quad (4.2)$$

The systematic lattice identities from Section 3 give $\Lambda_0 = \mathbb{Z}^h$ and $\det \Lambda_m = q^m$, as well as the displayed formula for $\mathcal{D}$. The appended coordinates are chosen by the algorithm, so the observation count remains h .

Assume a known joint energy bound and coordinate bound

$$\begin{aligned} \|\mathbf{s}\|^2 + \|\mathbf{e}\|^2 &\leq nV, & |s_i| &\leq B, & V &\geq 1, \\ 1 &\leq B \leq \lfloor \sqrt{nV} \rfloor, & R^2 &= nV + 3B^2. \end{aligned} \quad (4.3)$$

Here V is rational and B is an integer. The energy bound itself provides $B = \lfloor \sqrt{nV} \rfloor$. Any smaller valid bound reduces the number of candidates. The case $B = 0$ is immediate. If the energy bound holds with probability $1 - p_E$, we condition on it for now and include p_E in the final failure bound. Choose any integer lift of $\mathbf{y}$. The scores depend only on its residue modulo q .

Lemma 4.1 (Coordinate energy gap). *For $i \in [n]$ and a candidate $c \in [-B, B] \cap \mathbb{Z}$, define*

$$\mathbf{v}_{i,c} = (\mathbf{y}, c\mathbf{e}_i), \quad \mathbf{E}_{i,c} = (\mathbf{e}, c\mathbf{e}_i - \mathbf{s}). \quad (4.4)$$

Here $\mathbf{e}_i$ is the i th standard basis vector of $\mathbb{Z}^n$. Then, simultaneously for all candidates,

$$\mathbf{v}_{i,c} = \mathbf{F}^T \mathbf{s} + \mathbf{E}_{i,c} \pmod{q},$$

$$\|\mathbf{E}_{i,c}\|^2 \leq R^2, \quad (4.5)$$

$$\|\mathbf{E}_{i,c}\|^2 - \|\mathbf{E}_{i,s_i}\|^2 = (c - s_i)^2. \quad (4.6)$$

Proof. The first statement is the public congruence and the identity block. Expansion gives

$$\|\mathbf{E}_{i,c}\|^2 = \|\mathbf{e}\|^2 + \|\mathbf{s}\|^2 + c^2 - 2cs_i.$$

The last two terms sum to at most $3B^2$. Substituting $c = s_i$ and subtracting proves the exact gap. $\square$

4.2 A score estimate on the entire energy ball

The distance gap in Lemma 4.1 is hidden because $\mathbf{s}$ and $\mathbf{e}$ are unknown. We reveal it by averaging a periodic score over short lattice vectors. The periodic Gaussian estimator of Aharonov and Regev [AR05], also used in decoding with preprocessing [DRS14], connects this average to Gaussian mass around the error vector. Provable dual attacks analyze such scores through lattice geometry [PS24, QX26]. Here we need an estimate uniform over the candidate ball, including its boundary, so that one list can test every label consistent with the energy bound. For $f > 0$, define the ideal score

$$\mu(\mathbf{v}) = \mathbb{E}_{\mathbf{X} \sim D_{\Lambda, q/f}} \left[\cos \left(\frac{2\pi}{q} \langle \mathbf{X}, \mathbf{v} \rangle \right) \right].$$

The phase of $\mathbf{F}^T \mathbf{a}$ is an integer multiple of 2π on Λ . Thus Poisson summation in (3.4) gives, for every $\mathbf{a} \in \mathbb{Z}^n$ and $\mathbf{E} \in \mathbb{Z}^{h+n}$,

$$\mu(\mathbf{F}^T \mathbf{a} + \mathbf{E}) = \frac{\rho_f(\mathcal{D} - \mathbf{E})}{\rho_f(\mathcal{D})}. \quad (4.7)$$

To measure the score and its coordinate gap, put

$$\tau = e^{-\pi R^2/f^2}, \quad g = \tau(1 - e^{-\pi/f^2}). \quad (4.8)$$

Here τ bounds the leading Gaussian score from below on the candidate ball, while g is a lower bound for the gap between the leading scores of the correct and any incorrect candidate.

Theorem 4.2 (Uniform periodic Gaussian score). *Suppose the fixed operator satisfies*

$$\rho_{\sqrt{2}f}(\mathcal{D} \setminus \{0\}) \leq \zeta, \quad 0 < \zeta \leq \tau g/64. \quad (4.9)$$

For every integer vector $\mathbf{a} \in \mathbb{Z}^n$ and every integer vector $\mathbf{E} \in \mathbb{Z}^{h+n}$ with $\|\mathbf{E}\| \leq R$, simultaneously,

$$|\mu(\mathbf{F}^T \mathbf{a} + \mathbf{E}) - e^{-\pi\|\mathbf{E}\|^2/f^2}| \leq \zeta/\tau \leq g/64. \quad (4.10)$$

Each correct coordinate candidate has Gaussian score exceeding every incorrect candidate's score by at least $31g/32$.

Proof. Put $A_{\mathbf{E}} = \pi\|\mathbf{E}\|^2/f^2$ and write numerator and denominator as $e^{-A_{\mathbf{E}}} + R_{\mathbf{E}}$ and $1+r$. Gaussian monotonicity gives $0 \leq r \leq \zeta$. The inequality

$$\|\mathbf{z} - \mathbf{E}\|^2 \geq \frac{1}{2}\|\mathbf{z}\|^2 - \|\mathbf{E}\|^2$$

gives $0 \leq R_{\mathbf{E}} \leq \zeta e^{A_{\mathbf{E}}}$, whence

$$\left| \frac{e^{-A_{\mathbf{E}}} + R_{\mathbf{E}}}{1+r} - e^{-A_{\mathbf{E}}} \right| = \frac{|R_{\mathbf{E}} - e^{-A_{\mathbf{E}}}r|}{1+r} \leq \max\{R_{\mathbf{E}}, e^{-A_{\mathbf{E}}}r\} \leq \zeta e^{A_{\mathbf{E}}} \leq \zeta/\tau.$$

By (4.6), the difference between the two leading Gaussian terms is at least $\tau(1 - e^{-\pi/f^2}) = g$. Subtracting their approximation errors proves the score gap. $\square$

The sampling construction will establish the sufficient condition

$$\eta_{\zeta}(\Lambda) \leq q/(4f). \quad (4.11)$$

Indeed, the definition of smoothing and $\mathcal{D} = q\Lambda^*$ give $\rho_{4f}(\mathcal{D} \setminus \{0\}) = \rho_{4f/q}(\Lambda^* \setminus \{0\}) \leq \zeta$. Gaussian monotonicity then implies (4.9).

Lemma 4.3. *Under (4.9), $\lambda_1(\mathcal{D}) > 2R$, and each coset of $\mathcal{D}$ contains at most one vector of norm at most R . In particular, a fixed public label $\mathbf{y}$ determines at most one integer secret satisfying (4.1) and (4.3), and $q > 2R$.*

Proof. A nonzero $\mathbf{z} \in \mathcal{D}$ with $\|\mathbf{z}\| \leq 2R$ would contribute at least $e^{-2\pi R^2/f^2} = \tau^2 > \zeta$ to the mass in (4.9). Two representatives in the same coset and ball would have such a difference. If $(\mathbf{s}_1, \mathbf{e}_1)$ and $(\mathbf{s}_2, \mathbf{e}_2)$ yield the same label, then

$$(\mathbf{e}_1 - \mathbf{e}_2, \mathbf{s}_2 - \mathbf{s}_1) \in \mathcal{D}, \quad \|(\mathbf{e}_1 - \mathbf{e}_2, \mathbf{s}_2 - \mathbf{s}_1)\| \leq 2\sqrt{nV} \leq 2R.$$

Distinct secrets give a nonzero vector, which is impossible. Since $q\mathbb{Z}^{h+n} \subseteq \mathcal{D}$, its shortest vector has length at most q . $\square$

4.3 One list for all public labels

We estimate all coordinate scores using the same list. Conditional similarity, as defined in Section 3, supplies concentration even when the list entries are dependent. Taking a union bound over public labels also makes the recovery event valid for secrets and residues chosen as functions of $\mathbf{M}$.

Theorem 4.4 (One list for all bounded-energy inputs). *Fix an operator satisfying (4.9), and let $0 < \delta < 1$. Assume that:*

1. a reference list of length N is strongly conditionally α -similar to $D_{\Lambda, q/f}^N$, where $\alpha \leq g/64$.
2. the implemented list is within total variation distance ν of the reference law.
3. the list length satisfies

$$N \geq \frac{512}{g^2} \ln \frac{2n(2B+1)q^h}{\delta}. \quad (4.12)$$

Then, with probability at least $1 - \delta - \nu$, that same list recovers $\mathbf{s}$ from (4.1) for every pair $(\mathbf{s}, \mathbf{e})$ satisfying (4.3). The procedure `LINEARSCORERECOVER` uses scores of the form

$$\hat{\mu}_{i,c} = \frac{1}{N} \sum_{j=1}^N \cos\left(\frac{2\pi}{q} \langle \mathbf{X}_j, (\mathbf{y}, c\mathbf{e}_i) \rangle\right), \quad (4.13)$$

and locates each coordinate's unique maximizing candidate by adjacent-score binary search. Given the list and a public label, the procedure uses

$$O(N[h + n \log(2B+1)])$$

scalar operations and cosine evaluations. Certified error at most $g/64$ in each evaluated score preserves this conclusion.

Proof. For the proof, consider all labels $\mathbf{y} \in (\mathbb{Z}/q\mathbb{Z})^h$ and all $n(2B+1)$ candidates. There are $H = n(2B+1)q^h$ associated functions, all bounded in $[-1, 1]$. Apply Lemma 3.1 to each function and take a union bound:

$$\Pr\left[\max_{a \leq H} |\hat{\mu}_a - \mu_a| > 2\alpha + t\right] \leq 2He^{-Nt^2/2}.$$

Use $t = g/16$ and $2\alpha \leq g/32$. Equation (4.12) bounds the failure by δ . The stochastic score error is at most $6g/64$. Adding the Gaussian error and the permitted evaluation error leaves at most $8g/64$ error from each leading Gaussian term. Their separation is at least g , so the correct candidate is the unique maximum. For adjacent candidates,

$$\|\mathbf{E}_{i,c+1}\|^2 - \|\mathbf{E}_{i,c}\|^2 = 2(c - s_i) + 1.$$

Its absolute value is at least one. The leading Gaussian scores therefore differ by at least g , with the larger score at $c+1$ exactly when $s_i \geq c+1$. The same error bound preserves this sign, so binary search retains the half-interval containing s_i . The joint event is simultaneous over every admissible label. Transfer this one event to the implemented list at cost ν . $\square$

Algorithm 2: LINEARSCORERECOVER: coordinatewise secret recovery

Input: Modulus q , public label $\mathbf{y} \in (\mathbb{Z}/q\mathbb{Z})^h$, candidate bound B , and a shared list $\mathbf{X}_1, \dots, \mathbf{X}_N \in \mathbb{Z}^{h+n}$.

Output: The integer vector $\hat{\mathbf{s}} \in [-B, B]^n$.

```

1 Write  $\mathbf{X}_j = (\mathbf{u}_j, \mathbf{t}_j)$  with  $\mathbf{u}_j \in \mathbb{Z}^h$  and  $\mathbf{t}_j \in \mathbb{Z}^n$ ;
2 Compute  $a_j \leftarrow \langle \mathbf{u}_j, \mathbf{y} \rangle \bmod q$  for every  $j$ ;
3 for  $i \leftarrow 1$  to  $n$  do
4   Set  $\ell \leftarrow -B$  and  $r \leftarrow B$ ;
5   while  $\ell < r$  do
6     Set  $c \leftarrow \lfloor (\ell + r)/2 \rfloor$ ;
7     Evaluate  $\hat{\mu}_{i,c}$  and  $\hat{\mu}_{i,c+1}$  by (4.13);
8     if  $\hat{\mu}_{i,c+1} > \hat{\mu}_{i,c}$  then
9        $\ell \leftarrow c + 1$ ;
10    else
11       $r \leftarrow c$ ;
12   $\hat{s}_i \leftarrow \ell$ ;
13 return  $\hat{\mathbf{s}}$ ;
```

For the given label, write $\mathbf{X}_j = (\mathbf{u}_j, \mathbf{t}_j)$ and compute $\langle \mathbf{u}_j, \mathbf{y} \rangle \bmod q$ once per sample. The candidate phases are then $\langle \mathbf{u}_j, \mathbf{y} \rangle + c(\mathbf{t}_j)_i \bmod q$, giving the cost in Theorem 4.4. If the observed alphabet has size at most A_y per coordinate, q^h in (4.12) can instead be replaced by A_y^h .

The concentration event still covers all $2B+1$ candidates, so it also covers the adaptively chosen binary search queries.

Corollary 4.5. *Suppose the public geometry and energy bound fail with probabilities at most ω_n and p_E , respectively. If the list guarantee of Theorem 4.4 holds for each good fixed operator and good auxiliary choice, then recovery with fresh randomness fails with probability at most*

$$\omega_n + p_E + \delta + \nu,$$

plus the probabilities of auxiliary or finite-execution failures. This holds for any conditional law of $(\mathbf{s}, \mathbf{e})$ given $\mathbf{M}$ that satisfies the energy bound.

Proof. For each good fixed operator, Theorem 4.4 is uniform over all admissible pairs. Intersect that event with the energy event and then average over the joint law of the input, adding the failure probabilities by a union bound. $\square$

4.4 Energy bounds for the secret and error

The recovery theorem uses the sum $\|\mathbf{s}\|^2 + \|\mathbf{e}\|^2$. The following estimates let us supply this one quantity throughout the applications. For example, separate bounds $\Pr[\|\mathbf{s}\|^2 > n\alpha_n^2] \leq p_{s,n}$ and $\Pr[\|\mathbf{e}\|^2 > n\beta_n^2] \leq p_{e,n}$ give

$$\begin{aligned} V_n &= \max\{1, \alpha_n^2 + \beta_n^2\}, & p_n &= p_{s,n} + p_{e,n}, \\ \Pr[\|\mathbf{s}\|^2 + \|\mathbf{e}\|^2 > nV_n] &\leq p_n. \end{aligned} \tag{4.14}$$

This union bound allows dependence between the two vectors. Bounded coefficients give a deterministic instance: if $h \leq n$, $\|\mathbf{s}\|_\infty \leq H_s$, and $\|\mathbf{e}\|_\infty \leq H_e$, take $V_n = \max\{1, H_s^2 + H_e^2\}$ and $p_n = 0$.

Lemma 4.6. Let $\mathbf{Y} = \boldsymbol{\mu} + \mathbf{X} \in \mathbb{R}^N$, with deterministic $\boldsymbol{\mu}$, and suppose a positive semidefinite matrix $\boldsymbol{\Sigma}$ satisfies

$$\mathbb{E}[e^{\mathbf{u}^T \mathbf{X}}] \leq e^{\mathbf{u}^T \boldsymbol{\Sigma} \mathbf{u}/2} \quad (\mathbf{u} \in \mathbb{R}^N).$$

For $t > 0$, put

$$R_t = \text{tr } \boldsymbol{\Sigma} + 2\sqrt{t \text{tr}(\boldsymbol{\Sigma}^2)} + 2t\|\boldsymbol{\Sigma}\|_{\text{op}}.$$

Then

$$\Pr[\|\mathbf{Y}\|^2 > (\|\boldsymbol{\mu}\| + \sqrt{R_t})^2] \leq e^{-t}.$$

In particular, for $\mathbf{Y} = (\mathbf{s}, \mathbf{e})$, any rational $V_n \geq 1$ with $nV_n \geq (\|\boldsymbol{\mu}\| + \sqrt{R_t})^2$ supplies an energy bound with failure probability e^{-t} .

Proof. The quadratic form inequality of Hsu, Kakade, and Zhang [HKZ12, Theorem 2.1] gives $\Pr[\|\mathbf{X}\|^2 > R_t] \leq e^{-t}$. For a singular $\boldsymbol{\Sigma}$, the assumed moment-generating function bound first implies that $\mathbf{X}$ lies in $\text{range}(\boldsymbol{\Sigma})$ almost surely. Whitening on that subspace reduces to their isotropic hypothesis. Finally apply $\|\boldsymbol{\mu} + \mathbf{X}\| \leq \|\boldsymbol{\mu}\| + \|\mathbf{X}\|$. $\square$

For Gaussian and centered binomial coefficients, concentration gives an explicit energy threshold. Write $Z_\eta = \sum_{a=1}^\eta (B_a - B'_a)$ for the centered binomial distribution, with independent fair Bernoulli variables.

Corollary 4.7. Let $Y_1, \dots, Y_N$ be independent centered coordinates. For a Gaussian coordinate $Y_i \sim D_{\mathbb{Z}, w_i}$ put $v_i^2 = w_i^2/(2\pi)$. For a centered binomial coordinate $Y_i \sim Z_{\eta_i}$ put $v_i^2 = \eta_i/2$. For every $t > 0$,

$$\Pr\left[\sum_i Y_i^2 > \sum_i v_i^2 + 2\sqrt{t \sum_i v_i^4} + 2t \max_i v_i^2\right] \leq e^{-t}.$$

In particular, for n secret coordinates of parameter η_s and n error coordinates of parameter η_e , one may take

$$V_n = \max\left\{1, \frac{\eta_s + \eta_e}{2} + \sqrt{\eta_s^2 + \eta_e^2} + \max(\eta_s, \eta_e)\right\}, \quad p_n = e^{-n}, \quad (4.15)$$

rounding V_n upward to a rational number when needed. The deterministic alternative is $V_n = \max\{1, \eta_s^2 + \eta_e^2\}$ and $p_n = 0$.

Proof. Completing the square and using the maximum of a shifted theta series at an integer shift gives, for $Z \sim D_{\mathbb{Z}, w}$,

$$\mathbb{E}[e^{uZ}] = e^{w^2 u^2/(4\pi)} \frac{\rho_w(\mathbb{Z} - w^2 u/(2\pi))}{\rho_w(\mathbb{Z})} \leq e^{w^2 u^2/(4\pi)}.$$

For the centered binomial distribution, $\mathbb{E}[e^{uZ_\eta}] = \cosh^{2\eta}(u/2) \leq e^{\eta u^2/4}$. Independence combines these bounds into the hypothesis of Lemma 4.6 with $\boldsymbol{\mu} = 0$ and $\boldsymbol{\Sigma} = \text{diag}(v_i^2)$. This proves the stated tail bound. Setting $t = n$ for the two centered binomial vectors proves (4.15). The deterministic alternative uses $|Z_\eta| \leq \eta$. $\square$

In every case, the energy threshold is fixed before the public matrix is sampled and must satisfy the recovery theorem's parameter window.

5 Constructing the Gaussian List

The recovery theorem reduces the algorithmic problem to producing a Gaussian list at width q/f . We start from the easy lattice $\mathbb{Z}^h$ and add the secret coordinates in blocks. At each step, Gaussian lifting adds a block and pairing removes an auxiliary modulus. The lifting and pairing method is due to DEL, with the quantitative LWE construction developed by HGH [DEL25, HGH26]. We use the width and block-capacity organization of HGH [HGH26, Sec. 3 and Appendix B], while checking the prefix conditions for the present recovery setting. Sections 3.6 and 3.7 recall these transformations and their sampling guarantees in the notation used below.

The main choice is how many coordinates to add at each width. Early steps have little capacity, while later steps can handle larger blocks. We show that their combined capacity reaches n while keeping the list length $2^{(1/2+o(1))n/D}$. The public distribution enters through one condition: smoothing at the selected coordinate prefixes.

5.1 The unified recovery statement

A sampling schedule specifies a tolerance ε_0 , widths $s_0, \dots, s_r = q/f$, and consecutive prefixes $0 = m_0 < m_1 < \dots < m_r = n$. At level i , the construction adds $b_i = m_i - m_{i-1}$ coordinates and uses the geometric threshold $t_i = s_{i-1}/\sqrt{2}$. These quantities depend only on n, q, V, B and are fixed before the public matrix is drawn. The next subsection constructs the schedule. The theorem needs a smoothing bound at these prescribed prefixes.

Theorem 5.1 (Low-energy linear relation recovery). *Let $n \rightarrow \infty$, $1 \leq h \leq n$, and $q \geq 2$. Fix rational V and integer B as in (4.3), with polynomial-length binary representations, before drawing $\mathbf{M}$. Put*

$$\begin{aligned} U = \sqrt{R^2/n} &\geq 1, & Q = \ln(q/\sqrt{n}) &> 0, \\ D = \ln \frac{Q}{1 + \ln U} &\rightarrow \infty, & \ln q &\leq (\ln n)^{c_q}, \end{aligned} \tag{5.1}$$

where c_q is fixed, and use the prescribed schedule below. Assume that:

1. the input satisfies $\mathbf{y} = \mathbf{M}\mathbf{s} + \mathbf{e} \pmod{q}$, and the energy and candidate bounds (4.3) fail with probability at most p_E .
2. the scheduled prefixes (4.2) satisfy

$$\Pr_{\mathbf{M}} [\exists i \in [r] : \eta_{\varepsilon_0}(\Lambda_{m_i}) > t_i/2] \leq \omega_n = o(1). \tag{5.2}$$

Then there is a randomized algorithm that recovers the integer vector $\mathbf{s}$ from the h labels with probability at least $1 - p_E - \omega_n - o(1)$, in expected time and space

$$2^{(1/2+o(1))n/D}. \tag{5.3}$$

The algorithm constructs one Gaussian list and applies LINEARSCORERECOVER to all coordinates using that list. The joint law of $(\mathbf{s}, \mathbf{e})$ may depend on $\mathbf{M}$. The algorithm uses fresh internal randomness independent of the input.

All references to the prescribed schedule use the finite values obtained by the rounding rules in Section A.2. Those rules preserve the bounds proved below, and their checks pass for every sufficiently large member of the family. The implementation attaining (5.3) uses Lemma 3.3 for

each Gaussian lift. The proof below verifies its width condition and accumulated sampling error. Direct enumeration of the auxiliary cosets gives a second finite implementation with time and space

$$2^{O(n/D)}. \quad (5.4)$$

Section A.2 specifies finite arithmetic and stopping rules for both implementations. When $h = n - o(n)$, either complexity exponent can also be expressed in terms of the observation dimension up to a $1 + o(1)$ factor.

Remark 5.2. If only $\|\mathbf{s}\|^2 + \|\mathbf{e}\|^2 \leq nV$ is known, take $B = \lfloor \sqrt{nV} \rfloor$. Then $nV \leq R^2 \leq 4nV$, so replacing $U = R/\sqrt{n}$ by $\sqrt{V}$ changes D by at most $\ln 2$. For $q = n^{\kappa+o(1)}$, fixed $\kappa > 1/2$, and polylogarithmic V , this gives $D \sim \ln \ln n$.

5.2 Widths and block sizes

The construction starts at a small width on $\mathbb{Z}^h$, where sampling is easy. At these early widths, a list can enforce only small blocks of new coordinate constraints. Lifting and pairing increase the width while preserving the constraints already imposed, so later levels can add larger blocks. The factor D in the exponent comes from adding these capacities across successive widths. At threshold t_i , a list of length N can add about $\ln N / \ln(q/t_i)$ coordinates. The thresholds grow geometrically, so the denominators decrease in equal steps. Summing their reciprocals produces a logarithmic factor, asymptotic to D . Thus a list with $\log_2 N \sim n/(2D)$ has enough total capacity. The parameters below also ensure that its scores can be estimated accurately.

Fix a rational constant $K > 4\pi/\ln 2$ and first define the ideal parameters

$$\begin{aligned} f^2 &= K(R^2/n)D, & \tau &= e^{-\pi R^2/f^2}, \\ g &= \tau(1 - e^{-\pi/f^2}), & \varepsilon &= \frac{\tau g}{256q^4}, \\ \varepsilon_0 &= \varepsilon/5, & L &= \ln(4n/\varepsilon), & L_0 &= L + \ln 5. \end{aligned} \quad (5.5)$$

For a sufficiently large absolute constant C_0 , put

$$W = \ln \ln(q/\sqrt{L}) - \ln(1 + \ln f), \quad N = 2^{\lceil n/[2(W-C_0-1)] \rceil}. \quad (5.6)$$

The denominator is eventually positive. Section A.2 rounds these parameters in directions that preserve the inequalities below. All geometric hypotheses and the list construction use that actual finite schedule. The ideal parameters give

$$L_0 = (2\pi/K + o(1))n/D, \quad W = (1 + o(1))D, \quad (5.7)$$

$$\log_2 N = (1/2 + o(1))n/D, \quad \gamma_{\text{cap}} := \frac{\ln(1/\varepsilon_0)}{\ln(N/2)} = \frac{4\pi}{K \ln 2} + o(1) < 1. \quad (5.8)$$

Here $D = O(\ln \ln n)$ follows from $U \geq 1$ and the modulus bound. Moreover $\ln U = o(Q)$ and $Q \leq (\ln n)^{c_q}$, so $\ln f = O((\ln n)^{c_q} + \ln \ln n) = o(n/D)$. To check the potentially slow divergence in (5.7), write $z = 1 + \ln U$ and $Q = ze^D$. Then

$$\ln f = \ln U + \frac{1}{2} \ln D + O(1), \quad \ln(q/\sqrt{L}) = Q + \frac{1}{2} \ln D + O(1).$$

Taking logarithms gives $W = D + O(1 + \ln(1 + \ln D)) = (1 + o(1))D$ even if D grows slowly. The same calculations give

$$(\ln q)^3 = o(\ln N), \quad \ln N = o(n), \quad q/(f\sqrt{L_0}) \rightarrow \infty. \quad (5.9)$$

Use the binary schedule

$$\begin{aligned} r &= \lfloor 2 \log_2(q/f) - \log_2(512L_0) \rfloor, \\ s_i &= q/(f2^{(r-i)/2}) \quad (0 \leq i \leq r), \\ t_i &= s_{i-1}/\sqrt{2}, \quad x_i = \ln(q/t_i) \quad (1 \leq i \leq r). \end{aligned} \tag{5.10}$$

Then $s_r = q/f$, $s_0^2 \geq 512L_0$, $t_1 \geq 16\sqrt{L_0}$, $t_r = q/(2f)$, and the x_i decrease with spacing $\ln 2/2$. Also $s_0^2 < 1024L_0$ and $r = O(\ln q)$.

Lemma 5.3 (Block allocation and prefix bounds). *We follow the block-allocation argument of HGH [HGH26, Lemma B.1], verifying below the parameter conditions needed here. For all sufficiently large n , there are a prime $P \in (N/2, N)$ and positive integers $b_1, \dots, b_r$ summing to n such that*

$$\begin{aligned} P^{1/b_i} &> 16, \quad \sqrt{b_i} P^{1/b_i} \leq P, \\ 16qP^{-1/b_i} \sqrt{\max\{1, \ln(1/\varepsilon_0)/b_i\}} &\leq t_i. \end{aligned} \tag{5.11}$$

The cumulative prefixes $m_i = \sum_{j \leq i} b_j$ obey

$$b_{\min} = \Omega(\ln N / \ln q), \quad b_{\max} = O(\ln N / (1 + \ln f)), \quad m_i x_i = O(\ln N \ln t_i). \tag{5.12}$$

The allocation can be fixed independently of $\mathbf{M}$.

Proof. We first bound the capacity of each level, then sum these bounds and choose the blocks. Let $U_0 = N/2$, and let $y_i > 1/2$ be the unique solution of

$$y_i = x_i + \frac{1}{2} \ln(1 + \gamma_{\text{cap}} y_i) + \ln 16, \quad B_i = \lfloor \ln U_0 / y_i \rfloor.$$

The function $y - \frac{1}{2} \ln(1 + \gamma_{\text{cap}} y)$ is strictly increasing on this interval. If $1 \leq b \leq B_i$, then $z = \ln U_0 / b \geq y_i$ and, for $P \geq U_0$,

$$16qP^{-1/b} \sqrt{\max\{1, \ln(1/\varepsilon_0)/b\}} \leq 16qe^{-z} \sqrt{1 + \gamma_{\text{cap}} z} \leq t_i.$$

For every $b \leq B_i$ and all sufficiently large n ,

$$b \leq B_i \leq \frac{\ln U_0}{y_i} < \frac{\ln U_0}{\ln 16} < U_0 < P.$$

Thus, for $b \geq 2$, $P^{1/b} \leq \sqrt{P}$ and $\sqrt{b} < \sqrt{P}$, proving the other conditions. For $b = 1$ the second condition is an equality.

To sum the capacities, use $B_i \geq \ln U_0 / y_i - 1$:

$$\sum_i B_i \geq \ln U_0 \sum_i \frac{1}{y_i} - r.$$

Uniformly $y_i = x_i + O(\ln(1 + x_i))$, and therefore

$$\frac{1}{y_i} = \frac{1}{x_i} - O\left(\frac{\ln(1 + x_i)}{x_i^2}\right).$$

The x_i have spacing $\ln 2/2$, so the integrability of $\ln(1 + x)/x^2$ on $[\ln 2, \infty)$ makes the accumulated error $O(1)$. The same spacing and an integral comparison give

$$\sum_i \frac{1}{x_i} = \frac{2}{\ln 2} \ln \frac{x_1}{x_r} + O(1), \quad \ln \frac{x_1}{x_r} = W + O(1).$$

Consequently there is an absolute constant C_{cap} such that, for all sufficiently large n , the harmonic capacity satisfies

$$\sum_i B_i \geq 2 \log_2 N (W - C_{\text{cap}}) - C_{\text{cap}} r \geq n.$$

Finally fix $C_0 \geq C_{\text{cap}} + 10$ to absorb the summation and rounding errors. First assign $b_* := \lfloor c_* \ln N / \ln q \rfloor$ coordinates to each level, with sufficiently small fixed $c_* > 0$. Each B_i is at least $2b_*$, and $rb_* = O(\ln N) = o(n)$. Assign the remaining coordinates in increasing level order without exceeding B_i . This fixes positive blocks with the first two bounds in (5.12). Moreover,

$$\begin{aligned} m_i x_i &\leq \ln U_0 \left(1 + \frac{2}{\ln 2} x_i \ln(x_1/x_i) \right) \\ &\leq \ln U_0 \left(1 + \frac{2}{\ln 2} (x_1 - x_i) \right) = O(\ln N \ln t_i), \end{aligned}$$

using $\ln u \leq u - 1$ and $x_1 - x_i = \ln(t_i/t_1)$. A prime in $(U_0, 2U_0)$ can be found by deterministic primality tests in $N \text{ poly}(\ln N)$ work. All choices depend only on the specified numerical parameters. $\square$

The subsequent geometric proofs need only the following consequences of the schedule:

$$\begin{aligned} \varepsilon_0 &= 4ne^{-L_0}, \quad L_0 = \Theta(n/D), \quad \ln n = o(L_0), \quad L_0 = o(n), \\ T_i &:= t_i/2 \geq 8\sqrt{L_0}, \quad T_i \leq q, \quad \max_i \frac{m_i \ln(q/T_i)}{n \ln T_i} = o(1). \end{aligned} \tag{5.13}$$

For the last estimate, $\ln(q/T_i) = x_i + \ln 2 \leq 2x_i$ because $x_i \geq \ln 2$. Since $\ln T_i = \ln t_i - \ln 2$, (5.12) and $\ln N = o(n)$ give

$$\frac{m_i \ln(q/T_i)}{n \ln T_i} \leq \frac{2m_i x_i}{n \ln T_i} = O\left(\frac{\ln N \ln t_i}{n \ln T_i}\right) = o(1)$$

uniformly in the prescribed levels. In particular $\ln(n \ln q/L_0) = o(\ln L_0)$, when this additional entropy comparison is convenient in a structured geometric argument. The remaining task for each public family is to establish (5.2) at these scales.

5.3 Lifting and pairing

Fix an integer representative $\widehat{\mathbf{M}} \in \mathbb{Z}^{h \times n}$ of $\mathbf{M}$. For each level define the column index block

$$I_i = \{m_{i-1} + 1, \dots, m_i\}, \quad b_i = m_i - m_{i-1}.$$

Let $\mathbf{M}^{(i)} \in \mathbb{Z}^{h \times b_i}$ consist of the columns of $\widehat{\mathbf{M}}$ indexed by I_i , in increasing order. This fixed lift supplies the integer values used in extension blocks and shifted sampling centers.

To add a block, we first enlarge its lattice by a cyclic group of order P . This makes Gaussian lifting possible. Pairing vectors with the same quotient label then returns their differences to the desired prefix lattice. Independently of $\mathbf{M}$, sample each $\mathbf{z}_i$ uniformly from $\mathbb{F}_P^{b_i} \setminus \{0\}$, mutually independently over $i \in [r]$, and define

$$\Gamma_i = q\mathbb{Z}^{b_i} + (q/P)\mathbf{z}_i\mathbb{Z}, \quad \Lambda'_i = \Lambda_{m_i} + (0 \oplus \Gamma_i). \tag{5.14}$$

Deletion of the newest coordinates gives

$$0 \longrightarrow \Gamma_i \longrightarrow \Lambda'_i \longrightarrow \Lambda_{m_{i-1}} \longrightarrow 0, \quad \Lambda'_i/\Lambda_{m_i} \simeq \mathbb{Z}/P\mathbb{Z}. \tag{5.15}$$

Indeed a previous vector $(\mathbf{u}, \mathbf{v})$ can be extended by $-(\mathbf{M}^{(i)})^T \mathbf{u}$. The kernel after deletion is Γ_i . The second quotient is the class of Γ_i modulo $q\mathbb{Z}^{b_i}$, of order exactly P because $\mathbf{z}_i \neq 0$. The construction works over every integer modulus.

For this fixed schedule, write

$$\mathcal{G}_n = \{\mathbf{M} : \eta_{\varepsilon_0}(\Lambda_{m_i}(\mathbf{M})) \leq t_i/2 \text{ for every } i \in [r]\}. \quad (5.16)$$

Equation (5.2) bounds $\Pr_{\mathbf{M}}[\mathbf{M} \notin \mathcal{G}_n]$ by ω_n .

Algorithm 3: GAUSSIANLISTENGINE: constructing the shared list

Input: A public operator $\mathbf{M}$, the fixed prefixes and widths $(m_i, s_i)_{i=0}^r$, the auxiliary prime P , and target length N .

Output: A list of N vectors in the final lattice Λ_n .

- 1 Initialize a list $L \leftarrow 3^r N$ independent samples from $D_{\mathbb{Z}^h, s_0}$;
 - 2 **for** $i \leftarrow 1$ **to** r **do**
 - 3 Draw $\mathbf{z}_i$ uniformly from $\mathbb{F}_P^{b_i} \setminus \{0\}$, independently of the input and previous choices, and form Γ_i, Λ'_i as in (5.14);
 - 4 Independently lift every vector in L to Λ'_i using the shifted Gaussian on Γ_i of width s_{i-1} ;
 - 5 $L \leftarrow \text{BUCKETANDCOMBINE}(L, \Lambda'_i, \Lambda_{m_i})$, using Algorithm 1;
 - 6 **return** L ;
-

Theorem 5.4 (Gaussian list construction). *Fix a public operator $\mathbf{M} \in \mathcal{G}_n$. With probability at least $1 - 2r2^{-b_{\min}}$ over the independent uniform choices $(\mathbf{z}_i)_{i=1}^r$, the following hold:*

1. *Exact shifted sampling produces exactly N vectors in Λ , strongly conditionally α -similar to $D_{\Lambda, q/f}$, with $\alpha \leq g/256$.*
2. *Using the finite initialization in Section A.2 and Lemma 3.3 for the shifted draws produces a list within total variation distance $\nu_{\text{DGS}} \leq \exp[-\Omega(b_{\min}^2)] = o(1)$ of this reference list. Its expected time and space satisfy (5.3).*

The expected resource bound is uniform over all public inputs.

Proof. The random auxiliary lattice bound (3.13), with (5.11), gives $\eta_{\varepsilon_0}(\Gamma_i) \leq t_i$ for all i except with probability at most $2r2^{-b_{\min}}$. On the resulting auxiliary-good event,

$$s_{i-1} > \eta_{\varepsilon_0}(\Gamma_i), \quad s_{i-1} > \sqrt{2}\eta_{\varepsilon_0}(\Lambda_{m_i}).$$

These are the smoothing conditions on the auxiliary and target lattices required by the lifting transformation (3.9) and Lemma 3.2.

Concretely, for $(\mathbf{u}, \mathbf{v}) \in \Lambda_{m_{i-1}}$ put $\mathbf{c}_i = (\mathbf{M}^{(i)})^T \mathbf{u}$, sample $\mathbf{z} \leftarrow D_{\Gamma_i, s_{i-1}, \mathbf{c}_i}$, and append $\mathbf{w} = \mathbf{z} - \mathbf{c}_i$. Thus $\mathbf{w} + (\mathbf{M}^{(i)})^T \mathbf{u} = \mathbf{z} \in \Gamma_i$. For a coordinate j with $z_{i,j} \neq 0 \pmod{P}$, the quotient label of the appended block is

$$z_{i,j}^{-1} \frac{P(\mathbf{w} + (\mathbf{M}^{(i)})^T \mathbf{u})_j}{q} \pmod{P}.$$

The displayed numerator divided by q is an integer, and equal labels give differences in Λ_{m_i} . The input length $M_i = 3^{r-i+1}N \geq 3N > 3P$ meets the quotient-size condition of Lemma 3.2, which supplies the exact output length and conditional similarity in (3.10). Initialization uses the product Gaussian on $\mathbb{Z}^h$, whose width condition follows from $s_0^2 \geq 512L_0 \gg \ln(h+2)$. For the finite

implementation, the scalar procedure in Section A.2 generates the initial list with total statistical error at most $\nu_{\text{init}} = 2^{-n^2}$ and cost $N^{1+o(1)}$.

The recurrence (3.11) gives length $3^{r-i}N$ and width $s_i = 2^{i/2}s_0$ after level i . At the final level,

$$\alpha = 5(4^r - 1)\varepsilon_0 < q^4\varepsilon \leq \tau g/256 \leq g/256, \quad (5.17)$$

since $4^r = (q/(fs_0))^4 < q^4$.

For the implementation, reduce $\mathbf{c}_i$ modulo $q\mathbb{Z}^{b_i}$ first. Translation back preserves the lift, since this product lattice is contained in Γ_i . The reduced center is within distance $q\sqrt{b_i}/2$ of Γ_i for every history. The bounds (5.9) and (5.12) imply $\sqrt{b_{\min}} \gg \ln q$, so

$$s_{i-1} > q\sqrt{b_i}/(2e^{\sqrt{b_i}}).$$

Thus Lemma 3.3 applies to every center encountered by the algorithm. Summing the input lengths over all levels gives $J = (3^{r+1} - 3)N/2$ shifted calls, each using fresh random bits. Lemma 3.4 gives

$$\nu_{\text{DGS}} \leq \nu_{\text{init}} + J2^{-c_{\text{DGS}}b_{\min}^2}. \quad (5.18)$$

In particular,

$$\nu_{\text{DGS}} \leq \exp\left[-\Omega\left(\frac{n^2}{D^2(\ln q)^2}\right)\right] = o(1),$$

because $\ln J = O(n/D + \ln q) = o(b_{\min}^2)$ and $\nu_{\text{init}} = 2^{-n^2}$ is smaller than the same exponential scale.

Each shifted call costs $2^{b_i+o(b_i)}$, by Lemma 3.3. As $b_{\max} = O(\ln N/(1 + \ln f)) = o(\ln N)$, all lifting, pairing, initialization, quotient arithmetic, and primality search have expected cost at most

$$N^{1+O(1/(1+\ln f))+o(1)} \text{poly}(n, \ln q, \ln(1/\varepsilon_0)) = 2^{(1/2+o(1))n/D}.$$

The bound holds uniformly because the schedule and center-distance estimate hold for every public input. $\square$

Section A.2 converts this uniform bound on expected cost into a deterministic time limit by stopping after $n^{K_t}\bar{T}_n$ steps, with at most n^{-K_t} additional failure probability. The truncated algorithm retains the exponent in (5.3) and supplies the time bound used in Section D.

5.4 From the list to recovery

Proof of Theorem 5.1. On the public good event, the final prefix of (5.2) gives $\eta_{\varepsilon_0}(\Lambda) \leq t_r/2 = q/(4f)$ and hence $\rho_{4f}(\mathcal{D} \setminus \{0\}) \leq \varepsilon_0$. This implies the mass hypothesis of Theorem 4.2, since $\varepsilon_0 \leq \tau g/(1280q^4)$. The list from Theorem 5.4 has conditional similarity error at most $g/256$.

The list is also long enough to estimate every score. Indeed,

$$\ln(Ng^2) = \left(\frac{\ln 2}{2} - \frac{2\pi}{K} + o(1)\right)n/D = \Omega(n/D), \quad (5.19)$$

and, writing $H = n(2B + 1)q^h$, we have

$$\ln H = O(n \ln q), \quad \ln \ln H = O(\ln n + \ln \ln q) = o(n/D).$$

Thus $Ng^2/\ln H \rightarrow \infty$, as required by Theorem 4.4. The estimate for H uses $B \leq R$ and $\ln R = \frac{1}{2} \ln n + \ln U = O(\ln q + \ln n)$, as follows from (5.1). A conservative concentration failure bound is

$2n(2B+1)q^h \exp(-Ng^2/2048) = o(1)$. Theorems 4.2 and 4.4 therefore identify every coordinate on the energy event. Lemma 4.3 identifies the original integer representative.

Combining the public, auxiliary, implementation, and scoring events gives the failure bound

$$p_E + \omega_n + 2r2^{-b_{\min}} + \delta_{\text{impl}} + 2n(2B+1)q^h e^{-Ng^2/2048}. \quad (5.20)$$

For the ideal reference process $\delta_{\text{impl}} = 0$. Section A.2 proves $\delta_{\text{impl}} = o(1)$ for both finite implementations, including sampling, truncation, and stopping errors. The score computation costs $O(N[h + n \log(2B+1)])$ operations. Since $\ln B = O(\ln q + \ln n) = o(n/D)$, it preserves the exponent in (5.3). The implementations in Section A.2 give the claimed resource bounds. Corollary 4.5 allows the secret and residue to depend on the public operator. $\square$

6 Ternary LWE: Provable Subexponential Recovery

We first demonstrate the power of our linear recovery framework on unstructured LWE. In particular, we consider ternary LWE as studied by May [May21], who gave meet-in-the-middle attacks based on representations with heuristic exponential complexity. In independent and concurrent work, Chen, Jiang, Wang, and Zhang [CJWZ26] give a provable subexponential-time decision algorithm for narrow LWE, including ternary secrets and errors, over odd prime polynomial moduli using sublinear samples.

We obtain provable subexponential recovery in the common unstructured model

$$\begin{aligned} \mathbf{M} &\leftarrow \text{Unif}(\mathbb{F}_q^{n \times n}), & \mathbf{y} &= \mathbf{M}\mathbf{s} + \mathbf{e} \pmod{q}, & \mathbf{s}, \mathbf{e} &\in \{-1, 0, 1\}^n, \\ q &= n^{\kappa+o(1)} \text{ prime}, & \kappa &> 1/2 \text{ fixed.} \end{aligned} \quad (6.1)$$

Here the entries of $\mathbf{M}$ are independent and uniform. For comparison with May, we take $(\mathbf{s}, \mathbf{e})$ independent of $\mathbf{M}$. Coordinates within $(\mathbf{s}, \mathbf{e})$ may be dependent, as for a prescribed Hamming weight. The recovery result below also allows dependence on $\mathbf{M}$. May writes $\mathbf{M}\mathbf{s} = \mathbf{y} + \mathbf{e}$. Replacing his error by its negative gives our convention. Both attacks use the original n equations [May21, Sections 2.1–2.3].

Let $\mathbf{E}_m : \mathbb{F}_q^m \hookrightarrow \mathbb{F}_q^n$ append zero coordinates, so $\mathbf{M}_m = \mathbf{M}\mathbf{E}_m$ consists of the first m columns of $\mathbf{M}$. The prefixes used by the algorithm are

$$\Lambda_m(\mathbf{M}) = \{(\mathbf{u}, \mathbf{v}) \in \mathbb{Z}^{n+m} : \mathbf{M}_m^T \mathbf{u} + \mathbf{v} = 0 \pmod{q}\}. \quad (6.2)$$

For every fixed nonzero $\mathbf{z} \in \mathbb{F}_q^m$, the vector $\mathbf{M}_m \mathbf{z} = \mathbf{M}\mathbf{E}_m \mathbf{z}$ is uniform in $\mathbb{F}_q^n$: each independent uniform row of $\mathbf{M}$ takes a uniform value on $\mathbf{E}_m \mathbf{z}$. Lemma 3.5 and Corollary 3.7 therefore control the dual mass and smoothing of the prescribed prefixes.

Dual Gaussian mass is also central to smoothing analyses of random orthogonal lattices [KNSW20] and of codes and lattices [DADRT23]. Here the modular residues index dual cosets, and their total nonzero Gaussian mass controls the prescribed prefixes.

Corollary 6.1. *Let $q > 2$ be prime, $\mathbf{M} \leftarrow \text{Unif}(\mathbb{F}_q^{n \times n})$, and*

$$\mathbf{y} = \mathbf{M}\mathbf{s} + \mathbf{e} \pmod{q}, \quad \mathbf{s} \in \{-1, 0, 1\}^n, \quad \mathbf{e} \in \mathbb{Z}^n.$$

Suppose a deterministic rational $V_n \geq 1$ of polynomial bit length satisfies

$$\Pr[\|\mathbf{s}\|^2 + \|\mathbf{e}\|^2 > nV_n] \leq p_E. \quad (6.3)$$

With $U = \sqrt{V_n + 3/n}$, assume

$$D = \ln \frac{\ln(q/\sqrt{n})}{1 + \ln U} \longrightarrow \infty, \quad \ln q \leq (\ln n)^{c_q} \quad (6.4)$$

for a fixed constant c_q . Then the algorithm of Theorem 5.1 recovers $\mathbf{s}$ from the n labels with probability $1 - p_E - o(1)$, in expected time and space

$$2^{(1/2+o(1))n/D}. \quad (6.5)$$

The secret and error may depend on $\mathbf{M}$.

Proof. Take $h = n$, $V = V_n$, and $B = 1$, so that $R^2 = nV_n + 3$ and $U = \sqrt{V_n + 3/n}$. For each prescribed prefix and each fixed nonzero $\mathbf{z} \in \mathbb{F}_q^{m_i}$, the independent uniform rows of $\mathbf{M}$ make $\mathbf{M}_{m_i}\mathbf{z}$ uniform in $\mathbb{F}_q^n$. Lemma 3.5 therefore gives the expected dual mass in (3.16). The actual finite schedule satisfies (5.13), so Corollary 3.7 applies with $T_i = t_i/2$ and $\varepsilon_0 = 4ne^{-L_0}$, giving

$$\Pr_{\mathbf{M}}[\exists i \in [r] : \eta_{\varepsilon_0}(\Lambda_{m_i}) > t_i/2] \leq \frac{r}{2n} e^{-2L_0} = o(1).$$

This is the geometry condition (5.2). The energy bound (6.3) and $|s_j| \leq 1$ give the certificate (4.3) with failure probability at most p_E . Combine these failure events by a union bound under the joint input law. Together with (6.4), Theorem 5.1 gives the stated success probability and resource bounds. $\square$

In the square ternary model (6.1), $\|\mathbf{s}\|^2 + \|\mathbf{e}\|^2 \leq 2n$ deterministically. Thus Corollary 6.1 applies with

$$\begin{aligned} V_n &= 2, & B &= 1, & p_E &= 0, & U &= \sqrt{2 + 3/n}, \\ D &= \ln \frac{\ln(q/\sqrt{n})}{1 + \ln U} = \ln \ln n + O(1). \end{aligned} \quad (6.6)$$

Indeed, $\ln(q/\sqrt{n}) = (\kappa - 1/2 + o(1)) \ln n$, and the growth condition on $\ln q$ in (6.4) also holds. The original secret is therefore recovered from the given n equations with probability $1 - o(1)$, in expected time and space

$$2^{(1/2+o(1))n/\ln \ln n} = 2^{o(n)}. \quad (6.7)$$

This energy bound holds for every ternary pair $(\mathbf{s}, \mathbf{e})$.

For a prescribed ternary type of fixed positive entropy rate $\mathcal{H} > 0$, the secret space has size

$$S = 2^{\mathcal{H}n+o(n)}. \quad (6.8)$$

May's representation-based attacks have heuristic cost $S^{c+o(1)}$ for fixed $c > 0$ (roughly $S^{0.25}$ in the summary of [May21]). In the common square model with $(\mathbf{s}, \mathbf{e})$ independent of $\mathbf{M}$, these are exponential estimates, whereas (6.7) is a proved subexponential upper bound.

The energy corollary also allows larger errors. If $q = n^{\kappa+o(1)}$ with $\kappa > 1/2$ and $\|\mathbf{e}\|_\infty \leq B_n = (\ln n)^{O(1)}$, choose $V_n = 1 + B_n^2$ and $p_E = 0$. Then $D = (1 + o(1)) \ln \ln n$. A squared-norm bound $\Pr[\|\mathbf{e}\|^2 > R_n] \leq \nu_n$ works just as directly, with $V_n = 1 + R_n/n$ and $p_E = \nu_n$.

7 FrodoKEM: Recovery over Power-of-Two Moduli

Let $q = 2^{b_q}$ and write $R_q = \mathbb{Z}/q\mathbb{Z}$. A nonzero vector over this ring may have additive order smaller than q , so its image under a uniform matrix can be confined to a proper subgroup. Grouping dual vectors by their additive order restores an exact Gaussian mass calculation. For $q = n^{\kappa+o(1)}$, $\kappa > 1/2$, and polylogarithmically bounded coefficients, this yields recovery in expected time and space $2^{(1/2+o(1))n/\ln \ln n}$. We prove the smoothing bound, then apply it to the matrix relation used by FrodoKEM.

7.1 An exact additive order identity

For a uniform scalar matrix over a field, every nonzero label has a uniform image. For a power of two modulus, the image instead depends on the additive order of the label. Grouping labels by this order separates the output mass from the systematic mass and gives an exact theta series identity for the nonzero dual mass.

As before, $\theta(t) = \rho_t(\mathbb{Z})$ and lengths are measured in the ordinary coefficient norm.

Lemma 7.1. *Let $h, d \geq 1$, let $a = h + d$, and sample $\mathbf{U} \leftarrow (\mathbb{Z}/q\mathbb{Z})^{d \times h}$ uniformly. For $\Lambda_{\mathbf{U}} = \Lambda_q^\perp([\mathbf{U} \mid \mathbf{I}_d])$ and $T > 0$,*

$$\mathbb{E}_{\mathbf{U}} \left[\rho_{1/T}(\Lambda_{\mathbf{U}}^* \setminus \mathbb{Z}^a) \right] = \sum_{\substack{Q=2^j \\ 1 \leq j \leq b_q}} Q^{-h} \theta(Q/T)^h (\theta(Q/T)^d - \theta(Q/(2T))^d). \quad (7.1)$$

Proof. The systematic coordinates identify distinct dual cosets with $\mathbf{z} \in (\mathbb{Z}/q\mathbb{Z})^d$. A nonzero $\mathbf{z}$ has additive order $Q = 2^j$ precisely when $\mathbf{z} = (q/Q)\mathbf{u}$, with $\mathbf{u} \in (\mathbb{Z}/Q\mathbb{Z})^d$ having an odd coordinate. A uniform column of $\mathbf{U}$ sends $\mathbf{z}/q$ uniformly to $Q^{-1}\mathbb{Z}/\mathbb{Z}$, and the h columns are independent. For each such output coordinate, averaging the Gaussian coset mass gives $Q^{-1}\theta(Q/T)$. Summing the systematic coordinate mass over $\mathbf{u}$ with an odd coordinate gives the difference $\theta(Q/T)^d - \theta(Q/(2T))^d$. The subtracted vectors are exactly the all-even vectors. Multiplying these contributions and summing over all additive orders proves the formula. $\square$

Lemma 7.2 (Rectangular dyadic prefix smoothing). *Suppose $1 \leq d \leq 3h/2$, $a = h + d$, $0 < \zeta \leq 1$, and $\ln(1/\zeta) \leq h/8$. If*

$$T \geq 8 \max \left\{ \sqrt{\ln(2a/\zeta)}, q^{d/a} \zeta^{-1/a} \right\}, \quad (7.2)$$

then $\Pr_{\mathbf{U}}[\eta_\zeta(\Lambda_{\mathbf{U}}) > T] \leq 8e^{-h/4}$.

Proof. For $0 < u \leq 1$, direct Gaussian series bounds give $\theta(u) \leq 1 + 3e^{-\pi/u^2}$ and $\theta(u) \leq e^{1/8}$. Poisson summation gives $\theta(u) = u\theta(1/u)$. Let F_Q denote a summand in (7.1). For $Q \leq T$,

$$\sum_{Q \leq T} F_Q \leq \frac{e^{a/8} 2^{-h}}{1 - 2^{-h}} \leq \frac{e^{-(\ln 2 - 5/16)h}}{1 - 2^{-h}}.$$

For $Q > T$, transform all theta factors and use the determinant threshold:

$$\sum_{Q > T} F_Q \leq e^{a/8} T^{-a} \frac{q^d}{1 - 2^{-d}} \leq \frac{\zeta e^{-(\ln 8 - 1/8)a}}{1 - 2^{-d}}.$$

The integer part is deterministic and satisfies

$$\rho_{1/T}(\mathbb{Z}^a \setminus \{0\}) = \theta(1/T)^a - 1 \leq e^{3ae^{-\pi T^2}} - 1 \leq \zeta/2.$$

Markov's inequality bounds the probability that the remaining mass is larger than $\zeta/2$ by

$$\frac{2e^{-(\ln 2 - 7/16)h}}{1 - 2^{-h}} + \frac{2e^{-(\ln 8 - 1/8)a}}{1 - 2^{-d}} \leq 8e^{-h/4}.$$

Here $\ln 2 - 7/16 > 1/4$, $a \geq h + 1$, and both denominators are at least $1/2$. The prefix $d = 0$ is the integer lattice used to initialize the algorithm. $\square$

7.2 Recovery from a square or nearly square system

For $U \geq 1$, write

$$Q_n = \ln(q/\sqrt{n}), \quad D_n(U) = \ln \frac{Q_n}{1 + \ln U}. \quad (7.3)$$

Throughout this section $Q_n > 0$ and $\ln q \leq (\ln n)^{c_q}$ for a fixed c_q . The recovery schedule is that of Theorem 5.1, fixed before drawing the public matrix.

Proposition 7.3. *Let $h = n - o(n)$ and $\mathbf{M} \leftarrow \text{Unif}(R_q^{h \times n})$. Let $\mathbf{M}_m$ consist of the first m columns of $\mathbf{M}$. For the scheduled prefixes $m_i = \sum_{j \leq i} b_j$ and*

$$\Lambda_{m_i} = \Lambda_q^\perp([\mathbf{M}_{m_i}^T \mid \mathbf{I}_{m_i}]),$$

the geometric condition (5.2) fails with probability at most $8re^{-h/4} = o(1)$.

Proof. Put $T_i = t_i/2$, $\zeta = \varepsilon_0 = 4ne^{-L_0}$, and $a_i = h + m_i$. The schedule gives $t_i \geq 16\sqrt{L_0}$ and $\ln(2a_i/\zeta) \leq L_0$, establishing the first threshold in (7.2). The second is equivalent to

$$a_i \ln t_i - m_i \ln q \geq \ln(1/\zeta) + a_i \ln 16.$$

Write $x_i = \ln(q/t_i)$. After subtracting $a_i \ln 16$, the left side is

$$h(\ln t_i - \ln 16) - m_i(x_i + \ln 16).$$

The allocation bound (5.12) gives $m_i x_i = O(\ln N \ln t_i)$, with $x_i \geq \ln 2$ and $\ln N = o(h)$. Also $\ln t_i \geq (1/2 + o(1)) \ln n$ and $\ln(1/\zeta) = O(n/D) = o(n)$. Hence this expression is eventually at least $(h/4) \ln t_i \gg \ln(1/\zeta)$. Finally, $m_i \leq n \leq 3h/2$ and $\ln(1/\zeta) \leq h/8$ eventually. Apply Lemma 7.2 to each prefix and sum its failure bound over the $r = O(\ln q)$ prescribed prefixes. $\square$

Theorem 7.4 (Dyadic recovery). *Let $h = n - o(n)$, $\mathbf{M} \leftarrow \text{Unif}(R_q^{h \times n})$, and $\mathbf{y} = \mathbf{M}\mathbf{s} + \mathbf{e} \pmod{q}$. Suppose a deterministic rational $V_n \geq 1$ of polynomial bit length satisfies*

$$\Pr[\|\mathbf{s}\|^2 + \|\mathbf{e}\|^2 > nV_n] \leq p_n, \quad D_n(\sqrt{V_n}) \rightarrow \infty.$$

Then Theorem 5.1 recovers the original integer vector $\mathbf{s}$ from the h labels with probability $1 - p_n - o(1)$, in expected time and space

$$2^{(1/2 + o(1))n/D_n(\sqrt{V_n})}. \quad (7.4)$$

The joint law of $\mathbf{s}, \mathbf{e}$ may depend on $\mathbf{M}$.

Proof. Set $V = V_n$ and $B = \lfloor \sqrt{nV_n} \rfloor$. On the stated energy event, every secret coordinate satisfies $|s_j| \leq B$, so the certificate (4.3) holds. The candidate radius satisfies

$$nV_n \leq R^2 = nV_n + 3B^2 \leq 4nV_n, \quad \sqrt{V_n} \leq U = R/\sqrt{n} \leq 2\sqrt{V_n}.$$

As in Remark 5.2, this gives $D_n(U) = D_n(\sqrt{V_n}) + O(1) \sim D_n(\sqrt{V_n})$. Proposition 7.3 verifies the geometry condition (5.2) for the prescribed finite schedule, with failure probability at most $8re^{-h/4} = o(1)$. A union bound with the energy failure event, followed by Theorem 5.1, gives the stated recovery probability and resource bounds for the joint law of $\mathbf{M}, \mathbf{s}, \mathbf{e}$. $\square$

7.3 Several columns with the same public matrix

FrodoKEM uses the relation $\mathbf{B} = \mathbf{A}\mathbf{S} + \mathbf{E} \pmod{q}$, with a scalar matrix $\mathbf{A}$ and several short columns [Frodo20, Sections 2.2.2, 2.2.4, 2.2.5, and 2.2.9, Algorithm 12]. In our asymptotic family, $\mathbf{A} \leftarrow \text{Unif}(R_q^{n \times n})$ and $\mathbf{S}, \mathbf{E} \in \mathbb{Z}^{n \times t_n}$, where $1 \leq t_n \leq n^c$ for a fixed constant c . Thus

$$\mathbf{B} = \mathbf{A}\mathbf{S} + \mathbf{E} \pmod{q} \tag{7.5}$$

is a collection of relations with one shared operator. The list can therefore be reused.

Corollary 7.5. *Suppose a deterministic rational $V_n \geq 1$ of polynomial bit length satisfies*

$$\Pr \left[\max_{j \leq t_n} (\|\mathbf{s}^{(j)}\|^2 + \|\mathbf{e}^{(j)}\|^2) > nV_n \right] \leq p_n, \tag{7.6}$$

$$D_n(\sqrt{V_n}) \rightarrow \infty.$$

One Gaussian list recovers every column of $\mathbf{S}$ with probability $1 - p_n - o(1)$, with the same asymptotic time and space as Theorem 7.4, normalized by the dimension n of one column. Dependence among the columns and on $\mathbf{A}$ is allowed.

Proof. Use the common certificate $V = V_n$ and $B = \lfloor \sqrt{nV_n} \rfloor$ for every column. As in the proof of Theorem 7.4, Proposition 7.3 supplies the geometry event. On this event, Theorem 5.4 constructs one Gaussian list with the required conditional similarity and implementation error. The final prefix gives (4.9), and (5.19) ensures (4.12) for some $\delta = o(1)$, as verified in the proof of Theorem 5.1. The construction and its error bounds depend on the common operator and certificate. Theorem 4.4 therefore gives one event on which this list recovers every label whose secret and error satisfy that certificate. Intersect this event with the simultaneous energy event in (7.6) and apply a union bound, including the geometry, auxiliary, and implementation errors. The resulting success probability is $1 - p_n - o(1)$ under the stated joint law of the columns and $\mathbf{A}$. Scoring the columns successively reuses the same storage and multiplies the scoring time by at most t_n . Since t_n is polynomial in n , the stated exponent is preserved. $\square$

For a concrete sequence, take

$$b_{q,n} = \lceil \kappa \log_2 n \rceil, \quad q = 2^{b_{q,n}}, \quad \kappa > 1/2. \tag{7.7}$$

If the entries of $\mathbf{S}$ and $\mathbf{E}$ are bounded in absolute value by public positive integers $H_s, H_e = (\ln n)^{O(1)}$, respectively, use

$$B = H_s, \quad V = \max\{1, H_s^2 + H_e^2\}, \quad V_n = V, \quad p_n = 0.$$

These bounds hold simultaneously for all columns. Using $B = H_s$ in Theorem 5.1, the normalized candidate radius is $U = \sqrt{V + 3B^2/n} = \sqrt{V}(1 + O(1/n))$, since $B^2 \leq V$. Thus $D_n(U) \sim D_n(\sqrt{V}) \sim \ln \ln n$, and the expected time and space are

$$2^{(1/2+o(1))n/\ln \ln n}. \quad (7.8)$$

This covers any prescribed family of coefficient tables with such support. The same complexity also holds for centered subgaussian coefficients of polylogarithmic width: the energy estimates of Section 4.4, followed by a union bound over columns, give a polylogarithmic V_n and $p_n = o(1)$. Packing and unpacking the public matrix preserve the residues in (7.5).

Corollary 7.6 (FrodoKEM family specialization). *Let $q = 2^{\lceil \kappa \log_2 n \rceil}$ with fixed $\kappa > 1/2$, and let $t_n \leq n^c$ be the prescribed number of columns. Suppose a public polylogarithmic $V_n \geq 1$ gives the simultaneous joint energy bound nV_n with failure probability $o(1)$. For uniform $\mathbf{A} \in (\mathbb{Z}/q\mathbb{Z})^{n \times n}$, one Gaussian list recovers all columns of $\mathbf{S}$ from $\mathbf{B} = \mathbf{AS} + \mathbf{E}$ with probability $1 - o(1)$, in expected time and space $2^{(1/2+o(1))n/\ln \ln n}$.*

Proof. The candidate-energy substitution of Remark 5.2 gives $U_n = \Theta(\sqrt{V_n})$ and $1 \leq U_n = (\ln n)^{O(1)}$. Since $\ln(q/\sqrt{n}) = (\kappa - 1/2) \ln n + O(1)$,

$$D_n = \ln \frac{\ln(q/\sqrt{n})}{1 + \ln U_n} = \ln \ln n + O(\ln \ln \ln n) \sim \ln \ln n.$$

Corollary 7.5 gives simultaneous recovery on the single energy event. The polynomial number of columns changes neither the leading exponent nor the reusable list space. This proves Corollary 1.2. $\square$

8 Kyber/ML-KEM: Recovery over Prime Cyclotomic Modules

Multiplication in a polynomial ring creates dependencies among matrix entries. A nonzero vector can now have an image supported on a proper ideal, so the uniform-image calculation of Section 6 no longer applies directly. We group the dual vectors by their image ideal. Large ideals give enough randomness. Small ideals force long integer representatives. These two estimates establish the same prefix condition as before. For fixed module rank and splitting degree, polynomial moduli and polylogarithmically bounded coefficients then give recovery in expected time and space $2^{(1/2+o(1))n/\ln \ln n}$.

8.1 The public operator and the modulus restriction

Definition 8.1 (Cyclotomic modules over a prime modulus). Fix a positive integer k , let d tend to infinity through powers of two, and put $n = kd$. For an odd prime q , write

$$O = \mathbb{Z}[X]/(X^d + 1), \quad R_q = O/qO, \quad \delta = \text{ord}_{2d}(q).$$

Choose a family with constant δ and $\ln q \leq (\ln n)^c$ for a fixed c . Sample $\mathbf{A} \leftarrow \text{Unif}(R_q^{k \times k})$. For module vectors $\mathbf{s}_R, \mathbf{e}_R \in O^k$ and $\mathbf{y}_R \in R_q^k$, write $\mathbf{s} = \text{coeff}(\mathbf{s}_R)$, $\mathbf{e} = \text{coeff}(\mathbf{e}_R)$, and $\mathbf{y} = \text{lift}_q^+(\text{coeff}(\mathbf{y}_R))$, concatenating the coefficients of the k polynomial components. Let $\mathbf{M}$ be the $n \times n$ coefficient matrix of multiplication by $\mathbf{A}$. Thus the k ring equations

$$\mathbf{y}_R = \mathbf{As}_R + \mathbf{e}_R \pmod{q} \quad (8.1)$$

are precisely $\mathbf{y} = \mathbf{M}\mathbf{s} + \mathbf{e} \pmod{q}$ in coefficient coordinates. Order the columns of $\mathbf{M}$ first by secret polynomial and then by coefficient degree. All norms are Euclidean norms of these integer coefficient vectors.

For $k = 1$, this is the algebraic form of the ring-LWE relation [LPR13a]. Ring vectors and matrices give the module form [LS15]. The same public relation underlies Kyber/ML-KEM [FIPS203, Section 5.1, Algorithm 13]. Its bounded coefficient distributions will enter after the geometric argument. Fixing the splitting degree forces $q = \Omega(n)$. We prove this restriction in Lemma 8.5, where it determines the complexity of the bounded-coefficient case.

8.2 Ideal images in the coefficient basis

A coefficient prefix can end inside a polynomial block. Its image under ring multiplication is controlled by an ideal, but its allowed inputs need not form a submodule. The mass bound must therefore track both the image ideal and the dimension of a partial block. We use the ring decomposition to count ideals and the original coefficient basis to measure lengths.

The Gaussian mass calculation is related to ring regularity [LPR13b, Theorem 4.1], cyclotomic leftover hashing [BL26], and its Gaussian module counterpart [AFLW26]. Here we need the estimate for every prescribed coefficient prefix, including those with an incomplete polynomial block.

For $0 \leq m \leq n$, let $\mathbf{M}_m$ consist of the first m columns of $\mathbf{M}$, and define the systematic prefix lattice

$$\Lambda_m = \{(\mathbf{u}, \mathbf{v}) \in \mathbb{Z}^n \times \mathbb{Z}^m : \mathbf{M}_m^T \mathbf{u} + \mathbf{v} = 0 \pmod{q}\}. \quad (8.2)$$

In particular, $\Lambda_0 = \mathbb{Z}^n$. The identity block makes the row map surjective and gives the disjoint dual coset decomposition

$$\Lambda_m^* = \bigcup_{\mathbf{z} \in \mathbb{F}_q^m} \left(\mathbb{Z}^{n+m} + \frac{1}{q}(\mathbf{M}_m \mathbf{z}, \mathbf{z}) \right). \quad (8.3)$$

For $\mathbf{z} \in \mathbb{F}_q^m$, let $\tilde{\mathbf{z}} \in R_q^k$ be the module vector obtained by padding $\mathbf{z}$ with zeros and grouping the coefficients into polynomial blocks. The first component $\mathbf{M}_m \mathbf{z}$ is $\text{coeff}(\mathbf{A}\tilde{\mathbf{z}})$.

For $T > 0$, put

$$a_T = \rho_{1/T}(\mathbb{Z}), \quad b_T = \rho_{q/T}(\mathbb{Z}).$$

If $V \leq \mathbb{F}_q^D$ is a subspace of dimension v , then

$$\sum_{\mathbf{z} \in V} \rho_{1/T}(\mathbb{Z}^D + \mathbf{z}/q) \leq a_T^{D-v} b_T^v. \quad (8.4)$$

Indeed, select v information coordinates on which projection is a bijection. Every remaining coordinate contributes a shifted Gaussian mass at most a_T , by one-dimensional Poisson summation and the triangle inequality. Summing the information coordinates over all their residues then gives b_T^v . We shall also use $b_T \leq 1 + q/T \leq 2q/T$ for $T \leq q$.

Lemma 8.2. *Let $I \triangleleft R_q$ be an ideal of dimension w over $\mathbb{F}_q$. If $z \in \mathbb{Z}[X]$ is nonzero, has degree less than d , and satisfies $z \pmod{q} \in I$, then*

$$\|\text{coeff}(z)\| \geq q^{1-w/d}. \quad (8.5)$$

Proof. Let $\mathbf{M}_z$ be the integral negacyclic multiplication matrix of z . Its image modulo q lies in I because I is an ideal, so its rank modulo q is at most w . Smith normal form implies $q^{d-w} \mid \det \mathbf{M}_z$.

As d is a power of two, $X^d + 1$ is irreducible over $\mathbb{Q}$, hence $\det \mathbf{M}_z \neq 0$. Each column of $\mathbf{M}_z$ is a signed permutation of the coefficient vector of z . Hadamard's inequality yields $q^{d-w} \leq |\det \mathbf{M}_z| \leq \|\text{coeff}(z)\|^d$. $\square$

Write $m = jd + s$, with $0 \leq s < d$, using $j = k, s = 0$ at $m = n$. For a nonzero prefix $\mathbf{z} \in \mathbb{F}_q^m$ with $\tilde{\mathbf{z}} = (z_1, \dots, z_k) \in R_q^k$, define

$$g = \gcd(X^d + 1, z_1, \dots, z_k), \quad w = d - \deg g, \quad h_g = jw + \max\{0, s - d + w\},$$

where the gcd is monic. A complete polynomial block in the ideal (g) has w free coefficients. A partial block of degree less than s is an ordinary multiple of g of degree less than s , and contributes $\max\{0, s - \deg g\}$ dimensions. Thus h_g is exactly the dimension of the subspace of prefixes all of whose blocks lie in (g) , and

$$h_g \leq mw/d, \quad h_g \leq kw. \quad (8.6)$$

For any fixed prefix with this exact gcd, each ring component of $\mathbf{A}\tilde{\mathbf{z}}$ is uniform in (g) , and the k components are independent. In fact, the map

$$R_q^k \longrightarrow R_q, \quad (a_1, \dots, a_k) \longmapsto \sum_{t=1}^k a_t z_t$$

has image (g) and equal-size fibres. Independence follows from the independent rows of the ring matrix $\mathbf{A}$.

Let $\mathcal{E}_g$ be the exact-gcd class of nonzero prefixes, and write

$$D_g(T) = \sum_{\mathbf{z} \in \mathcal{E}_g} \mathbb{E}_{\mathbf{A}} \left[\rho_{1/T}(\mathbb{Z}^n + \mathbf{M}_m \mathbf{z}/q) \rho_{1/T}(\mathbb{Z}^m + \mathbf{z}/q) \right]. \quad (8.7)$$

The uniform-image statement and (8.4) bound the expected first factor by $q^{-kw} a_T^{n-kw} b_T^{kw}$. Enlarge the systematic prefix class to its containing subspace of dimension h_g and apply (8.4) once more. For $T \leq q$ this proves

$$\begin{aligned} D_g(T) &\leq q^{-kw} a_T^{n+m-h_g-kw} b_T^{h_g+kw} \\ &\leq q^{h_g} a_T^{n+m-h_g-kw} (2/T)^{h_g+kw}. \end{aligned} \quad (8.8)$$

The first estimate uses the dimension of the image ideal to quantify the averaging over uniform outputs. Larger ideals provide more randomness and will be controlled by this bound. Small ideals instead force long nonzero integer representatives, giving a complementary geometric estimate.

For any point of a nonzero systematic coset, multiplying by q gives a nonzero integer polynomial block whose residue lies in (g) . Lemma 8.2 therefore bounds the norm of this point below by $q^{-w/d}$. Split the Gaussian exponent into two equal parts, use this norm bound on one part, and bound the other shifted mass by the unshifted mass. Each first-component shifted mass is at most a_T^n , and there are at most q^{h_g} prefixes. Consequently,

$$D_g(T) \leq q^{h_g} a_T^n a_{T/\sqrt{2}}^m \exp\left(-\frac{\pi}{2} T^2 q^{-2w/d}\right). \quad (8.9)$$

8.3 Smoothing every prescribed prefix

We now combine the two mass estimates. The logarithmic condition below ensures that the count of large ideals is absorbed by their Gaussian decay. The recovery schedule will satisfy it automatically.

Proposition 8.3. *Assume the public model above. Let L_0 satisfy*

$$\ln n = o(L_0), \quad L_0 = o(n), \quad \ln \left(\frac{n \ln q}{L_0} \right) = o(\ln L_0), \quad \varepsilon_0 = 4ne^{-L_0}. \quad (8.10)$$

Choose deterministic pairs $(m_i, T_i)_{i=1}^r$, independent of $\mathbf{A}$, with $1 \leq m_i \leq n$, $8\sqrt{L_0} \leq T_i \leq q$, and

$$\max_i \frac{m_i \ln(q/T_i)}{n \ln T_i} = o(1). \quad (8.11)$$

Then, for all sufficiently large n ,

$$\Pr_{\mathbf{A}}[\exists i : \eta_{\varepsilon_0}(\Lambda_{m_i}) > T_i] \leq \frac{r}{2n} e^{-2L_0}. \quad (8.12)$$

Proof. Fix one of the prescribed pairs and abbreviate it by (m, T) . Because $T^2 \geq 64L_0$ and $\ln n = o(L_0)$, every integer theta factor in (8.8) and (8.9) is $1 + o(1)$, uniformly in the ideal and the pair. For example, $a_{T/\sqrt{2}} = 1 + O(e^{-\pi T^2/2})$, and its exponent is at most $2n$. Equations (8.8), (8.6), and (8.11) imply

$$\begin{aligned} \ln D_g(T) &\leq o(1) + h_g \ln(q/T) - kw \ln T + (h_g + kw) \ln 2 \\ &\leq o(1) - (1 - o(1))kw \ln T. \end{aligned} \quad (8.13)$$

All error terms in this proof are uniform in the prescribed pairs.

First consider large image ideals. Set $w_* = 8L_0/(k \ln T) = o(d)$. The polynomial $X^d + 1$ is squarefree modulo q , and every irreducible factor has degree δ . The number of dimension- w ideals is therefore $\binom{d/\delta}{w/\delta}$ when $\delta \mid w$, and zero otherwise. For $w \geq w_*$, it is at most $\binom{d}{w}$, and

$$\ln \binom{d}{w} \leq w \ln(ed/w_*) \leq w \ln \left(\frac{en \ln q}{L_0} \right) = o(w \ln T).$$

Here the last equality follows from (8.10) and $\ln T \geq \frac{1}{2} \ln L_0 + \ln 8$. Summing (8.13) over ideals and at most d dimensions yields

$$\sum_{w \geq w_*} \sum_{d - \deg g = w} D_g(T) \leq e^{-6L_0}. \quad (8.14)$$

For the small ideals, suppose $0 < w < w_*$. A nonempty exact-gcd class has $h_g > 0$. The prefix dimension formula then forces $m > d - w > d/2$ eventually. Using (8.11) and fixed k gives

$$\ln(q/T) = o(\ln T), \quad \ln q = (1 + o(1)) \ln T,$$

and hence

$$q^{-2w/d} \geq \exp \left(-\frac{16L_0}{n} \frac{\ln q}{\ln T} \right) = 1 - o(1). \quad (8.15)$$

Since $2d \mid q^\delta - 1$, the number of dimension- w ideals is at most $(d/\delta)^{w/\delta} \leq q^w$. Along with $h_g \leq kw$, this bounds the logarithm of the product of the ideal count and the number of prefixes by

$$(k+1)w_* \ln q = \left(\frac{8(k+1)}{k} + o(1) \right) L_0 \leq (16 + o(1))L_0.$$

The negative exponent in (8.9) is at most $-(32\pi - o(1))L_0$, by (8.15). After summing over the possible dimensions, we obtain

$$\sum_{0 < w < w_*} \sum_{d - \deg g = w} D_g(T) \leq e^{-80L_0}. \quad (8.16)$$

Together, (8.3), (8.14), and (8.16) give

$$\mathbb{E}_{\mathbf{A}} \left[\rho_{1/T}(\Lambda_m^* \setminus \mathbb{Z}^{n+m}) \right] \leq e^{-3L_0}.$$

Lemma 3.6 now proves (8.12). $\square$

8.4 Recovery and the Kyber/ML-KEM relation

The geometry now connects to the recovery theorem in one step. We state the consequence for arbitrary low-energy inputs, then specialize the parameters and coefficient distributions.

Theorem 8.4 (Recovery over a prime modulus). *In Definition 8.1, let a deterministic rational $V \geq 1$ of polynomial bit length satisfy*

$$\Pr[\|\mathbf{s}\|^2 + \|\mathbf{e}\|^2 > nV] \leq p_E. \quad (8.17)$$

Set $U = \sqrt{V}$ and assume

$$D = \ln \frac{\ln(q/\sqrt{n})}{1 + \ln U} \longrightarrow \infty. \quad (8.18)$$

The algorithm of Theorem 5.1 recovers the original integer vector $\mathbf{s}$ from the k ring equations with probability $1 - p_E - o(1)$. Its expected time and space are

$$2^{(1/2+o(1))n/D}. \quad (8.19)$$

The joint law of $\mathbf{s}, \mathbf{e}$ may depend on $\mathbf{A}$.

Proof. Remark 5.2 gives the candidate bound and the same asymptotic depth.

For the schedule of Theorem 5.1, the bounds from (5.5)–(5.13) give $L_0 = \Theta(n/D)$, $T_i = t_i/2 \geq 8\sqrt{L_0}$, and (8.11). Since $\ln q \leq (\ln n)^c$ and $U \geq 1$, we have $D = O(\ln \ln n)$ and

$$\ln \left(\frac{n \ln q}{L_0} \right) = O(\ln \ln n) = o(\ln L_0).$$

Thus all hypotheses of Proposition 8.3 hold for this schedule. Its failure probability is at most $re^{-2L_0}/(2n) = o(1)$. Theorem 5.1 gives recovery uniformly over the stated energy event. $\square$

To specialize the exponent, we first make explicit the restriction imposed by the fixed splitting degree.

Lemma 8.5. *In this family, $q = \Omega(n)$. Consequently, if*

$$\ln(q/\sqrt{n}) = (\ln n)^{\beta+o(1)} \quad (8.20)$$

for a fixed $\beta > 0$, then $\beta \geq 1$.

Proof. Write $2d = 2^u$. The order δ is a power of two. If $\delta = 1$, then $q \equiv 1 \pmod{2d}$, and $q \geq 2d + 1$. Otherwise write $\delta = 2^a$, with $a \geq 1$. The elementary valuation identity gives

$$u \leq v_2(q^{2^a} - 1) = v_2(q - 1) + v_2(q + 1) + a - 1.$$

Exactly one of $v_2(q - 1)$ and $v_2(q + 1)$ equals one. It follows that $q \equiv \pm 1 \pmod{2d/\delta}$ and $q \geq 2d/\delta - 1$. Both k and δ are fixed, so $q = \Omega(n)$. Therefore $\ln(q/\sqrt{n}) \geq \frac{1}{2} \ln n - O(1)$, which excludes $\beta < 1$. $\square$

For fixed coefficient bounds and $\ln(q/\sqrt{n}) = (\ln n)^{\beta+o(1)}$, the expected time and space are

$$2^{(1/(2\beta)+o(1))n/\ln \ln n} \quad (8.21)$$

with $\beta \geq 1$ by Lemma 8.5. A useful broader case has $\ln(q/\sqrt{n}) = \Theta(\ln n)$ and $\|\mathbf{s}\|_\infty + \|\mathbf{e}\|_\infty = (\ln n)^{O(1)}$. The bounded-coefficient estimate in Section 4.4 gives $D = (1 + o(1)) \ln \ln n$, and hence expected time and space

$$2^{(1/2+o(1))n/\ln \ln n}. \quad (8.22)$$

Corollary 8.6. *Fix k . For every power of two $d \geq 2$, let q_d be the least prime satisfying*

$$q_d \equiv d + 1 \pmod{2d}. \quad (8.23)$$

Then Definition 8.1 holds with $\delta = 2$. Polylogarithmically bounded coefficients give the expected time and space in (8.22).

Proof. The residue class is coprime to $2d$, and the least-prime theorem of [Xyl11, Theorem 1.1] gives $d + 1 \leq q_d \leq C(2d)^{5.18}$ for an effective absolute constant C . Moreover $(d + 1)^2 \equiv 1 \pmod{2d}$ and $d + 1 \not\equiv 1 \pmod{2d}$, so $\text{ord}_{2d}(q_d) = 2$ and all irreducible factors have degree two. For $n = kd$, these bounds give $\ln(q_d/\sqrt{n}) = \Theta(\ln n)$. The preceding specialization applies. Enumerating candidates up to the polynomial upper bound and testing primality also constructs this modulus sequence in polynomial time. $\square$

For centered binomial secret and error coefficients with public positive integer parameters $\eta_s, \eta_e = (\ln n)^{O(1)}$, the respective supports are $[-\eta_s, \eta_s]$ and $[-\eta_e, \eta_e]$. Take

$$B = \eta_s, \quad V = \max\{1, \eta_s^2 + \eta_e^2\}.$$

The joint secret and error energy is then deterministically at most nV . The normalized candidate radius in (5.1) is $U = \sqrt{V + 3B^2/n} = \sqrt{V}(1 + O(1/n))$. When $\ln(q/\sqrt{n}) = \Theta(\ln n)$, we obtain $D = (1 + o(1)) \ln \ln n$ and (8.22). For independent coordinates, Corollary 4.7 also gives a probabilistic energy bound. Kyber/ML-KEM key generation uses $\eta_s = \eta_e = \eta_1$. The parameter η_2 is used for encryption errors.

Public data in an invertible transform representation are first returned to coefficient form, as in [FIPS203, Sections 2.4.5 and 4.3]. This preserves the uniform distribution of the public matrix and provides the coefficient representation used throughout the recovery argument.

Corollary 8.7 (Kyber/ML-KEM family specialization). *Let $n = kd$, with fixed k and d growing through powers of two. Take prime $q = n^{\kappa+o(1)}$, fixed $\kappa \geq 1$, and fixed $\text{ord}_{2d}(q)$. For uniform $\mathbf{A} \in R_q^{k \times k}$, suppose a public polylogarithmic $V \geq 1$ gives a joint secret and error energy bound nV with probability $1 - o(1)$. Then the original coefficient secret is recovered from $\mathbf{t} = \mathbf{A}\mathbf{s}_R + \mathbf{e}_R \pmod{q}$ with probability $1 - o(1)$, in expected time and space $2^{(1/2+o(1))n/\ln \ln n}$.*

Proof. The fixed order is the fixed splitting degree in Definition 8.1. With $U = \sqrt{V}$, $1 + \ln U = O(\ln \ln n)$ and $\ln(q/\sqrt{n}) = (\kappa - 1/2 + o(1)) \ln n$. Hence

$$D = \ln \frac{\ln(q/\sqrt{n})}{1 + \ln U} = \ln \ln n + O(\ln \ln \ln n) \sim \ln \ln n.$$

Theorem 8.4 applies with $p_E = o(1)$ to the k ring equations, which supply exactly n coefficient observations. This proves Corollary 1.3. $\square$

9 SABER: Recovery from Rounded Module Products

Rounding a module product gives a bounded integer residue, so the common recovery theorem applies once we prove smoothing for its coefficient prefixes. The new issue is the modulus: modulo two, $X^d + 1 = (X + 1)^d$, and the ideals form a single nested chain. We use this chain to count prefix vectors and bound their Gaussian mass directly in coefficient coordinates.

9.1 The rounded relation and its energy

Fix a positive module rank k , let $d = 2^\nu$ tend to infinity, and put $n = kd$. Set

$$O = \mathbb{Z}[x]/(x^d + 1), \quad R_q = O/qO, \quad q = 2^{b_q}, \quad \mathbf{A} \leftarrow \text{Unif}(R_q^{k \times k}).$$

For the secret module vector $\mathbf{s}_R \in O^k$, write $\mathbf{s} = \text{coeff}(\mathbf{s}_R) \in \mathbb{Z}^n$, concatenating the coefficients of its k polynomial components. Let $\mathbf{M}$ be the coefficient matrix of multiplication by $\mathbf{A}^T$, so that $\text{coeff}(\mathbf{A}^T \mathbf{s}_R) = \mathbf{M}\mathbf{s} \pmod{q}$. Coordinates are ordered by polynomial block and then coefficient degree. All norms below use this basis.

For $p = 2^{b_p}$ with $0 \leq b_p \leq b_q$, set $\Delta = q/p$ and $h_{\text{rnd}} = \lfloor \Delta/2 \rfloor$. The public module vector is $\mathbf{b} \in R_p^k$, with $R_p = O/pO$. Write $\mathbf{b}^+ = \text{lift}_p^+(\text{coeff}(\mathbf{b})) \in \mathbb{Z}^n$. Writing $[z]_q$ for the least nonnegative residue, define its coefficients by

$$b_j^+ = \left\lfloor \frac{[(\mathbf{M}\mathbf{s})_j + h_{\text{rnd}}]_q}{\Delta} \right\rfloor, \quad 0 \leq b_j^+ < p. \quad (9.1)$$

For $\Delta \geq 2$, this is the public rounding map in the SABER specification [SaberSpec, Section 2.3 and Algorithm 1]. The case $\Delta = 1$ is the unrounded extension.

Lemma 9.1. *The computable integer lift $\mathbf{y} = \Delta \mathbf{b}^+$ satisfies*

$$\mathbf{y} = \mathbf{M}\mathbf{s} + \mathbf{e}_{\text{rnd}} \pmod{q}, \quad e_{\text{rnd},j} = h_{\text{rnd}} - [(\mathbf{M}\mathbf{s})_j + h_{\text{rnd}}]_{\Delta}. \quad (9.2)$$

For $\Delta \geq 2$,

$$e_{\text{rnd},j} \in \{1 - \Delta/2, \dots, \Delta/2\}, \quad \|\mathbf{e}_{\text{rnd}}\|^2 \leq n\Delta^2/4. \quad (9.3)$$

For $\Delta = 1$, the residue is zero. More generally, an offset $0 \leq h_{\text{rnd}} < \Delta$ gives $h_{\text{rnd}} - \Delta + 1 \leq e_{\text{rnd},j} \leq h_{\text{rnd}}$.

Proof. Euclidean division gives $[(\mathbf{M}\mathbf{s})_j + h_{\text{rnd}}]_q = \Delta b_j^+ + r_j$ with $0 \leq r_j < \Delta$. Since $\Delta \mid q$, its remainder is $r_j = [(\mathbf{M}\mathbf{s})_j + h_{\text{rnd}}]_{\Delta}$. Rearranging proves the identity and bounds. $\square$

Thus, if $|s_j| \leq B$ and $\|\mathbf{s}\|^2 \leq nV_s$, the common recovery construction uses

$$V = \max\{1, V_s + r_{\text{rnd}}^2\}, \quad R^2 = nV + 3B^2, \quad r_{\text{rnd}} = \lfloor \Delta/2 \rfloor. \quad (9.4)$$

The residue bound holds for every matrix and secret. In particular, the specification's CBD_η secret has $B = \eta$ (its parameter $\mu = 2\eta$), so $V_s = \eta^2$ is a deterministic choice. Independent coefficients also admit the energy bounds of Corollary 4.7.

The resulting expected recovery time and space are $2^{(1/2+o(1))n/\ln \ln n}$ when $q = n^{\kappa+o(1)}$, $\kappa > 1/2$, and both the secret bound and rounding step are polylogarithmic. The remaining task is to prove prefix smoothing for this public matrix distribution.

9.2 Ideals and coefficient prefixes

The element $1-x$ generates the unique maximal ideal of R_q . We first record its ideal powers and the dimensions seen by a partial polynomial block. The underlying ring structure is classical [Dinh05, LM17]. The prefix counts will determine the Gaussian estimate.

Lemma 9.2. *There is a unit $u \in O^\times$ with $2 = u(1-x)^d$. The ideals of R_q are exactly*

$$I_j = (1-x)^j R_q \quad (0 \leq j \leq b_q d), \quad |I_j| = 2^{b_q d - j}, \quad I_{b_q d} = 0.$$

Every nonzero element has a unique valuation $j < b_q d$, meaning that it is $(1-x)^j$ times a unit. For $j = ad + r$ with $0 \leq r < d$,

$$(1-x)^j O = 2^a (1-x)^r O, \quad I_j = 2^a (1-x)^r R_q. \quad (9.5)$$

Proof. If ζ is the image of x in the cyclotomic field, then

$$2 = \Phi_{2d}(1) = \prod_{\substack{1 \leq b < 2d \\ b \text{ odd}}} (1 - \zeta^b).$$

For odd b , $(1 - \zeta^b)/(1 - \zeta)$ is an integral geometric sum of norm one, hence a unit. This proves $2 = u(1-x)^d$. In R_q , $1-x$ is nilpotent and $R_q/(1-x) = \mathbb{F}_2$. Every element outside $(1-x)R_q$ has the form $1 + (1-x)b$ and is invertible by a finite geometric series. Repeated division by $1-x$ therefore writes every nonzero element as a power of $1-x$ times a unit. An element of least valuation generates any nonzero ideal. Multiplication by $1-x$ on O has determinant of absolute value two, so $[O : (1-x)^j O] = 2^j$. Together with $qO = (1-x)^{b_q d} O$, this proves the cardinalities and uniqueness of the valuation. $\square$

The quotient and remainder a, r in (9.5) give a convenient description of this single chain. Define the binary code

$$C_r = (1+x)^r \mathbb{F}_2[x] / ((1+x)^d) \subseteq \mathbb{F}_2^d.$$

We identify ideals and codes with their coefficient embeddings when viewing them as lattices or subsets of coefficient spaces. For $0 \leq r \leq d$, the code has dimension $d-r$ and

$$(1-x)^r O = \{\mathbf{z} \in \mathbb{Z}^d : \mathbf{z} \bmod 2 \in C_r\}. \quad (9.6)$$

Indeed, both lattices contain $2O$, have the same binary image, and have index 2^r in O .

Lemma 9.3. Let pr_t select the first t coefficients, and identify $\mathbb{F}_2^t$ with vectors whose last $d - t$ coordinates are zero. For $0 \leq r < d$ and $0 \leq t \leq d$,

$$\dim \text{pr}_t C_r = \min(t, d - r), \quad \dim(C_r \cap \mathbb{F}_2^t) = \max(0, t - r). \quad (9.7)$$

Consequently, for $j = ad + r < b_q d$,

$$\begin{aligned} |\text{pr}_t I_j| &= 2^{(b_q - a)t - \max(0, t - d + r)}, \\ |I_j \cap (\mathbb{Z}/q\mathbb{Z})^t| &= 2^{(b_q - a)t - \min(t, r)}. \end{aligned} \quad (9.8)$$

Proof. The polynomials $(1 + x)^r x^b$, $0 \leq b < d - r$, form a basis of C_r . For $t \leq d - r$, their first t coefficient rows contain a triangular $t \times t$ submatrix with diagonal entries one. This gives the projection rank, which stays $d - r$ for larger t . A polynomial of degree less than t belongs to C_r exactly when it is $(1 + x)^r h$ with $\deg h < t - r$, giving the intersection rank. By (9.6), a vector of I_j has coefficients divisible by 2^a , its next bit belongs to C_r , and its remaining bits are free. Each allowed pattern on t coordinates has $2^{(b_q - a - 1)t}$ higher-bit lifts, proving the cardinalities. $\square$

Let $V_m \subseteq R_q^k$ consist of vectors supported on the first m coefficients. Write $m = hd + t$, $0 \leq t < d$, with $h = k$, $t = 0$ when $m = n$. For $j = ad + r < b_q d$, set

$$\begin{aligned} v_j(m) &= h(d - r) + \max(0, t - r), \\ c_j(m) &= m - v_j(m) = hr + \min(t, r), \\ H_j(m) &= (b_q - a)m - c_j(m). \end{aligned} \quad (9.9)$$

Here $v_j(m)$ counts the free bits in the lowest allowed binary layer of the prefix, $c_j(m)$ counts its independent binary constraints, and $H_j(m)$ counts all free bits, including the unrestricted higher layers. The preceding lemma gives

$$\begin{aligned} |I_j^k \cap V_m| &= 2^{H_j(m)}, \\ v_j(m) &\leq \frac{m(d - r)}{d} \leq k(d - r), \quad H_j(m) \leq m(b_q - j/d). \end{aligned} \quad (9.10)$$

The two inequalities follow from $\max(0, t - r) \leq t(d - r)/d$ and $\min(t, r) \geq tr/d$. These counts include prefixes ending inside a polynomial.

Lemma 9.4. If $\mathbf{z}_R = (z_1, \dots, z_k) \in R_q^k$ has minimum component valuation j , then the k components of $\mathbf{A}^T \mathbf{z}_R$ are independent and uniform in I_j .

Proof. The map $\mathbf{a} = (a_1, \dots, a_k) \mapsto \sum_b a_b z_b$ has image equal to the ideal generated by the z_b , namely I_j , and has equally sized fibres. Apply this to the independent columns of $\mathbf{A}$. $\square$

9.3 Gaussian mass of the prescribed prefixes

Let $\mathbf{M}_m$ consist of the first m columns of $\mathbf{M}$. The prefix lattice and its dual are

$$\begin{aligned} \Lambda_m &= \{(\mathbf{u}, \mathbf{v}) \in \mathbb{Z}^n \times \mathbb{Z}^m : \mathbf{M}_m^T \mathbf{u} + \mathbf{v} = 0 \pmod{q}\}, \\ \Lambda_m^* &= \mathbb{Z}^{n+m} + q^{-1} \begin{pmatrix} \mathbf{M}_m \\ \mathbf{I}_m \end{pmatrix} \mathbb{Z}^m. \end{aligned} \quad (9.11)$$

For a coefficient prefix $\mathbf{z} \in (\mathbb{Z}/q\mathbb{Z})^m$, let $\tilde{\mathbf{z}} \in V_m$ be the module vector obtained by padding with zeros and grouping the coefficients into polynomial blocks. The systematic coordinates distinguish

the dual cosets, and $\mathbf{M}_m \mathbf{z} = \text{coeff}(\mathbf{A}^T \tilde{\mathbf{z}})$. Thus the preceding ideal image lemma applies directly. Write coeff_m for the first m coordinates of the coefficient embedding, so $\text{coeff}_m(\tilde{\mathbf{z}}) = \mathbf{z}$.

Write $\theta(s) = \rho_s(\mathbb{Z})$. We separate the Gaussian mass of the prefix vectors from the average mass of one output polynomial:

$$G_{j,m}(T) = \sum_{\mathbf{z}_R \in I_j^k \cap V_m} \rho_{1/T}(\mathbb{Z}^m + \text{coeff}_m(\mathbf{z}_R)/q),$$

$$A_j(T) = |I_j|^{-1} \rho_{q/T}((1-x)^j O) \quad (j < b_q d).$$

Lemma 9.5. *For $T > 0$,*

$$\mathbb{E}_{\mathbf{A}} \left[\rho_{1/T}(\Lambda_m^* \setminus \mathbb{Z}^{n+m}) \right] = \sum_{j=0}^{b_q d - 1} A_j(T)^k (G_{j,m}(T) - G_{j+1,m}(T)). \quad (9.12)$$

Here $G_{b_q d, m}(T) = \theta(1/T)^m$. For $j = ad + r < b_q d$ and $Q = 2^{b_q - a}$,

$$A_j(T)^k \leq 2^{kr} Q^{-n} \theta(Q/T)^{n-kr} \theta(Q/(2T))^{kr},$$

$$G_{j,m}(T) \leq \theta(Q/T)^{v_j(m)} \theta(Q/(2T))^{c_j(m)}. \quad (9.13)$$

Proof. For prefix vectors of exact valuation j , the expected Gaussian mass of each output polynomial is

$$|I_j|^{-1} \sum_{y \in I_j} \rho_{1/T}(\mathbb{Z}^d + \text{coeff}(y)/q)$$

$$= |I_j|^{-1} \rho_{q/T}((1-x)^j O) = A_j(T).$$

The k output polynomials are independent. The systematic mass on these vectors is $G_{j,m} - G_{j+1,m}$. Summing proves the identity.

For the bounds, write a coefficient vector in $(1-x)^j O$ as $2^a \mathbf{w}$ with $\mathbf{w} \bmod 2 \in C_r$. Its even and odd coordinate masses are

$$\alpha = \theta(Q/(2T)), \quad \gamma = \theta(Q/T) - \alpha \leq \alpha,$$

where the inequality is the maximum property of a shifted Gaussian theta series. Choose $d - r$ coordinates on which the binary code projects bijectively. Summing over them contributes $(\alpha + \gamma)^{d-r}$. Each remaining coordinate contributes at most α . Hence

$$\rho_{q/T}((1-x)^j O) \leq \theta(Q/T)^{d-r} \theta(Q/(2T))^r.$$

Divide by $|I_j| = Q^d/2^r$ and raise to the k th power for the first bound. Apply the same argument to the binary prefix code $C_r^h \times (C_r \cap \mathbb{F}_2^t)$, of dimension $v_j(m)$, for the second. All these masses use the original coefficient norm. $\square$

For a prefix vector of exact valuation $j = ad + r$, its additive order is $Q = 2^{b_q - a}$: after division by 2^a , at least one coefficient is odd. Denote the corresponding summand in (9.12) by $D_j(m, T)$. Dropping the subtraction and using (9.13) gives

$$D_j(m, T) \leq 2^{kr} Q^{-n} \theta(Q/T)^{n+m-kr-c_j(m)} \theta(Q/(2T))^{kr+c_j(m)}. \quad (9.14)$$

We next sum over the ideal chain, separating the layers into three additive order ranges. The order two range is exceptional. A nonzero systematic label of order two has a coordinate equal to $q/2$, so its normalized prefix is at distance at least $1/2$ from $\mathbb{Z}^m$. This distance gives Gaussian decay directly, while the intermediate and large orders are controlled by (9.14).

Theorem 9.6 (Smoothing for dyadic module prefixes). *Let k be fixed, $n = kd \rightarrow \infty$ with d a power of two, and $q = 2^{b_q}$. Choose L_0 with*

$$L_0 \rightarrow \infty, \quad L_0 = o(n), \quad \ln(nb_q) = o(L_0). \quad (9.15)$$

Before drawing $\mathbf{A}$, prescribe pairs $(m_i, T_i)_{i=1}^r$ satisfying

$$1 \leq m_i \leq n, \quad 8\sqrt{L_0} \leq T_i \leq q, \quad \max_i \frac{m_i \ln(q/T_i)}{n \ln T_i} = o(1). \quad (9.16)$$

For all sufficiently large n , uniformly in these pairs,

$$\mathbb{E}_{\mathbf{A}} \left[\rho_{1/T_i}(\Lambda_{m_i}^* \setminus \mathbb{Z}^{n+m_i}) \right] \leq e^{-3L_0}. \quad (9.17)$$

Consequently, with $\varepsilon_0 = 4ne^{-L_0}$,

$$\Pr_{\mathbf{A}}[\exists i : \eta_{\varepsilon_0}(\Lambda_{m_i}) > T_i] \leq \frac{r}{2n} e^{-2L_0}. \quad (9.18)$$

Proof. Fix (m, T) . Write each index as $j = ad + \rho$, with $0 \leq \rho < d$, and set $Q = 2^{b_q - a}$. There are at most $b_q d$ nonzero layers, which we divide into three ranges.

Order two. For $Q = 2$, put $u = d - \rho$ and $v = v_j(m)$. The output factor in (9.13) is

$$2^{-ku} \theta(2/T)^{ku} \theta(1/T)^{k\rho} = 2^{-ku} (1 + o(1)).$$

A nonzero coefficient prefix has the form $\mathbf{z} = (q/2)\mathbf{w}$ with a nonzero binary vector $\mathbf{w}$. Every point of $\mathbb{Z}^m + \mathbf{w}/2$ has norm at least $1/2$. Splitting its Gaussian exponent into equal halves yields

$$\rho_{1/T}(\mathbb{Z}^m + \mathbf{w}/2) \leq e^{-\pi T^2/8} \theta(\sqrt{2}/T)^m. \quad (9.19)$$

The containing prefix code has 2^v vectors, and $v \leq ku$ by (9.10). Their count cancels the factor 2^{-ku} , giving

$$D_j(m, T) \leq (1 + o(1)) e^{-\pi T^2/8} \leq (1 + o(1)) e^{-8\pi L_0}. \quad (9.20)$$

The factors $1 + o(1)$ are uniform, since $\theta(c/T) = 1 + O(e^{-\pi T^2/c^2})$ for fixed $c > 0$, $T^2 \geq 64L_0$, and $\ln n = o(L_0)$.

Intermediate orders. For $4 \leq Q \leq T$, use $\theta(s) \leq e^{1/8}$ for $0 < s \leq 1$ in (9.14) to obtain

$$D_j(m, T) \leq 2^{k\rho} Q^{-n} e^{(n+m)/8} \leq e^{-(\ln 2 - 1/4)n} = e^{-\omega(L_0)}. \quad (9.21)$$

Here $k\rho \leq n$, $m \leq n$, and $L_0 = o(n)$.

Large orders. For $Q > T$, Poisson summation gives $\theta(Q/T) \leq 2Q/T$, and monotonicity bounds the smaller theta factor by the larger one. Thus

$$\begin{aligned} D_j(m, T) &\leq 2^{3n} Q^m T^{-(n+m)}, \\ \ln D_j(m, T) &\leq 3n \ln 2 + m \ln(q/T) - n \ln T = -(1 - o(1))n \ln T. \end{aligned} \quad (9.22)$$

This is $e^{-\omega(L_0)}$ uniformly by the prescribed-pair conditions.

Summing over at most $b_q d$ layers proves the expected mass bound, since $\ln(b_q d) = o(L_0)$. Lemma 3.6 gives the simultaneous smoothing bound. $\square$

Corollary 9.7. Assume (9.15). A deterministic schedule satisfying

$$\begin{aligned} m_i &= \sum_{j \leq i} b_j \leq n, & 16\sqrt{L_0} \leq t_i \leq q, \\ m_i \ln(q/t_i) &= O(\ln N \ln t_i), & \ln N = o(n) \end{aligned}$$

obeys (9.18) with $T_i = t_i/2$.

Proof. The width bound gives $T_i \geq 8\sqrt{L_0}$, and

$$\frac{m_i \ln(q/T_i)}{n \ln T_i} \leq O(\ln N/n) \frac{\ln t_i}{\ln T_i} + \frac{\ln 2}{\ln T_i} = o(1).$$

Apply Theorem 9.6. □

9.4 Recovery and the effect of rounding

Corollary 9.8. Consider the rounded relation (9.1), with fixed module rank k , $d \rightarrow \infty$ through powers of two, and $\ln q \leq (\ln n)^c$ for a fixed c . Fix a rational $V \geq 1$ and an integer $1 \leq B \leq \lfloor \sqrt{nV} \rfloor$ of polynomial bit length before drawing $\mathbf{A}$, and assume

$$\Pr[\|\mathbf{s}\|_\infty > B \text{ or } \|\mathbf{s}\|^2 + \|\mathbf{e}_{\text{rnd}}\|^2 > nV] \leq p_E.$$

Assume also that

$$R^2 = nV + 3B^2, \quad U = \sqrt{R^2/n}, \quad D = \ln \frac{\ln(q/\sqrt{n})}{1 + \ln U} \rightarrow \infty. \quad (9.23)$$

Then the original integer secret is recovered from exactly k rounded ring equations with probability at least $1 - p_E - o(1)$, in expected time and space $2^{(1/2+o(1))n/D}$. The secret may depend on $\mathbf{A}$.

Proof. Lemma 9.1 supplies the integer relation. For the schedule of Theorem 5.1, the bounds in (5.13) give $L_0 = \Theta(n/D)$, $L_0 = o(n)$, and $\ln(nb_q) = o(L_0)$. Corollary 9.7 therefore supplies its prefix smoothing with failure probability at most $re^{-2L_0}/(2n) = o(1)$. Theorem 5.1 now gives recovery uniformly over the stated energy event. □

For the SABER-type support $|s_j| \leq \eta$ and $\Delta \geq 2$, take $B = \eta$, $V = \eta^2 + \Delta^2/4$, and $p_E = 0$. In particular,

$$\begin{aligned} q &= n^{\kappa+o(1)}, \quad \kappa > 1/2, \quad \eta + \Delta = (\ln n)^{O(1)} \\ \implies D &= (1 + o(1)) \ln \ln n. \end{aligned} \quad (9.24)$$

These families are nonempty: choose $q = 2^{\lceil \kappa \log_2(kd) \rceil}$ and a fixed or polylogarithmic power-of-two divisor $\Delta \geq 2$ of q . The resulting expected time and space are $2^{(1/2+o(1))n/\ln \ln n}$. To quantify coarser rounding, keep $q = n^{\kappa+o(1)}$ with fixed $\kappa > 1/2$, and assume $\eta = O(\Delta)$. Then $U = \Theta(\Delta)$. If $\ln \Delta = (\ln n)^{a+o(1)}$ for fixed $0 < a < 1$, the depth is $D = (1 - a + o(1)) \ln \ln n$. If instead $\Delta = n^{a+o(1)}$ for fixed $a > 0$, then

$$D = \ln \frac{\kappa - 1/2}{a} + o(1).$$

Corollary 9.9 (SABER family specialization). Let $n = kd$, with fixed k and power-of-two $d \rightarrow \infty$, and take power-of-two moduli $p \mid q$ with $q = n^{\kappa+o(1)}$, fixed $\kappa > 1/2$. If $|s_j| \leq \eta$ and $\Delta = q/p \geq 2$ are polylogarithmic, then the original integer secret is recovered from the k rounded ring equations for uniform $\mathbf{A} \in R_q^{k \times k}$ with probability $1 - o(1)$, in expected time and space $2^{(1/2+o(1))n/\ln \ln n}$.

Proof. The public lift has an error satisfying $\|\mathbf{e}_{\text{rnd}}\|^2 \leq n\Delta^2/4$. Thus $B = \eta$, $V = \eta^2 + \Delta^2/4$, and $U^2 = V + 3\eta^2/n$ satisfy the energy condition deterministically. Here $1 \leq U = (\ln n)^{O(1)}$ and $\ln(q/\sqrt{n}) = (\kappa - 1/2 + o(1)) \ln n$, so $D = \ln \ln n + O(\ln \ln \ln n) \sim \ln \ln n$. Corollary 9.8 supplies the result, including the dependence of the rounding error on the public matrix and secret. This proves Corollary 1.4. $\square$

10 NTRU LPRime: Recovery from Rounded Field Products

We use p for the ring degree and $n = p$ for the recovery dimension. A single ring product contains n coefficient observations, but all entries of its multiplication matrix depend on the same public multiplier. In this case the dependence is easy to handle: over a field, multiplying any fixed nonzero element by a uniform element gives a uniform result. This is exactly the property required by Lemma 3.5.

10.1 The public multiplier and its Gaussian mass

Let $p \geq 1$ be an integer, let $q \geq 5$ be prime, and assume that $X^p - X - 1$ is irreducible modulo q . Set $O = \mathbb{Z}[X]/(X^p - X - 1)$ and $R_q = O/qO = \mathbb{F}_q[X]/(X^p - X - 1)$, and draw the ring element G uniformly in R_q . In the basis $(1, X, \dots, X^{p-1})$, let $\mathbf{M}_G$ be the coefficient matrix of multiplication by G . Thus $\text{coeff}(Gc) = \mathbf{M}_G \text{coeff}(c)$ for $c \in R_q$. If $\mathbf{E}_m$ appends zeros to a vector of length m , let $\mathbf{M}_m$ consist of the first m columns of $\mathbf{M}_G$. Then

$$\mathbf{M}_m = \mathbf{M}_G \mathbf{E}_m, \quad \mathbf{M}_m \mathbf{z} = \mathbf{M}_G \mathbf{E}_m \mathbf{z}. \quad (10.1)$$

Thus the prefix lattice is

$$\Lambda_m(G) = \{(\mathbf{u}, \mathbf{v}) \in \mathbb{Z}^n \times \mathbb{Z}^m : \mathbf{M}_m^T \mathbf{u} + \mathbf{v} = 0 \pmod{q}\}. \quad (10.2)$$

Lemma 10.1. *For each prescribed $1 \leq m \leq n$ and each fixed $0 \neq \mathbf{z} \in \mathbb{F}_q^m$, the vector $\mathbf{M}_m \mathbf{z}$ is uniform in $\mathbb{F}_q^n$.*

Proof. The polynomial represented by $\mathbf{E}_m \mathbf{z}$ has degree less than p and is nonzero. It is therefore a nonzero element of the field R_q . Multiplication by it permutes R_q . $\square$

Applying Lemma 3.5 with $\mathbf{A}_m = \mathbf{M}_m$ gives

$$\mathbb{E}_G \left[\rho_{1/T}(\Lambda_m(G)^* \setminus \mathbb{Z}^{n+m}) \right] = q^{-n} \theta(q/T)^n (\theta(q/T)^m - \theta(1/T)^m). \quad (10.3)$$

Corollary 3.7 now supplies the smoothing bound for the sampler's prescribed prefixes. The argument takes place entirely in coefficient coordinates and applies to any irreducible defining polynomial.

10.2 Rounding gives a small error

The NTRU LPRime Core relation [NTRU20b, Sections 2.1 and 2.4.2] uses a ternary secret of weight w , with $0 \leq w \leq p$:

$$\mathcal{S}_w = \{a \in O : \text{coeff}(a) \in \{-1, 0, 1\}^n, \|\text{coeff}(a)\|_0 = w\}. \quad (10.4)$$

For $a \in \mathcal{S}_w$, write $\mathbf{a} = \text{coeff}(a)$. Products with G use the reduction of a modulo q . For an integer z , let $\text{cen}_q(z)$ denote its representative in $[-(q-1)/2, (q-1)/2]$. Define nearest rounding to a multiple of three by

$$\text{rnd}_q(z) = 3 \left\lfloor \frac{\text{cen}_q(z) + 1}{3} \right\rfloor. \quad (10.5)$$

For a ring element $c \in R_q$, define $\text{Round}(c) \in O$ by applying rnd_q to each coefficient. Thus $\text{Round} : R_q \rightarrow O$ has an integer ring output. The public ring element and its integer coefficient observation are

$$A = \text{Round}(aG) \in O, \quad \mathbf{y} = \text{coeff}(A) \in \mathbb{Z}^n. \quad (10.6)$$

Lemma 10.2. *Let $x_j = \text{cen}_q((\mathbf{M}_G \mathbf{a})_j)$ and let $[\cdot]_3$ denote the least nonnegative residue modulo three. Then*

$$\mathbf{y} = \mathbf{x} + \mathbf{e} = \mathbf{M}_G \mathbf{a} + \mathbf{e} \pmod{q}, \quad e_j = 1 - [x_j + 1]_3 \in \{-1, 0, 1\}. \quad (10.7)$$

In particular,

$$\|\mathbf{a}\|^2 + \|\mathbf{e}\|^2 \leq w + n. \quad (10.8)$$

Proof. Write $x_j + 1 = 3t_j + [x_j + 1]_3$. Rounding returns $y_j = 3t_j$, so $y_j - x_j = 1 - [x_j + 1]_3$. Summing squares proves the bound. $\square$

We retain the rounded integer output when forming $\mathbf{e}$. For $q \equiv 5 \pmod{6}$, its extreme values can lie one unit outside the centered interval. Reducing $\mathbf{y}$ modulo q leaves the same bounded error valid in the modular equation.

10.3 Recovery from the given product

The energy bound supplies the common theorem with

$$V = 1 + w/n, \quad B = 1, \quad R^2 = n + w + 3. \quad (10.9)$$

Corollary 10.3. *Fix a parameter sequence (p, q, w) as above, with $n \rightarrow \infty$, before drawing G , and assume*

$$D_0 = \ln \ln(q/\sqrt{n}) \rightarrow \infty, \quad \ln q \leq (\ln n)^c$$

for a fixed c . Given G and $A = \text{Round}(aG)$, the algorithm recovers the integer coefficient vector $\mathbf{a}$, and hence the ring element a , with probability $1 - o(1)$, in expected time and space $2^{(1/2+o(1))n/D_0}$. The secret may depend on G , provided $a \in \mathcal{S}_w$.

Proof. Lemma 10.2 gives the lifted relation with $h = n$ and the deterministic certificate $V = 1 + w/n$, $B = 1$. Thus $R^2 = n + w + 3$ and $U^2 = 1 + w/n + 3/n = O(1)$, giving $D = D_0 - O(1) \sim D_0$. For every prescribed prefix, Lemma 10.1 gives a uniform image for each fixed nonzero label. The schedule bounds (5.13) therefore allow Corollary 3.7 to be applied with $T_i = t_i/2$. Its geometry failure probability is at most $re^{-2L_0}/(2n) = o(1)$. Theorem 5.1 gives the stated recovery probability and resource bounds, including secrets that depend on G . $\square$

In particular, $q = \Theta(n)$ gives $D_0 \sim \ln \ln n$.

For the LPRime family, the supplied degree p is also prime and the parameters satisfy the specification's arithmetic requirements. For prime p , irreducibility of $f = X^p - X - 1$ is checked by $f \mid X^{q^p} - X$ and $\gcd(f, X^q - X) = 1$, using modular polynomial arithmetic.

Corollary 10.4 (NTRU LPRime family specialization). *Let (p, q, w) be a supplied growing family satisfying the LPRime arithmetic requirements, with primes p, q , $q = \Theta(p)$, $X^p - X - 1$ irreducible modulo q , and $0 \leq w \leq p$. For uniform $G \in R_q$ and $a \in \mathcal{S}_w$, the original ternary secret is recovered from $A = \text{Round}(aG)$ with probability $1 - o(1)$, in expected time and space $2^{(1/2+o(1))n/\ln \ln n}$, where $n = p$.*

Proof. Lemma 10.2 gives deterministic joint energy $w + n$. The candidate radius in (10.9) has $U^2 = 1 + w/n + 3/n = O(1)$. Since $q = \Theta(n)$,

$$D_0 = \ln \ln(q/\sqrt{n}) = \ln \ln n + O(1), \quad D = D_0 - O(1) \sim \ln \ln n.$$

Corollary 10.3 applies to the n coefficient observations in this one rounded product. It recovers the integer coefficient vector, hence a itself. This proves Corollary 1.5. $\square$

11 Dilithium/ML-DSA: Recovery from Compressed Public Keys

The ML-DSA public key contains the high bits of a noisy module product. Its discarded low bits become part of the error in the lifted relation, so the same prime-module geometry used for Kyber/ML-KEM applies. Selecting a square block of public equations recovers the original vector $\mathbf{s}_1$; the remaining short algebraic components can then be completed coefficientwise.

11.1 The public relation and its integer lift

We use the key generation relation and the **Power2Round** map of [FIPS204, Algorithms 6 and 35]. Write m for the ring degree and d for the number of discarded public-key bits, and set

$$R = \mathbb{Z}[X]/(X^m + 1), \quad R_q = R/qR, \quad \Delta = 2^d.$$

In the specified parameter sets, $m = 256$, $d = 13$, and the module dimensions (k, ℓ) are $(4, 4), (6, 5), (8, 7)$. The growing family below keeps the module dimensions fixed while the ring degree and modulus grow.

Definition 11.1 (ML-DSA public family). Fix integers $k \geq \ell \geq 1$. Let $m \rightarrow \infty$ through powers of two, put $n = \ell m$, and take a supplied prime modulus $q \equiv 1 \pmod{2m}$ with $q \leq n^{C_q}$ for fixed C_q . Choose public integers $\eta \geq 1$, $d \geq 1$, with $\eta, \Delta = 2^d = (\ln n)^{O(1)}$, before the public draw. The bit count d may be fixed; if it grows, it satisfies $d = O(\ln \ln n)$. Draw $\mathbf{A} \leftarrow \text{Unif}(R_q^{k \times \ell})$ and let $\mathbf{s}_1 \in R^\ell$, $\mathbf{s}_2 \in R^k$ have coefficient absolute values at most η . They may depend on $\mathbf{A}$. The public data are $(\mathbf{A}, \mathbf{t}_1)$, where

$$\mathbf{t} = \mathbf{A}\mathbf{s}_1 + \mathbf{s}_2 \pmod{q}, \quad (\mathbf{t}_1, \mathbf{t}_0) = \text{Power2Round}_d(\mathbf{t}) \quad (11.1)$$

coefficientwise, with $\mathbf{t} \in R_q^k$ and integer outputs $\mathbf{t}_1, \mathbf{t}_0 \in R^k$. The map Power2Round_d is the standard function parameterized by its discarded-bit count d , with the scalar convention below.

For $r \in \mathbb{Z}_q$, let $r^+ \in \{0, \dots, q-1\}$ be its least nonnegative representative and define

$$r_1 = \left\lfloor \frac{r^+ + \Delta/2 - 1}{\Delta} \right\rfloor, \quad r_0 = r^+ - \Delta r_1, \quad \text{Power2Round}_d(r) = (r_1, r_0). \quad (11.2)$$

Thus $-\Delta/2 < r_0 \leq \Delta/2$, equivalently $-2^{d-1} < r_0 \leq 2^{d-1}$, and $r = \Delta r_1 + r_0 \pmod{q}$, including the tie convention in the specification. The first output is stored as an integer. In the encoded public key, $\mathbf{A}$ is reconstructed from its public seed; an NTT representation is returned to coefficient form.

Lemma 11.2 (Lifted error bound). Define $\mathbf{y}_R := 2^d \mathbf{t}_1$ and $\mathbf{e}_R := \mathbf{s}_2 - \mathbf{t}_0$. Then

$$\mathbf{y}_R = \mathbf{A}\mathbf{s}_1 + \mathbf{e}_R \pmod{q}, \quad \|\mathbf{e}_R\|_\infty \leq \eta + \Delta/2. \quad (11.3)$$

Fix any set $J \subseteq [k]$ of ℓ rows before drawing $\mathbf{A}$, and let $\mathbf{A}_J$ denote the corresponding restriction. Set $\mathbf{y} := \text{coeff}((\mathbf{y}_R)_J)$, $\mathbf{e} := \text{coeff}((\mathbf{e}_R)_J)$, and $\mathbf{s} := \text{coeff}(\mathbf{s}_1)$. Then $\mathbf{A}_J$ is uniform in $R_q^{\ell \times \ell}$, and its coefficient matrix $\mathbf{M}_J$ gives $n = \ell m$ observations with deterministic certificate

$$\begin{aligned} \mathbf{y} &= \mathbf{M}_J \mathbf{s} + \mathbf{e} \pmod{q}, \\ \|\mathbf{s}\|_2^2 + \|\mathbf{e}\|_2^2 &\leq nV, \quad V = \eta^2 + (\eta + \Delta/2)^2, \quad B = \eta. \end{aligned} \quad (11.4)$$

Proof. Subtract $\mathbf{t}_0$ from (11.1) and use (11.2). The coefficient bound on $\mathbf{s}_2$ and $|r_0| \leq \Delta/2$ give (11.3). A predetermined restriction to ℓ rows of a uniform ring matrix is a uniform square ring matrix. Both $\mathbf{s}_1$ and the restricted error have n coefficients, so summing their squared bounds gives (11.4). These statements hold for every realized secret and rounding output. $\square$

11.2 Recovery of the original secret component

Theorem 11.3 (ML-DSA secret recovery). *For the family in Definition 11.1, a randomized algorithm recovers the original integer coefficient vector of $\mathbf{s}_1$ from $(\mathbf{A}, \mathbf{t}_1)$ with probability $1 - o(1)$ over the public matrix and independent algorithmic randomness. Its expected time and space are*

$$2^{(1/2+o(1))n/\ln \ln n}, \quad n = \ell m. \quad (11.5)$$

Proof. Fix $J = \{1, \dots, \ell\}$, so the algorithm uses ℓ of the available k ring equations. Lemma 11.2 supplies the uniform square module operator and its energy certificate with $p_E = 0$. The modulus congruence gives splitting degree $\text{ord}_{2m}(q) = 1$, so Theorem 8.4 applies with its degree parameter d set to the present m and its module rank set to ℓ . The dependence of $\mathbf{e}$ on the public matrix and the secret is included in that theorem.

For the depth in Theorem 8.4, put

$$U = \sqrt{V}, \quad D = \ln \frac{\ln(q/\sqrt{n})}{1 + \ln U}.$$

The bounds $q \geq 2m + 1$ and $q \leq n^{C_q}$ imply $\ln(q/\sqrt{n}) = \Theta(\ln n)$, since ℓ is fixed. Moreover $1 \leq U = (\ln n)^{O(1)}$, whence

$$D = \ln \ln n + O(\ln \ln \ln n) \sim \ln \ln n.$$

The theorem therefore recovers the original $\mathbf{s}_1$ with the stated probability and resource bound. Forming the public lift, selecting the rows, and reading the remaining public data cost only polynomial time. This proves Corollary 1.6. $\square$

11.3 Completion of the short algebraic components

The recovered vector also determines a compatible choice of the other short algebraic components of the public-key relation.

Proposition 11.4 (Coefficientwise completion). *Given $(\mathbf{A}, \mathbf{t}_1)$ and the recovered original $\mathbf{s}_1$, a deterministic polynomial-time algorithm finds $\mathbf{s}'_2, \mathbf{t}'_0 \in R^k$ such that*

$$\begin{aligned} \|\mathbf{s}'_2\|_\infty &\leq \eta, \quad -\Delta/2 < (t'_0)_j \leq \Delta/2 \quad \text{for every coefficient } j, \\ \text{Power2Round}_d(\mathbf{A}\mathbf{s}_1 + \mathbf{s}'_2) &= (\mathbf{t}_1, \mathbf{t}'_0), \\ \mathbf{A}\mathbf{s}_1 + \mathbf{s}'_2 &= \Delta \mathbf{t}_1 + \mathbf{t}'_0 \pmod{q}. \end{aligned} \quad (11.6)$$

Thus the tuple $(\mathbf{s}_1, \mathbf{s}'_2, \mathbf{t}'_0)$ has the specified coefficient bounds and reproduces the same compressed public key. Its construction and all checks use polynomial bit complexity.

Proof. Compute $\mathbf{a} = \mathbf{A}\mathbf{s}_1 \pmod{q}$. For each coefficient position j , enumerate $z \in \{-\eta, \dots, \eta\}$ and retain the first one for which the first component of $\text{Power2Round}_d(a_j + z)$ equals $(t_1)_j$. The original coefficient $(s_2)_j$ is a valid choice, so this search always succeeds. Set $(s'_2)_j = z$ and let $(t'_0)_j$ be the second component of the same rounding operation. Equation (11.2) gives all the asserted bounds and identities. There are km positions and $2\eta + 1$ trials per position, with $O(\ln q + \ln \Delta)$ -bit rounding arithmetic. Since η is polylogarithmic, the work is polynomial. Recomputing the rounded product verifies the output exactly. $\square$

12 Recovering Short Pairs from Public Quotients

For a public matrix $\mathbf{H} \in (\mathbb{Z}/q\mathbb{Z})^{m \times m}$, define

$$\Lambda_{\mathbf{H}} = \ker_q[\mathbf{H} \mid -\mathbf{I}_m] = \{(\mathbf{u}, \mathbf{v}) \in \mathbb{Z}^{2m} : \mathbf{H}\mathbf{u} = \mathbf{v} \pmod{q}\}.$$

The Falcon parameter family uses the multiplication matrix of h . Cyclic NTRU uses that of $\hat{h} = 3^{-1}h \pmod{q}$. In both cases, recovery seeks a nonzero short pair passing public algebraic checks.

Algorithmic picture. The algorithm fills the coordinates of $\mathbf{u}$ one at a time, completes $\mathbf{v}$ from the public relation, and checks the resulting pair. Dual Gaussian lists estimate the weights used to choose each coordinate. The public mass bound sets the required accuracy and number of repetitions.

An anchor is a fixed positive coordinate $u_i = a$, with $1 \leq a \leq \lfloor B \rfloor$. Since $B < q/2$, every nonzero pair of norm at most B has $\mathbf{u} \neq \mathbf{0}$, so either it or its negative agrees with some anchor. For each position i , fix an order for the other coordinates. These m orders use at most m sets of free coordinates each, hence at most m^2 sets in total. Each set has its own dual Gaussian list, reused across coordinate values and repetitions.

Suppose the current free set is S and the next coordinate is u_j . For a candidate value b , let A_b contain all lattice pairs that agree with the coordinates already fixed and with $u_j = b$. Its Gaussian mass sums the weights of all possible completions. The ideal choice is

$$\Pr[u_j = b \mid \text{fixed coordinates}] \propto \rho_{s_{\text{search}}}(A_b),$$

over integers $b \in [-\lfloor B \rfloor, \lfloor B \rfloor]$ whose fixed coordinates have squared norm at most B^2 . All these child slices have the direction lattice for $S \setminus \{j\}$. Their estimates use the same list with different shifts in the cosine average. The common mass denominator cancels when the weights are normalized. The algorithm samples one value from the estimated weights and continues along that path.

Once $\mathbf{u}$ is complete, set $\mathbf{v} = \text{lift}_q^{\text{cen}}(\mathbf{H}\mathbf{u})$. Retain the pair and its negative if the lattice, nonzero, and norm checks pass. Repeat the search T times for every anchor using the same lists and fresh choices. Scan the resulting table with the public verifier to obtain a pair satisfying the application's algebraic conditions.

For a fixed short witness, the ideal transition probabilities are bounded below by ratios of successive slice masses. Their product telescopes to a lower bound from the witness's Gaussian weight and the anchor mass. The estimates preserve this bound up to a constant factor, and repetition gives high success probability. Algorithm 4 gives the search procedure. Sections B.1 and B.2 of Appendix B give the list construction and search analysis.

12.1 From Gaussian mass to verifiable search

The following criterion combines the public mass bound with the list construction, search, and verification steps above.

Fix computable rational constants $0 < c < 1$, $1/2 < \beta < 1$, and $A_q, A_0, C_B, c_{\text{search}} > 0$. Use the finite schedule of Theorem B.2, with

$$0 \leq \ln \ln m - D_Q \leq 2^{-m^2}, \quad (1 - 2^{-m^2})q/m^\beta \leq s_{\text{mass}}^2 \leq q/m^\beta, \\ s_{\text{search}}^2 = c_{\text{search}} q D_Q / m.$$

The quantities D_Q, s_{mass}^2 are positive dyadic rationals. The widths s_{mass} and s_{search} control the mass bound and the search distribution, respectively. The constant $c_{\text{search}} > 0$ is a fixed scheduling constant.

The supplied public bound A_{mass} on $\rho_{s_{\text{mass}}}(\Lambda_{\mathbf{H}})$ sets the list accuracy and number of search repetitions.

Theorem 12.1 (Mass criterion). *Let $m \rightarrow \infty$ and*

$$2m + 1 \leq q \leq m^{A_q}, \quad \ln(q/m) \leq (\ln m)^{1-c}.$$

Given $\mathbf{H}$, public rational $A_{\text{mass}} \geq 1$ and B_Q^2 of polynomial bit length, and a polynomial-time verifier $\mathbf{V}$, suppose

$$\rho_{s_{\text{mass}}}(\Lambda_{\mathbf{H}}) \leq A_{\text{mass}}, \quad \ln A_{\text{mass}} \leq A_0 m / D_Q, \\ s_{\text{mass}}^2 \geq 8s_{\text{search}}^2,$$

and $1 \leq B_Q \leq C_B \sqrt{q} < q/2$. If $\mathbf{V}$ accepts some nonzero $\mathbf{z} \in \Lambda_{\mathbf{H}}$ with $\|\mathbf{z}\| \leq B_Q$, then for every fixed $K > 0$ a finite randomized algorithm outputs such a vector with probability at least $1 - m^{-K}$, in time and space $2^{O(m/D_Q)} = 2^{O(m/\ln \ln m)}$.

Proof. Monotonicity gives $\rho_{s_{\text{search}}}(\Lambda_{\mathbf{H}}) \leq A_{\text{mass}}$. For the search of Theorem B.3, take $A = A_{\text{mass}}$, $B = B_Q$, and

$$p_B = A_{\text{mass}}^{-1} e^{-\pi B_Q^2 / s_{\text{search}}^2} \geq \exp\left(-(A_0 + \pi C_B^2 / c_{\text{search}})m / D_Q\right).$$

Allocate $m^{-(K+2)}$ to each of preprocessing, list implementation, categorical sampling, query error, and missed-witness error. With $H = \lfloor B_Q \rfloor$, set $\xi = p_B / [200m(2H + 1)]$. Equations (B.17) and (B.18) allow a list length $N = p_B^{-2} \text{poly}(m)$ and a repetition count $T = p_B^{-1} \text{poly}(m)$ at these error budgets. Each slice query scans N samples. There are at most $m^2 H(2H + 1)T$ queries, giving total query cost $NT \text{poly}(m)$, including finite precision. Since $\ln(1/p_B) = O(m/D_Q)$ and H is polynomial in m , this cost is $\exp(O(m/D_Q))$, and $\ln(1/\xi) = O(m/D_Q)$. Choose the constants a, A_N in Theorem B.2 so that its lists meet these bounds and $\delta_{\text{out}} \leq \xi/16$. Section B.2 implements these bounds by rounding accuracy thresholds downward and list sizes and repetition counts upward, with fixed constant margins, using the finite sampling tools of Section A.1.

Fix one accepted witness for the analysis. Theorem B.3 puts it and its negative in the candidate table except with the sum of the five budgets, at most m^{-K} for large m . Scan the table and return the first entry accepted by $\mathbf{V}$. Lattice membership, nonzeroness, and the norm bound are checked exactly. Retaining both signs permits an arbitrary verifier. The table and all checks cost $\exp(O(m/D_Q))$. $\square$

Algorithm 4 uses the prepared dual lists to produce the candidate table.

Algorithm 4: AFFINESLICESearch: short pair candidate table

Input: A public matrix $\mathbf{H} \in (\mathbb{Z}/q\mathbb{Z})^{m \times m}$, dual Gaussian lists, a norm bound $1 \leq B < q/2$, and repetition count T .

Output: A table $\mathcal{C}(\mathbf{H})$ of nonzero short pairs in $\Lambda_{\mathbf{H}}$.

```
1 Initialize  $\mathcal{C}(\mathbf{H}) \leftarrow \emptyset$ ;
2 foreach coordinate  $i \in [m]$  and positive anchor  $a \in [1, \lfloor B \rfloor]$  do
3   for  $t \leftarrow 1$  to  $T$  do
4     Set  $u_i = a$ , leave the other  $\mathbf{u}$  coordinates free, and use fresh randomness for this run;
5     while some  $\mathbf{u}$  coordinate remains free do
6       Evaluate the affine slice masses for next values in  $[-\lfloor B \rfloor, \lfloor B \rfloor]$  whose fixed
7         coordinates have squared norm at most  $B^2$ ;
8       if the total nonnegative certified mass is zero then
9         | Declare this run failed and continue with the next repetition;
9       | Sample the next value proportionally to the nonnegative certified masses;
10    Set  $\mathbf{v} \leftarrow \text{lift}_q^{\text{cen}}(\mathbf{H}\mathbf{u})$ ;
11    if  $(\mathbf{u}, \mathbf{v}) \in \Lambda_{\mathbf{H}} \setminus \{\mathbf{0}\}$  and  $\|(\mathbf{u}, \mathbf{v})\| \leq B$ , checked exactly then
12      | Add  $(\mathbf{u}, \mathbf{v})$  and its negative to  $\mathcal{C}(\mathbf{H})$ ;
13 return  $\mathcal{C}(\mathbf{H})$ ;
```

12.2 Falcon: recovery from the key generation quality bound

In this subsection, the Falcon parameter family uses $R = \mathbb{Z}[x]/(x^n + 1)$ for a power of two n and a prime $q \equiv 1 \pmod{2n}$. A quality-qualified public key admits the algebraic and quality witness in (1.2). The quality quantity γ_q is defined in (B.27).

Theorem 12.2 (Falcon recovery). *Fix rational constants $C \geq 1$ and $0 < c < 1$. Let $n = 2^k \rightarrow \infty$ and let $q \equiv 1 \pmod{2n}$ be prime, with*

$$\ln(q/n) \leq (\ln n)^{1-c}.$$

For every fixed $K > 0$, there is a classical randomized algorithm with the following guarantee for all sufficiently large n . Given only (n, q, h) , C , and the prescribed public parameters, suppose $h \in R_q$ admits $f, g, F_0, G_0 \in R$ satisfying

$$\begin{aligned} f \bmod q &\in R_q^\times, & h &= gf^{-1} \pmod{q}, \\ fG_0 - gF_0 &= q, & \gamma_q(f, g)^2 &\leq C^2 q. \end{aligned}$$

The algorithm returns $u, v, F, G \in R$ such that

$$\begin{aligned} u \bmod q &\in R_q^\times, & vu^{-1} &= h \pmod{q}, \\ uG - vF &= q, & \gamma_q(u, v)^2 &\leq C^2 q, \end{aligned} \tag{12.1}$$

with probability at least $1 - n^{-K}$ over its own randomness, uniformly for every such public key. In particular, $\|(u, v)\|_2 \leq C\sqrt{q}$, and the completed coefficient basis has maximum Gram–Schmidt norm at most $C\sqrt{q}$. Its time and space are $2^{O(n/\ln \ln n)}$. The hidden constants may depend on C, c, K and the fixed public schedule constants.

Proof. The quality witness supplies a full basis with Gram–Schmidt lengths at most $C\sqrt{q}$. The public lattice satisfies $q\Lambda_h^* = \mathbf{J}\Lambda_h$ for an orthogonal map $\mathbf{J}$. Its reversed dual basis therefore supplies another basis of Λ_h with Gram–Schmidt lengths at least $\sqrt{q}/C$. Successive one-dimensional Gaussian sums give

$$\rho_s(\Lambda_h) \leq \theta(Cs/\sqrt{q})^{2n}, \quad \theta(t) = \sum_{a \in \mathbb{Z}} e^{-\pi a^2/t^2}.$$

At $s_{\text{mass}}^2 \leq q/n^\beta$, this gives the public mass budget $A_{\text{mass}} = 2$ for all sufficiently large n . Theorem 12.1, with exact algebraic and quality verification, then proves recovery. Appendix B.3.2 gives the full proof. Its shared ingredients are the lattice duality in Lemma B.4 and the integer basis completion procedure in Appendix B.3.1. Since $D_Q \sim \ln \ln n$, this is the probability and resource bound of Corollary 1.8, with the same algebraic and quality guarantees on the output. $\square$

Every distribution supported on quality-qualified public keys inherits the same success probability and complexity. This includes KeyGen outputs satisfying the exact algebraic and quality checks above.

12.3 Cyclic NTRU: distributional short key recovery

Use the NTRU-HPS/HRSS families in Section 1.2, with the raw and accepted laws of Definition B.22. The full cyclic lattice may contain additional points in the rational span of the short pair. Their contribution is measured by

$$\Delta_{\text{sat}} = [R_N : fR_N + gR_N + kR_N], \quad k = (\hat{h}f - g)/q.$$

Proposition B.16 gives

$$\rho_{s_{\text{mass}}}(\Lambda_h^{\text{cyc}}) \leq \Delta_{\text{sat}} \theta(s_{\text{mass}}/\sqrt{G_{\min}})^N \theta(s_{\text{mass}}\sqrt{G_{\max}}/q)^N.$$

Here $G_{\min}, G_{\max}$ are scalar spectral extrema of $|f(\zeta)|^2 + |g(\zeta)|^2$ over $\zeta^N = 1$, not ring elements. The search needs only a total logarithmic budget $O(N/D_N)$. Corollaries B.17 and B.18 make this pointwise criterion explicit. For instance, it is enough that

$$G_{\min} \geq (q/N^\beta) \ln D_N, \quad G_{\max} \leq qN^\beta / \ln D_N, \quad \ln \Delta_{\text{sat}} \leq C_{\text{sat}} N/D_N.$$

For the HPS and HRSS samplers, the analysis establishes

$$G_{\min} \geq q/N^\alpha, \quad G_{\max} \leq C_{\text{sp}} q \ln N, \quad \ln \Delta_{\text{sat}} \leq N/D_N, \quad 2/3 < \alpha < \beta < 1.$$

These bounds satisfy the sufficient conditions above for all sufficiently large N and give a fixed public mass budget. The proof combines the spectral estimates with a bound on the logarithm of the saturation index, then conditions on key generation acceptance. Appendix B.4.6 collects this joint event and its failure probability. Fix rational $2/3 < \alpha < \beta < 1$, positive integers K, K_{sp} , and write

$$\begin{aligned} r_{\text{sp}}(N) &= C \left(N^{1-3\alpha/2} + N^{-\alpha/2} + N^{-K_{\text{sp}}} \right), \\ r_{\text{sat}}(N) &= 64D_N \ln(3N) e^{-\sqrt{\ln N}/4} + 12N^{-1/2}, \quad a = 86/256. \end{aligned} \quad (12.2)$$

Here D_N is the prescribed positive dyadic approximation to $\ln \ln N$. The effective constant C depends only on the fixed modulus window and the chosen spectral parameters.

Theorem 12.3 (HRSS recovery). *Let N run along the supplied arithmetic sequence (B.55), and use its HRSS modulus and correctness conditions. Generate h from the accepted law μ_N^{HRSS} . For all sufficiently large N , a randomized algorithm using only (N, q, h) and the fixed public parameters returns a pair accepted by $\text{Valid}_{\text{HRSS}}$ and its short key core with probability at least*

$$1 - \frac{r_{\text{sp}}(N) + r_{\text{sat}}(N)}{(1 - a^{N-1})^2} - N^{-K} = 1 - o(1). \quad (12.3)$$

Its time and space are $2^{O(N/\ln \ln N)}$.

Proof. Proposition B.34 proves that, except with the first failure term in (12.3), the generating pair satisfies $G_{\min} \geq q/N^\alpha$, $G_{\max} \leq C_{\text{sp}} q \ln N$, and $\ln \Delta_{\text{sat}} \leq N/D_N$. Because $\alpha < \beta$, these imply the weak spectral conditions for all sufficiently large N : $N^{\beta-\alpha} \geq \ln D_N$ and $C_{\text{sp}} \ln N \ln D_N \leq N^\beta$. Corollary B.18 therefore supplies the public integer budget

$$A_{\text{mass}} = 4^{\lceil 3N/D_N \rceil}, \quad \ln A_{\text{mass}} \leq 6N/D_N. \quad (12.4)$$

It has $O(N/D_N)$ bits and is fixed before the key is sampled.

Apply Theorem 12.1 with $m = N$, $c_{\text{search}} = 1$, $A_0 = 6$, $B_Q^2 = C_{\text{len}}^2 q$, and the actual dyadic width s_{mass}^2 from the finite schedule. The modulus $q = \Theta(N)$ meets its window, and

$$\frac{s_{\text{mass}}^2}{s_{\text{search}}^2} \geq (1 - 2^{-N^2}) \frac{N^{1-\beta}}{D_N} \rightarrow \infty.$$

Lemma B.15 supplies a fixed rational C_{len} , a valid nonzero witness, and the exact verifier. The radius satisfies $C_{\text{len}} \sqrt{q} < q/2$ eventually. For each fixed witness on the good event, fresh algorithmic randomness succeeds with probability at least $1 - N^{-K}$. Averaging over the accepted input law proves (12.3). The algorithm uses the public budget, radius, and quotient. $\square$

Theorem 12.4 (HPS recovery). *Use the supplied arithmetic sequence (B.55) with the HPS weight conditions, and generate h from the ideal accepted law μ_N^{HPS} based on uniform sampling of fixed type vectors. A randomized algorithm with the same public inputs returns a pair accepted by $\text{Valid}_{\text{HPS}}$ and its short key core with probability at least*

$$1 - \frac{r_{\text{sp}}(N) + r_{\text{sat}}(N)}{1 - a^{N-1}} - N^{-K} = 1 - o(1), \quad (12.5)$$

in time and space $2^{O(N/\ln \ln N)}$.

Proof. The independent ternary and uniform fixed type blocks satisfy the spectral bound of Proposition B.33 and the saturation bound of Proposition B.29. Lemma B.23 gives acceptance probability $1 - a^{N-1}$. Proposition B.34 therefore bounds the input failure probability by $(r_{\text{sp}}(N) + r_{\text{sat}}(N))/(1 - a^{N-1})$.

On the resulting event, the spectral and saturation bounds give the public budget (12.4), as in the proof of Theorem 12.3. Apply Theorem 12.1 with $m = N$, $c_{\text{search}} = 1$, $A_0 = 6$, and $B_Q^2 = C_{\text{len}}^2 q$. The modulus $q = \Theta(N)$ meets its window, and the finite widths satisfy $s_{\text{mass}}^2/s_{\text{search}}^2 \rightarrow \infty$. Lemma B.15 supplies the HPS witness and its exact verifier, with $C_{\text{len}} \sqrt{q} < q/2$ eventually. For every fixed input on this event, the search succeeds with probability at least $1 - N^{-K}$. Averaging over the accepted HPS law and adding the input failure proves (12.5) and the stated resource bounds. $\square$

Corollary 12.5 (HPS recovery with sorting labels). *For the same HPS parameter sequence, generate h using the sorting sampler with label length b_N and accepted law μ_{N,b_N}^{HPS} . The algorithm succeeds with probability at least*

$$1 - \frac{r_{\text{sp}}(N) + r_{\text{sat}}(N) + \varepsilon_{\text{FT}}(N)}{1 - a^{N-1}} - N^{-K}. \quad (12.6)$$

For $b_N = \lceil (K_{\text{tag}} + 3) \log_2 N \rceil$, with any fixed positive integer K_{tag} , this is $1 - o(1)$ and the time and space bound remains $2^{O(N/\ln \ln N)}$.

Proof. Lemma B.24 pays ε_{FT} once for the entire raw good event. Algebraic acceptance still depends only on the nonzero f block and has probability $1 - a^{N-1}$. Thus Proposition B.34 gives exactly the input failure term displayed above. The same public search budget and verifier give the remaining N^{-K} term. The chosen labels have $O(\log N)$ bits and $\varepsilon_{\text{FT}} = o(1)$.

For all three accepted laws, $r_{\text{sp}} + r_{\text{sat}} = o(1)$ and $D_N \sim \ln \ln N$. The two theorems and this corollary therefore prove Corollary 1.9 for the NTRU-HPS/HRSS family. $\square$

13 Conclusion and Further Discussion

We developed two frameworks for recovery in growing families associated with the seven lattice candidates in NIST’s third round. Gaussian estimates play a central role in both frameworks. For noisy or rounded linear relations, they reveal the energy difference created by a coordinate guess. The joint energy of the secret and error determines the sampling depth and the leading recovery exponent. For quotient relations, estimates of the mass in affine slices guide the search for a short pair satisfying the public checks. A bound on the total Gaussian mass controls the probability of reaching such a pair.

The common approach separates the recovery procedure from the geometry of the public operator. The applications supply this geometry through uniform images, ideals, additive orders, key quality, or spectral estimates. This separation gives a way to study further public relations by identifying the energy and mass bounds they require. Appendix D also examines how dimension and energy together determine the cost of composing a reduction with the recovery algorithm. We next describe an application to homomorphic encryption and several directions for further work.

13.1 Further Applications to Homomorphic Encryption

The framework for linear relations also applies to RLWE public keys in homomorphic encryption. BGV, BFV, and CKKS use closely related public relations [BGV12, FV12, CKKS17]. Let N be a power of two, and write

$$R = \mathbb{Z}[X]/(X^N + 1), \quad R_Q = R/QR.$$

Here Q is the modulus of the public key relation. It corresponds to q_L in the CKKS key generation described in [CKKS17, Section 3.4]. In a residue number system representation, write $Q = \prod_j q_j$ with distinct primes q_j . These primes are individual components, whereas the CKKS notation q_ℓ denotes the full modulus at level ℓ . More generally, the argument below uses any known prime divisor $q_i \mid Q$. Write $R_{q_i} = R/q_i R$ for the selected component.

Table 2 gives the three public relations and their normalization at the selected prime. In the table, $a, b \in R_Q$ retain their public key meanings, and a is uniform in R_Q . The parameter t is the plaintext modulus of BGV. We use $\tilde{a}$ for the normalized multiplier and y for the normalized public observation.

Table 2: Public key relations in homomorphic encryption and their normalization at a selected prime.

Scheme	Public relation modulo Q	Normalization modulo q_i
BGV	$b = as + te$	Set $\tilde{a} = t^{-1}(a \bmod q_i)$ and $y = t^{-1}(b \bmod q_i)$ when t is invertible modulo q_i .
BFV	$b = -(as + e)$	Set $\tilde{a} = a \bmod q_i$ and $y = -b \bmod q_i$.
CKKS	$b = -as + e$	Set $\tilde{a} = -a \bmod q_i$ and $y = b \bmod q_i$.

After reduction and normalization, each relation takes the form

$$y = \tilde{a}s + e \pmod{q_i}.$$

These transformations make $\tilde{a}$ uniform in R_{q_i} and retain the same integer secret and error. Reduction to a prime component introduces no additional rounding error. This gives a module instance of rank one to which the recovery theorem can be applied, with ring degree and secret coefficient dimension both equal to N . One ring equation supplies all N coefficient observations.

Let N grow through powers of two, take $q_i \equiv 1 \pmod{2N}$, and assume $\ln q_i \leq (\ln N)^c$ for a fixed c . Choose a public rational energy bound $V \geq 1$ of polynomial bit length before the public draw, with

$$\Pr[\|s\|_2^2 + \|e\|_2^2 > NV] \leq p_E, \quad D_i = \ln \frac{\ln(q_i/\sqrt{N})}{1 + \frac{1}{2} \ln V}.$$

When $D_i \rightarrow \infty$, Theorem 8.4 gives recovery of the original integer secret with probability $1 - p_E - o(1)$ in expected time and space

$$2^{(1/2+o(1))N/D_i}.$$

For $q_i = N^{\kappa+o(1)}$ with fixed $\kappa \geq 1$ and polylogarithmic V , this becomes $2^{(1/2+o(1))N/\ln \ln N}$. The selected prime determines the recovery depth. Reading the full public input adds polynomial work when the number of components and their bit lengths are polynomial in N .

In the ring LWE version of BGV's basic key generation, $t = 2$ [BGV12, Section 3.1]. The division by t in the table is performed modulo the selected prime, so the energy bound uses e . When this division is unavailable, set $\tilde{a} = a \bmod q_i$ and $y = b \bmod q_i$. The resulting relation has effective error te , and the energy certificate uses $\|s\|_2^2 + \|te\|_2^2$. For CKKS, the energy comes from the secret and the public key error in KeyGen [CKKS17, Section 3.4]. The plaintext encoding scale and later rescaling operations do not enter this public relation. Recovery identifies the secret of the selected public key, which can then be used with the corresponding decryption algorithm.

13.2 Further Directions

Implementation and concrete costs. Our algorithms use conservative parameter choices to ensure provable recovery. A natural next step is to implement these algorithms with suitable parameters and measure their performance on small instances. For linear recovery, the experiments can measure the score gap between adjacent candidates. For quotient recovery, they can compare estimated slice masses with direct calculations and record the number of trials needed to find a valid pair. These measurements would support a cost model for tuning block sizes, Gaussian widths, list lengths, and search repetitions in practice.

Heuristic improvements. Gaussian peak estimation could accelerate linear recovery by locating each secret coordinate from a few score evaluations, followed by a local search around the predicted value.

Extreme pruning could reduce the search in quotient recovery through tighter norm budgets at intermediate depths. Estimated Gaussian masses could then guide which of the remaining partial assignments to retain. Varying the pruning thresholds and randomizing branch selection across restarts could explore alternative paths, with every completed pair checked by the full public verifier.

Other public distributions and rings. Further geometric estimates could extend the frameworks to public operators generated by seed expansion, other coefficient distributions, and reducible polynomial rings beyond the cyclotomic cases treated here. Appendix C gives conditions for transferring a geometric guarantee between public distributions. Another route is to estimate the Gaussian mass directly under the new distribution. For linear relations, the main task is to control the required coefficient prefixes. For quotient relations, it is to obtain a useful mass bound and an efficient test for the desired output.

Gram matrix public keys. HAWK [DPPvW22] provides a contrast to the seven NIST lattice families considered here. Its public key has the form $\mathbf{Q} = \mathbf{B}^* \mathbf{B}$, where $\mathbf{B} \in \text{SL}_2(R)$ is a secret basis over a cyclotomic ring R whose degree is a power of two, and $*$ denotes conjugate transpose. Recovery seeks a basis $\mathbf{B}'$ with the same Gram matrix. The linear framework requires a noisy or rounded linear relation with a public operator, while HAWK publishes the quadratic Gram relation above. The quotient framework requires a formulation as a search for a short pair in a public modular lattice, together with a suitable Gaussian mass bound.

HAWK lies outside the scope of our two frameworks, but not beyond the reach of lattice cryptanalysis. Straznickas and Weis reduce HAWK key recovery in degree n to polynomially many calls to exact SVP in dimension at most $n/2 + 1$ [SW26, Theorem 6.1]. Following this attack, HAWK was withdrawn from NIST’s additional signature standardization process on 29 July 2026. It remains interesting to determine whether Gaussian mass estimates can also guide recovery from this quadratic public relation and what geometric information such a search would require.

Together, these directions connect the geometric framework with the concrete behavior of recovery algorithms and with further forms of public information.

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A Finite Implementation

We give finite sampling tools for both recovery frameworks, followed by the schedule and numerical precision for linear recovery. Sections B.1 and B.2 specify the calls and budgets for quotient recovery.

A.1 Common sampling tools

Finite random choices. To choose uniformly from $\{0, \dots, S-1\}$, draw $\lceil \log_2 S \rceil$ unbiased bits and reject values at least S . Each trial succeeds with probability at least $1/2$. For at most $J \geq 1$ calls and $0 < \nu_{\text{bits}} < 1$, a cap of $\lceil \log_2(J/\nu_{\text{bits}}) \rceil$ trials per call gives total abort probability at most ν_{bits} . Successful calls are exactly uniform. For $S = 1$, return zero. For K nonnegative rational weights with positive sum, partition $[0, 1)$ by their normalized cumulative sums and locate $U/2^a$ for a uniform integer $0 \leq U < 2^a$. This uses a random bits and has total variation error at most $K2^{-a}$. A zero sum is a failure.

Lemma A.1 (Scalar Gaussian masses and samples). *For rational $\lambda, a, c, s^2, \zeta$ with $\lambda, s^2 > 0$ and $0 < \zeta < 1/2$, put $\sigma = s/\lambda$ and $Z = \rho_s(a + \lambda\mathbb{Z} - c)$. One can compute a positive dyadic interval $[Z_-, Z_+]$ containing Z with $Z_+ - Z_- \leq \zeta Z_-$, and sample within total variation distance ζ of $D_{a+\lambda\mathbb{Z}, s, c}$, in*

$$\text{poly}(L_{\text{enc}}, 1 + \sigma + \sigma^{-1}, \ln(1/\zeta))$$

bit operations. Here L_{enc} bounds the input encoding lengths.

Proof. Write $(c-a)/\lambda = k + \delta$, with $k \in \mathbb{Z}$ and $|\delta| \leq 1/2$. Then $Z = \sum_j e^{-\pi(j-\delta)^2/\sigma^2}$. Set $A = \lceil \sigma^{-2} \rceil$ and $\ell = 2^{-2A}$. The nearest term and a Gaussian integral bound give, for $T \geq 1$,

$$Z \geq e^{-\pi/(4\sigma^2)} \geq \ell, \quad \sum_{|j| > T} e^{-\pi(j-\delta)^2/\sigma^2} \leq (1 + 4\sigma) e^{-\pi T^2/(8\sigma^2)}.$$

For the tail bound, use $|j - \delta| \geq |j|/2$ and split the exponent. Choose an integer $T = O(1 + \sigma\sqrt{A + \ln((1 + \sigma)/\zeta)})$ so the tail is at most $\zeta\ell/100$. Evaluate the retained terms in nonnegative dyadic intervals with a common denominator and total width at most $\zeta\ell/100$. Sum lower endpoints for Z_- and upper endpoints plus a tail bound of at most $2\zeta\ell/100$ for Z_+ . Then $Z_- \geq (1 - 2\zeta/100)\ell > 0$ and the required relative width follows.

Use the lower endpoints as sampling weights and allocate $\zeta/100$ to their finite random choice. Truncation and weight approximation contribute at most $3\zeta/100$, giving total error less than ζ . Return $a + \lambda(k + j)$, whose distance from c is at most $\lambda(T + 1/2)$. All bounds use outward rounding. The precision is $O(A + \ln(T + 1) + \ln(1/\zeta))$ bits. Certified exponential evaluation and the bound on T give the stated cost. $\square$

Lemma A.2 (Auxiliary Gaussian masses and samples). *Let $q \geq 2$ be an integer, P a prime, and $\Gamma = q\mathbb{Z}^b + (q/P)\mathbf{z}\mathbb{Z}$ for a nonzero $\mathbf{z} \in \mathbb{F}_P^b$. For rational $\mathbf{c}, s^2, \xi$ with $s^2 > 0$ and $0 < \xi < 1/2$, one can enclose $\rho_s(\Gamma - \mathbf{c})$ in a positive dyadic interval of relative width at most ξ , and sample within total variation distance ξ of $D_{\Gamma, s, \mathbf{c}}$, in*

$$P \text{poly}(b, L_{\text{enc}}, 1 + s/q + q/s, \ln P, \ln(1/\xi))$$

bit operations, where L_{enc} bounds the input encoding lengths.

Proof. The nonzero vector $\mathbf{z}$ has order P , so Γ is the disjoint union of the cosets $qk\mathbf{z}/P + q\mathbb{Z}^b$. Their masses are

$$W_k(\mathbf{c}) = \prod_{j=1}^b \rho_s(q\mathbb{Z} + qkz_j/P - c_j), \quad 0 \leq k < P. \quad (\text{A.1})$$

Evaluate each factor by Lemma A.1 at accuracy $\zeta = \xi/(100Pb)$. Products of the interval endpoints enclose each W_k with relative width at most $(1 + \zeta)^b - 1 \leq 2b\zeta$. Their sums give the mass interval. Select a coset using the lower weights, then sample its coordinates independently at accuracy ζ . With $\xi/10$ for the finite coset choice, the total error is at most $2b\zeta + \xi/10 + b\zeta < \xi$. There are Pb scalar mass calculations and b samples. A common binary precision keeps products and cumulative weights within the stated cost. $\square$

Both sampling bounds hold for every rational center and include truncation and finite random choices. Lemma 3.4 therefore bounds the error of adaptive calls with fresh randomness by the sum of their assigned accuracies.

A.2 Finite implementation for linear recovery

A deterministic finite schedule. All calculations precede the public draw and use certified intervals. A failed check reports that the parameters are outside the proved range. Choose an integer $f_{\text{sq}} \in [K(R^2/n)D, K(R^2/n)D + 2]$ by rounding the upper endpoint of an interval of width less than one. Check $f_{\text{sq}} \geq 4$, set $f^2 = f_{\text{sq}}$, and choose dyadic $\bar{\tau} \in [\tau/4, \tau]$, where $\tau = e^{-\pi R^2/f_{\text{sq}}}$. Put

$$\bar{g} = \bar{\tau}/f_{\text{sq}}, \quad \varepsilon = \frac{\bar{\tau}^2}{256f_{\text{sq}}q^4}, \quad \varepsilon_0 = \varepsilon/5.$$

Since $f_{\text{sq}}^{-1} \leq 1 - e^{-\pi/f_{\text{sq}}} \leq 4f_{\text{sq}}^{-1}$, we have $g/16 \leq \bar{g} \leq g$. Set $L = \ln(4n/\varepsilon)$ and $L_0 = L + \ln 5$, and check $L > 0$ and $q > \sqrt{L}$ before evaluating W . Recompute the schedule with these tolerances, to which the prefix condition (5.2) refers. For r , floor a certified lower bound within $1/4$ of (5.10). This can decrease r by one and gives $512L_0 \leq s_0^2 < 2048L_0$. Take a lower bound $\tilde{W}$ within $1/4$ of W , check $\tilde{W} > C_0 + 1$ and $1 \leq r \leq n$, and set

$$N = 2^{\lceil n/[2(\tilde{W} - C_0 - 1)] \rceil}, \quad U_0 = N/2.$$

Check $N \geq 4$. Compute rational bounds and capacities

$$0 \leq \tilde{y}_i - y_i \leq n^{-3}, \quad 0 \leq \ln U_0 - \tilde{\ell} \leq n^{-3}, \quad \tilde{B}_i = \left\lfloor \frac{\tilde{\ell}}{\tilde{y}_i} \right\rfloor,$$

$$\tilde{b}_* = \min \left\{ \left\lfloor \frac{1}{2} \min_i \tilde{B}_i \right\rfloor, \lfloor n/r \rfloor \right\}.$$

Check $\tilde{b}_* \geq 1$ and $\sum_i \tilde{B}_i \geq n$. Assign $\tilde{b}_*$ coordinates to each level and fill the rest without exceeding $\tilde{B}_i$. The rounding directions preserve the width inequalities, and $r\tilde{b}_* \leq n$ and $\tilde{B}_i \geq 2\tilde{b}_*$ hold by construction. The continuous approximation loses at most $O(rn^{-3} \ln U_0) = o(1)$ total capacity and the floors lose at most r . The margin $C_0 \geq C_{\text{cap}} + 10$ ensures that all checks pass for sufficiently large members of the family. Lemma 5.3 gives $\tilde{b}_* = \Omega(\ln N / \ln q)$. All squared widths $s_i^2 = q^2 2^{i-r} / f_{\text{sq}}$ are rational.

Coordinate size and score precision. Reduce centers modulo q and obtain a rational basis of Γ_i from the Hermite normal form of $[P\mathbf{I}_{b_i} \mid \mathbf{z}_i]$, scaled by q/P . Abort if an initial or centered shifted sample has norm exceeding $R_{\text{cap}} = 2^{n^2}$. The inclusions $q\mathbb{Z}^b \subseteq \Gamma_i \subseteq (q/P)\mathbb{Z}^b$ give the uniform ideal tail bound

$$e^{-\pi R_{\text{cap}}^2/(2s^2)}(1 + \sqrt{2}sP/q)^b e^{\pi b q^2/(4s^2)}.$$

For the denominator, use a point of $q\mathbb{Z}^b$ within $q\sqrt{b}/2$ of the center. For the numerator, split the exponent and sum over the containing fine product lattice. Poisson summation and $\theta(v) \leq 1 + v$ bound a shifted scalar mass at width to spacing ratio v by $1 + v$. The initial product Gaussian obeys the same type of bound. In the modulus window, all cutoff events together cost at most e^{-n^2} . Applying the same cutoff to both processes preserves their total variation bound. Retained coordinates have polynomial bit length through $r = O(\ln q)$ levels.

Compute phases modulo q exactly and each cosine to error $\bar{g}/64$ with a common binary denominator. Lemma 3.1, with $u = g/16$, gives failure at most $2He^{-Ng^2/512}$ for $H = n(2B + 1)q^h$. The errors from concentration, Gaussian approximation, conditional means, and cosine evaluation have total at most

$$g/16 + \bar{g}/1280 + \bar{g}/128 + \bar{g}/64 < g/8.$$

Thus two score errors leave a strict margin below the adjacent gap g , with precision $O(\ln(1/\bar{g})) = O(n/D)$ bits.

Auxiliary vectors and initialization. Choose a nonzero vector in $\mathbb{F}_P^{b_i}$ by expanding $U + 1$ in base P , with U uniform in $\{0, \dots, P^{b_i} - 2\}$. The rule of Section A.1, with independent random bits and total budget $\nu_z = n^{-10}$ over r levels, gives independent uniform vectors conditional on no abort. For the $M_{\text{init}} = 3^r h N$ initial coordinates, use Lemma A.1 with $\lambda = 1$, $a = c = 0$, and $\xi_{\text{init}} = 2^{-n^2}/(10M_{\text{init}})$. The total error is at most $\nu_{\text{init}} = 2^{-n^2}$. Since $s_0 = O(\sqrt{n/D})$ and $\ln(1/\xi_{\text{init}}) = O(n^2)$, initialization costs $M_{\text{init}} \text{poly}(n) = N^{1+o(1)}$.

The two implementations. Theorem 5.4 verifies the ADS15 width condition for every center and gives the error (5.18). The polynomial bit lengths above absorb the input factors in Lemma 3.3, preserving (5.3). For its computable uniform expected cost bound $\bar{T}_n$, stop after $n^{K_t} \bar{T}_n$ steps. Markov's inequality adds at most n^{-K_t} failure and preserves the exponent.

For direct sampling, apply Lemma A.2 to each Γ_i at width s_{i-1} . There are $J = (3^{r+1} - 3)N/2$ shifted calls. Give these calls and each initial scalar draw accuracy $\xi = \nu/[10(J + M_{\text{init}})]$, with $\nu = n^{-10}$. Lemma 3.4 gives total error at most ν . Since $1 \leq s_{i-1} \leq q$, each shifted call costs $P \text{poly}(n, q, \ln P, \ln(1/\xi))$. The numerical factor is absorbed by $\ln q = \text{polylog}(n) = o(n/D)$, and enumeration contributes the factor P , giving (5.4). The complete implementation budgets in (5.20) are

$$\delta_{\text{impl}} \leq \begin{cases} \nu_z + \nu_{\text{DGS}} + e^{-n^2} + n^{-K_t}, & \text{ADS15 sampler,} \\ \nu_z + \nu + e^{-n^2}, & \text{coset enumeration.} \end{cases}$$

B Proofs for Quotient Recovery

We first construct dual Gaussian lists and use them to search affine slices of a public lattice. We then bound its Gaussian mass and verify the output for the Falcon parameter family and cyclic NTRU. These are the three steps of the recovery method in Section 12.

In the generic construction and search proof below, n denotes the block length m of Section 12; the full pair lattice has dimension $2n$. We use $\mathbf{H}_h$ and $\mathcal{C}(h)$ as local notations for that section's

public matrix $\mathbf{H}$ and candidate table $\mathcal{C}(\mathbf{H})$, even for an arbitrary matrix. In the family applications, the matrix is coefficient multiplication by h for Falcon or by $\hat{h}$ for cyclic NTRU.

B.1 Persistent Gaussian lists

Let $\mathbf{H}_h$ be an integral representative of any public $n \times n$ matrix modulo q , and set $\Lambda_h = \ker_q[\mathbf{H}_h \mid -\mathbf{I}_n]$. One random rank-one extension will serve every coordinate prefix. This shared extension lets a single smoothing event control all lists. We remove it at the end by taking differences in predetermined pairs. The construction applies to both multiplication matrices used later.

For $S \subseteq [n]$, let $\mathbf{H}_{h,S}$ contain the columns of $\mathbf{H}_h$ indexed by S , and put

$$L_S = \{(\mathbf{u}_S, \mathbf{v}) \in \mathbb{Z}^{|S|+n} : \mathbf{H}_{h,S}\mathbf{u}_S - \mathbf{v} = \mathbf{0} \pmod{q}\}.$$

Let $\mathbf{U}_S : \mathbb{R}^{|S|} \oplus \mathbb{R}^n \rightarrow \mathbb{R}^n \oplus \mathbb{R}^{|S|}$ be the orthogonal block swap $\mathbf{U}_S(\mathbf{u}_S, \mathbf{v}) = (\mathbf{v}, \mathbf{u}_S)$, and define

$$K_S := \mathbf{U}_S(qL_S^*) = \ker_q[\mathbf{H}_{h,S}^T \mid \mathbf{I}_{|S|}] \subseteq \mathbb{Z}^{n+|S|}. \quad (\text{B.1})$$

Dualizing gives

$$qK_S^* = \mathbf{U}_S L_S, \quad \rho_{q/u}(qK_S^*) \leq \rho_{q/u}(\Lambda_h). \quad (\text{B.2})$$

The inequality uses the zero-padded inclusion $L_S \subseteq \Lambda_h$.

To remove an extension $\Lambda' \supseteq \Lambda$ of index $[\Lambda' : \Lambda] = P$, use the following fixed-pair procedure. Given a list $(\mathbf{X}_1, \dots, \mathbf{X}_{2M})$, pair entries $(\mathbf{X}_{2j-1}, \mathbf{X}_{2j})$. A pair succeeds when its difference lies in Λ . Scan pairs in an order fixed independently of their values and return the first N successful differences. If fewer than N pairs succeed, declare failure.

For the analysis, put $\mu = D_{\Lambda',s}$ and let K be the law of $\mathbf{X} - \mathbf{X}'$ conditioned on membership in Λ , for independent $\mathbf{X}, \mathbf{X}' \leftarrow \mu$. Complete a deficient output with independent K -samples to define a reference list of length N . This completion describes the reference law on the failure event.

Lemma B.1 (Quotient removal). *Suppose the input list is strongly conditionally δ -similar to μ^{2M} and*

$$\eta_\varepsilon(\Lambda') \leq s/\sqrt{2}, \quad 0 < \varepsilon \leq \frac{1}{2}.$$

Then the completed reference list is strongly conditionally

$$4\delta + \log \frac{1+\varepsilon}{1-\varepsilon} \quad (\text{B.3})$$

similar to $D_{\Lambda, \sqrt{2}s}^N$. Moreover, for $0 < \zeta < 1$,

$$M \geq 8e^{2\delta} P(N + \ln(1/\zeta)) \quad (\text{B.4})$$

implies that the finite procedure and completed reference disagree with probability at most ζ . The similarity conclusion also holds after conditioning on any positive-probability event determined by the vector of pair-success indicators, in particular on obtaining at least N successes.

Proof. Write $p_{\mathbf{c}} = \Pr[\mathbf{X} \in \Lambda + \mathbf{c}]$ for each quotient class represented by $\mathbf{c}$. The ideal success probability is $C = \sum_{\mathbf{c}} p_{\mathbf{c}}^2 \geq 1/P$. For $\mathbf{z} \in \Lambda$, completing the square gives

$$K(\mathbf{z}) \propto \rho_{\sqrt{2}s}(\mathbf{z}) \rho_{s/\sqrt{2}}(\Lambda' - \mathbf{z}/2).$$

Poisson summation and the smoothing hypothesis place the second factor, uniformly in $\mathbf{z}$, in one multiplicative interval of ratio $(1+\varepsilon)/(1-\varepsilon)$. Thus K has the stated pointwise distance from $D_{\Lambda, \sqrt{2}s}$.

Given all other original entries, each fixed pair is 2δ -similar to $\mu^{\otimes 2}$. Conditioning that pair on success pays the same normalization factor once more, so its successful difference is 4δ -similar to K . To pass to the output list, condition first on the complete vector of success indicators and on the original entries of all other pairs. The selected pair positions are now fixed, and conditioning the current pair on success gives exactly the bound just proved. Averaging over these conditions gives the same bound conditional on the other output differences. Padding entries are independent K -samples. An additional event determined by the success indicators is already fixed under this conditioning, so averaging over its allowed indicator vectors preserves the same bound. This proves (B.3) and its conditional form. Finally, each pair has conditional success probability at least $e^{-2\delta}/P$. The conditional Chernoff bound and (B.4) bound the shortage event by ζ . $\square$

For the list construction, fix the following finite widths and accuracy parameters. Fix computable rational constants $0 < c < 1$, $1/2 < \beta < 1$, and positive $a, K, A_q, A_0, A_N, A_S, c_{\text{search}}$. Let n, q satisfy

$$2n + 1 \leq q \leq n^{A_q}, \quad \ln(q/n) \leq (\ln n)^{1-c}.$$

Choose a positive dyadic rational D with $0 \leq \ln \ln n - D \leq 2^{-n^2}$ and a positive dyadic rational s_{mass}^2 with

$$(1 - 2^{-n^2})q/n^\beta \leq s_{\text{mass}}^2 \leq q/n^\beta,$$

and put

$$s_{\text{search}}^2 = c_{\text{search}}qD/n, \quad t = q/s_{\text{search}}.$$

Theorem B.2 (Persistent lists). *Use the parameters above. For a public matrix $\mathbf{H}_h$, suppose a computable rational upper bound A_{mass} is supplied with*

$$\rho_{s_{\text{mass}}}(\Lambda_h) \leq A_{\text{mass}}, \quad \ln A_{\text{mass}} \leq A_0 n/D, \quad s_{\text{mass}}^2 \geq 8s_{\text{search}}^2. \quad (\text{B.5})$$

Let $\mathcal{S}$ contain at most n^{A_S} coordinate sets, fixed before the sampler randomness, and let $N_{\text{out}} \geq 1$ be an integer with $\ln N_{\text{out}} \leq A_N n/D$. For all sufficiently large n , a finite randomized procedure produces an N_{out} -entry list in each L_S^ with the following guarantees.*

- List law. *Conditional on successful reference preprocessing, each reference list is strongly conditionally*

$$\delta_{\text{out}} \leq n^{-K} e^{-an/D}$$

similar to $D_{L_{S,1/s_{\text{search}}}^}^{N_{\text{out}}}$. Different lists may be dependent. Reference lists use the analytical completion in Lemma B.1.*

- Implementation. *Preprocessing failure and total variation error for the complete construction are each at most n^{-K} . Total time and space over all prescribed sets are $\exp(O(n/D))$.*

Proof. One mask for all prefixes. Write $u_0 = q/s_{\text{mass}}$. Choose an auxiliary prime $P_0 > q$, draw independently

$$\mathbf{a}_0 \leftarrow \mathbb{F}_{P_0}^n \setminus \{\mathbf{0}\}, \quad \mathbf{b}_0 \leftarrow \mathbb{F}_{P_0}^n,$$

and, using fixed integer representatives, set

$$\gamma_S = \frac{q}{P_0}(\mathbf{a}_0, (\mathbf{b}_0)_S), \quad \widetilde{K}_S = K_S + \mathbb{Z}\gamma_S.$$

The quotient $\widetilde{K}_S/K_S$ has order exactly P_0 . If $S_{i-1} \subseteq S_i$, deletion of the new coordinates maps $\widetilde{K}_{S_i}$ onto $\widetilde{K}_{S_{i-1}}$, with kernel $q\mathbb{Z}^{|S_i \setminus S_{i-1}|}$. Indeed, surjectivity follows by extending the integral $K_{S_{i-1}}$ part while retaining the same multiple of γ . If an element lies in the kernel, its first n coordinates

are an integer vector plus $kq\mathbf{a}_0/P_0$. Since $\mathbf{a}_0 \neq \mathbf{0}$, this forces $k = 0 \pmod{P_0}$, after which the systematic equation gives the stated kernel.

Fix j with $(\mathbf{a}_0)_j \neq 0 \pmod{P_0}$. For $\mathbf{x} = (\mathbf{x}_0, \mathbf{x}_S) \in \widetilde{K}_S$, its common quotient label is

$$\ell_S(\mathbf{x}) = (q(\mathbf{a}_0)_j)^{-1} P_0 (\mathbf{x}_0)_j \pmod{P_0}.$$

For the full set, direct duality gives

$$\widetilde{K}_{[n]}^* = \{\mathbf{y} \in K_{[n]}^* : \langle q\mathbf{y}, (\mathbf{a}_0, \mathbf{b}_0) \rangle = 0 \pmod{P_0}\}.$$

If $q\mathbf{y} \not\equiv \mathbf{0} \pmod{P_0}$, its survival probability is at most $1/P_0$: use uniform $\mathbf{b}_0$ if the last block is nonzero and otherwise uniform nonzero $\mathbf{a}_0$, for which the probability is $(P_0^{n-1} - 1)/(P_0^n - 1) < 1/P_0$. Since $qK_{[n]}^*$ is orthogonally equal to Λ_h ,

$$\mathbb{E}[\rho_{1/u_0}(\widetilde{K}_{[n]}^* \setminus \{0\})] \leq \frac{\rho_{s_{\text{mass}}}(\Lambda_h) - 1}{P_0} + \theta(s_{\text{mass}}/P_0)^{2n} - 1. \quad (\text{B.6})$$

Choose a positive dyadic δ and set $\varepsilon_\star$ by

$$\begin{aligned} \frac{1}{2}n^{-K-10}e^{-(a+10)n/D} &\leq \delta \leq n^{-K-10}e^{-(a+10)n/D}, \\ \varepsilon_\star &= \frac{\delta}{10^{44r}}, \end{aligned} \quad (\text{B.7})$$

where r is fixed below. Choose $P_0 = \exp(O(n/D))$ large enough that each term on the right of (B.6) is at most $\varepsilon_\star n^{-K-10}/2$. This is possible because $\ln A_{\text{mass}} \leq A_0 n/D$. A prime in the required constant-factor interval exists and may be chosen above the polynomially bounded q . Markov's inequality gives

$$\eta_{\varepsilon_\star}(\widetilde{K}_{[n]}) \leq u_0 \quad (\text{B.8})$$

except with probability n^{-K-10} . Projection of the full persistent lattice onto every $\widetilde{K}_S$ is surjective, so zero-padding embeds $\widetilde{K}_S^*$ into $\widetilde{K}_{[n]}^*$. The single event (B.8) therefore implies

$$\eta_{\varepsilon_\star}(\widetilde{K}_S) \leq u_0 \quad \text{for every } S \subseteq [n], \quad (\text{B.9})$$

without a union bound over coordinate sets.

Widths and block capacity. Choose the width schedule

$$r = \left\lceil \log_2 \frac{t^2}{4u_0^2} \right\rceil, \quad s_0 = \frac{t}{\sqrt{2}2^{r/2}}, \quad s_i = 2^{i/2}s_0, \quad u_i = s_{i-1}/\sqrt{2}.$$

The final condition in (B.5) makes $r \geq 1$. Then $s_0 \in [\sqrt{2}u_0, 2u_0)$, $s_r = t/\sqrt{2}$, and

$$r = (1 - \beta) \log_2 n - \log_2 D + O(1), \quad u_i \geq u_0.$$

For a compact capacity calculation put

$$x_i = \ln(q/u_i), \quad x_i \leq X_i \leq x_i + 2, \quad H_0 = \sum_{i=1}^r \frac{1}{X_i + 4}, \quad \ell = \ln(4/\varepsilon_\star).$$

Put

$$L_P = \max\{\ell, 4(n+r)/H_0, 2 \ln n + 20\}, \quad X_P = \lceil e^{L_P} \rceil.$$

By Bertrand's postulate, choose a prime P with $X_P < P < 2X_P$. Then $\ln P \geq L_P$ and $\ln P = L_P + O(1)$. Set

$$B_i = \left\lfloor \frac{\ln P}{2(X_i + 4)} \right\rfloor.$$

Then $\sum_i B_i \geq n$. Moreover, for every $1 \leq d \leq B_i$, writing $y = (\ln P)/d$ gives $y \geq 2(x_i + 4)$ and $\ell/d \leq y$, whence

$$16qP^{-1/d} \sqrt{\max\{1, \ell/d\}} \leq u_i, \quad P^{1/d} > 16, \quad \sqrt{d} P^{1/d} \leq P. \quad (\text{B.10})$$

The first inequality follows by taking logarithms. The bounds $y \geq 8$ and $d \leq n$ give the other two. The arithmetic progression x_i has endpoints

$$x_1 = \frac{1}{2}(\ln(q/n) + (1 - \beta) \ln n) + O(1), \quad x_r = \frac{1}{2}(\ln(q/n) + \ln D) + O(1).$$

Integral comparison and the modulus window give

$$H_0 = \Theta\left(\ln \frac{1 + \ln(q/n) + (1 - \beta) \ln n}{1 + \ln(q/n) + \ln D}\right) = \Theta(D). \quad (\text{B.11})$$

Since $\ell = O(n/D)$, this proves $\ln P = O(n/D)$.

Certified auxiliary lifts. For each S , choose a chain $\emptyset = S_0 \subseteq S_1 \subseteq \cdots \subseteq S_r = S$ with $d_i = |S_i \setminus S_{i-1}| \leq B_i$, allowing $d_i = 0$. Such a chain exists by the capacity bound. For $d_i > 0$, independently and uniformly draw $\mathbf{z}_i \leftarrow \mathbb{F}_P^{d_i} \setminus \{\mathbf{0}\}$ and define

$$\Gamma_i = q\mathbb{Z}^{d_i} + \frac{q}{P}\mathbb{Z}\mathbf{z}_i, \quad K'_i = \widetilde{K}_{S_i} + (0 \oplus \Gamma_i).$$

There are exact sequences

$$0 \longrightarrow \Gamma_i \longrightarrow K'_i \longrightarrow \widetilde{K}_{S_{i-1}} \longrightarrow 0, \quad K'_i / \widetilde{K}_{S_i} \simeq \mathbb{Z}/P\mathbb{Z}. \quad (\text{B.12})$$

The first map deletes the new block. The second assertion follows from $\Gamma_i/q\mathbb{Z}^{d_i} \simeq \mathbb{Z}/P\mathbb{Z}$.

Lemma A.2 implements each shifted draw on Γ_i at width s_{i-1} . Reduce the center modulo q and translate the output back. To certify smoothing, use Poisson summation:

$$F_i := \rho_{1/u_i}(\Gamma_i^* \setminus \{0\}) = \frac{q^{d_i}}{P u_i^{d_i}} \rho_{u_i}(\Gamma_i) - 1. \quad (\text{B.13})$$

Each shifted scalar mass is at most the centered mass, so $\theta(v) \leq 1 + v$ gives $F_i + 1 \leq (1 + q/u_i)^{d_i}$. Take a rational upper bound B_{cert} on this expression with $\ln B_{\text{cert}} = O(n \ln n)$. Enclose the mass and its scale factor in positive intervals of relative width at most $\varepsilon_*/(64B_{\text{cert}})$, using Lemma A.2 for the mass. Outward rounding gives an interval for F_i of width at most $\varepsilon_*/4$. Accept when its upper endpoint is at most ε_* . Every accepted lattice satisfies $\eta_{\varepsilon_*}(\Gamma_i) \leq u_i$, and every lattice with $F_i \leq \varepsilon_*/2$ passes.

For $d_i \geq 2$, the capacity condition (B.10) and (3.13) give $\eta_{\varepsilon_*/2}(\Gamma_i) \leq u_i$ with probability at least $1 - 2^{1-d_i} \geq 1/2$. For $d_i = 1$, $\Gamma_i = (q/P)\mathbb{Z}$. Put $v = u_i P/q$. The capacity condition gives $v \geq 16\sqrt{\ell}$, so

$$\rho_{1/u_i}(\Gamma_i^* \setminus \{0\}) = 2 \sum_{j \geq 1} e^{-\pi v^2 j^2} \leq \frac{2e^{-\pi v^2}}{1 - e^{-\pi v^2}} \leq \varepsilon_*/2.$$

Thus every such draw passes. A cap of $R_{\text{aux}} = O(\ln n)$ independent trials, chosen so that $|\mathcal{S}|r2^{-R_{\text{aux}}} \leq n^{-K-10}$, bounds total retry failure by n^{-K-10} . For $d_i = 0$, skip the lift and retain the prescribed number of differences in consecutive pairs. The quotient has size one.

List recursion and quotient removal. The initial persistent lattice is $J = \widetilde{K}_\emptyset = \mathbb{Z}^n + (q/P_0)\mathbb{Z}\mathbf{a}_0$. Choose k uniformly modulo P_0 and then sample the coordinates independently from $D_{\mathbb{Z}+qk(\mathbf{a}_0)_j/P_0, s_0}$, using the uniform rule and Lemma A.1 with spacing one. For the reference mixture, Poisson summation gives, uniformly in k ,

$$\rho_{s_0}(\mathbb{Z}^n + qk\mathbf{a}_0/P_0) = s_0^n(1 + O(ne^{-\pi s_0^2})).$$

Thus one reference vector is pointwise $\delta_0 = O(ne^{-\pi s_0^2})$ -similar to D_{J, s_0} . Independent repetition is strongly conditionally δ_0 -similar.

Put

$$\zeta = n^{-K-As-10}, \quad \bar{\delta}_i = 4^i \delta_0 + 5(4^i - 1)\varepsilon_\star.$$

The choices in (B.7) give $\bar{\delta}_r < \delta/16$ for all sufficiently large n . Choose an integer M_{pair} satisfying

$$M_{\text{pair}} \geq \max \{P, 16P_0(N_{\text{out}} + \ln(1/\zeta))\}. \quad (\text{B.14})$$

Round a certified upper bound on the right side upward, with total additive error at most two. Start with $2 \cdot 3^r M_{\text{pair}}$ initial vectors. At level i , use the primitive Gaussian lift along the first exact sequence in (B.12), with (A.1), and then use exactly the ordered quotient combination rule of Theorem 5.4 for $K'_i/\widetilde{K}_{S_i}$. Its hypotheses hold because

$$\eta_{\varepsilon_\star}(\Gamma_i) \leq u_i < s_{i-1}, \quad \eta_{\varepsilon_\star}(\widetilde{K}_{S_i}) \leq u_0 \leq u_i = s_{i-1}/\sqrt{2}.$$

The lifting and ordered pairing bounds (3.9)–(3.10) give, after level i , a list of length $2 \cdot 3^{r-i} M_{\text{pair}}$, width s_i , and strong conditional error at most $\bar{\delta}_i$. The same estimate covers $d_i = 0$. Hence the recursion produces $2M_{\text{pair}}$ vectors in $\widetilde{K}_S$ at width $s = t/\sqrt{2}$.

Apply Lemma B.1 with

$$\Lambda = K_S, \quad \Lambda' = \widetilde{K}_S, \quad P = P_0, \quad s = t/\sqrt{2}, \quad N = N_{\text{out}}, \quad \varepsilon = \varepsilon_\star.$$

The explicit condition $s_{\text{mass}}^2 \geq 8s_{\text{search}}^2$ gives

$$\eta_{\varepsilon_\star}(\widetilde{K}_S) \leq u_0 = q/s_{\text{mass}} \leq t/2 = s/\sqrt{2}.$$

Since $\bar{\delta}_r < 1/4$, the choice (B.14) implies (B.4). The completed reference list is therefore strongly conditionally

$$4\bar{\delta}_r + \log \frac{1 + \varepsilon_\star}{1 - \varepsilon_\star} < \delta$$

similar to $D_{K_S, t}^{N_{\text{out}}}$, and its mismatch with the finite list is at most ζ . A union bound over $|\mathcal{S}| \leq n^{As}$ charges all such shortages to the preprocessing failure. Applying $q^{-1}\mathbf{U}_S^{-1}$ and (B.1) gives the reference list in L_S^* at width $1/s_{\text{search}}$.

Successful reference preprocessing means the intersection of the global smoothing event (B.8), successful certified auxiliary selection within every retry cap, and the absence of a final fixed-pair shortage for every prescribed set. First fix the common mask and all selected auxiliary lattices. Conditional on these data, different list constructions use fresh independent randomness. The conditional form of Lemma B.1 preserves each list's strong conditional bound when its shortage event is excluded. Conditioning on the other lists' success has no further effect at fixed data.

Finally, mixing over these data preserves the uniform pointwise bound, since the target $D_{L_S^*, 1/s_{\text{search}}}$ is independent of them.

Finite precision and cost. Rational comparisons with powers of two determine r exactly, so all squared widths are rational. Bound X_i upward and choose X_P within two above e^{L_P} . Replace $\ln P$ in B_i by a rational lower bound within n^{-3} . This loses at most one per block from total capacity at least $2n + r$ and preserves (B.10).

The mask choices, auxiliary trials, initial coset and scalar choices, and complete shifted draws together number at most

$$J_{\text{calls}} = 2 + |\mathcal{S}|(rR_{\text{aux}} + 2(n + r + 1)3^r M_{\text{pair}}).$$

Choose dyadic $\nu \in [n^{-K-10}/2, n^{-K-10}]$. Give each Gaussian call accuracy $\nu/(2J_{\text{calls}})$ and use total uniform choice budget $\nu/2$ in Section A.1. Lemma 3.4 gives total implementation error at most ν . The bad mask, exhausted retries, and pair shortages have total reference probability at most $3n^{-K-10}$.

Initialization has $s_0^2 \in [2q^2/s_{\text{mass}}^2, 4q^2/s_{\text{mass}}^2)$ and spacing one. Lifting and certification have spacing q and widths s_{i-1} and u_i . These width ratios and their inverses have polynomial magnitude. Also $\ln P_0, \ln P, \ln M_{\text{pair}}, \ln(1/\varepsilon_\star), \ln(J_{\text{calls}}/\nu) = O(n/D)$. The scalar cutoffs and rational lattice denominators give polynomial bit lengths through $r = O(\ln n)$ levels. The extra certification precision has polynomial length as well. The total work is at most

$$2(1 + |\mathcal{S}|)r3^r M_{\text{pair}}P \text{ poly}(n, \ln q, \ln P, \ln P_0, \ln(1/\varepsilon_\star)),$$

and the final pair scan is linear in its input list. This proves the theorem. $\square$

B.2 Affine slice search and witness containment

For each anchor position $i \in [n]$, fix an order on $[n] \setminus \{i\}$. Let $\mathcal{S}$ collect the sets of coordinates left free along these orders. Each order has at most n stages, so $|\mathcal{S}| \leq n^2$. For each S , use a dual Gaussian list $\mathbf{Y}_1, \dots, \mathbf{Y}_N \in L_S^*$ from Theorem B.2.

Embed L_S in the subspace $\mathcal{U}_S \subseteq \mathbb{R}^{2n}$ of the free coordinates. Fixing the other coordinates gives a slice $A = \mathbf{s} + L_S$. Decompose $\mathbf{s} = \mathbf{s}_U + \mathbf{s}_\perp$ along $\mathcal{U}_S \oplus \mathcal{U}_S^\perp$. Poisson summation gives

$$\frac{\rho_{s_{\text{search}}}(A)}{\rho_{s_{\text{search}}}(L_S)} = e^{-\pi\|\mathbf{s}_\perp\|^2/s_{\text{search}}^2} \mathbb{E}_{\mathbf{Y} \leftarrow D_{L_S^*, 1/s_{\text{search}}}} [\cos(2\pi\langle \mathbf{Y}, \mathbf{s}_U \rangle)]. \quad (\text{B.15})$$

The corresponding sample estimate is

$$\hat{w}(A) = e^{-\pi\|\mathbf{s}_\perp\|^2/s_{\text{search}}^2} \frac{1}{N} \sum_{k=1}^N \cos(2\pi\langle \mathbf{Y}_k, \mathbf{s}_U \rangle).$$

For fixed S , changing the assigned coordinate values changes $\mathbf{s}$ while reusing the same samples. Candidate children share the denominator $\rho_{s_{\text{search}}}(L_S)$, which cancels when their weights are normalized. The procedure below evaluates this estimate with certified numerical error and clamps it at zero.

Since $q\mathbb{Z}^{n+|S|} \subseteq L_S$, we have $L_S^* \subseteq q^{-1}\mathbb{Z}^{n+|S|}$. Thus the cosine phase depends only on $\mathbf{s}_U \bmod q$. There are at most $n^2 q^{2n}$ phases across all lists. Simultaneous concentration uses the logarithm of this count to bound the required list length. The search evaluates the queries encountered on its sampled paths.

Define

$$\mathcal{W}_B(h) = \{\mathbf{0} \neq \mathbf{z} = (\mathbf{u}, \mathbf{v}) \in \Lambda_h : \|\mathbf{z}\| \leq B\}.$$

The search procedure. Fix $h, s_{\text{search}} > 0, A \geq 1, B > 0$, and $0 < \varepsilon_{\text{qry}}, \varepsilon_{\text{hit}} < 1$ such that

$$1 \leq B < q/2, \quad \rho_{s_{\text{search}}}(\Lambda_h) \leq A,$$

and put

$$H = \lfloor B \rfloor, \quad p_B = A^{-1} e^{-\pi B^2 / s_{\text{search}}^2}, \quad \xi = \frac{p_B}{200n(2H+1)}. \quad (\text{B.16})$$

Let $0 < \nu_{\text{list}}, \nu_{\text{cat}} < 1$ and put $\nu = \nu_{\text{list}} + \nu_{\text{cat}} < 1$. Given an N -entry list for each $S \in \mathcal{S}$, construct the table $\mathcal{C}(h)$ as follows.

1. Evaluate every use of (B.15) with certified numerical error at most $\xi/4$, compute rational estimates with that certified absolute error, and clamp them at zero.
2. For each $i \in [n]$ and $a \in [1, H]$, run the affine slice search from the anchor $u_i = a$. At every later coordinate retain only values in $[-H, H]$ consistent with the partial energy bound. Choose the next value with probability proportional to the resulting nonnegative rational weights. A zero total weight is a declared failure.
3. Run every anchored branch independently

$$T = \left\lceil C_3 p_B^{-1} \ln(1/\varepsilon_{\text{hit}}) \right\rceil \quad (\text{B.17})$$

times, completing all runs even if one has already found a verified output. There are at most $J_{\text{cat}} = n^2 HT$ adaptive choices from the rational weights. Use the finite weight rule in Section A.1 with accuracy $\xi_{\text{cat}} = \nu_{\text{cat}}/J_{\text{cat}}$ for each choice. Lemma 3.4 gives total error at most ν_{cat} .

4. Once all $\mathbf{u}$ coordinates are fixed, set $\mathbf{v} = \text{lift}_q^{\text{cen}}(\mathbf{H}_h \mathbf{u})$. Verify lattice membership, nonzeroness, and the norm bound exactly. Retain every verified pair and also its negative. Call the resulting table $\mathcal{C}(h)$.

Theorem B.3 (Witness containment). *Take $s_{\text{search}}^2, A, B^2$ and the error budgets to be positive rationals given in binary. Let L_{enc} bound their bit lengths and those of the list coordinates. There are absolute constants $C_2, C_3 > 0$ for which the procedure above has the following guarantee. Suppose that:*

1. *the lists have preprocessing failure at most ε_{pre} .*
2. *their reference laws are strongly conditionally δ -similar to the corresponding $D_{L_S^*, 1/s_{\text{search}}}^N$, and their joint implementation has total variation error at most ν_{list} .*
3. *the accuracy and list length satisfy*

$$\delta \leq \xi/16, \quad N \geq C_2 \xi^{-2} \left(2n \ln q + \ln \frac{2n^2}{\varepsilon_{\text{qry}}} \right). \quad (\text{B.18})$$

Then, for every fixed $\mathbf{z} \in \mathcal{W}_B(h)$,

$$\Pr[\mathcal{C}(h) \cap \{\mathbf{z}, -\mathbf{z}\} = \emptyset] \leq \varepsilon_{\text{pre}} + \nu + \varepsilon_{\text{qry}} + \varepsilon_{\text{hit}}. \quad (\text{B.19})$$

Moreover,

$$|\mathcal{C}(h)| \leq 2nHT. \quad (\text{B.20})$$

Writing T_{list} for the cost of constructing all the prescribed lists, the list construction plus all slice queries takes

$$T_{\text{list}} + \text{poly}(n, H, \ln q, L_{\text{enc}}, \ln T, \ln(1/\xi), \ln(1/\nu_{\text{cat}})) NT$$

bit operations. Its space is bounded by the list storage, the bit storage of the table, and polynomial working storage in these parameters.

Proof. Uniform query accuracy. Certified lower bounds for the accuracy thresholds and upper bounds for the required N, T give finite choices with fixed constant margins. In the public search schedule, L_{enc} and $\ln(1/\xi)$ are polynomial in n . For a reference list, Lemma 3.1, followed by a union bound over at most $n^2 q^{2n}$ phases shows that (B.18) makes every empirical cosine accurate within $3\xi/4$, except with probability ε_{qry} . Certified finite-precision evaluation uses the remaining $\xi/4$. The total variation bound ν transfers the complete reference analysis below to the implemented procedure.

A fixed witness path. Fix $\mathbf{z} = (\mathbf{u}, \mathbf{v}) \in \mathcal{W}_B(h)$. If $\mathbf{u} = \mathbf{0}$, then $\mathbf{v} \in q\mathbb{Z}^n$. Since $\|\mathbf{v}\| < q/2$, this would force $\mathbf{v} = \mathbf{0}$, a contradiction. Choose i with $u_i \neq 0$, and replace $\mathbf{z}$ by $-\mathbf{z}$ if necessary so that $a = u_i \in [1, H]$. The anchored branch (i, a) is therefore present.

Every prefix slice on the path to $\mathbf{z}$ contains $\mathbf{z}$. Since its direction lattice embeds in Λ_h , the correct child's normalized mass is at least

$$\frac{\rho_{s_{\text{search}}}(\mathbf{z})}{\rho_{s_{\text{search}}}(\Lambda_h)} \geq p_B.$$

If a level has at most $2H + 1$ children, additive error at most ξ changes the correct transition probability by a multiplicative factor at least

$$\frac{1 - \xi/p_B}{1 + (2H + 1)\xi/p_B}.$$

The product of these factors over at most n levels is bounded below by an absolute positive constant. To account explicitly for the retained candidates, let C_j be their set at level j , let A_{j-1} be the parent slice, and let A_j be the correct child. The child slices are disjoint subsets of the parent, so the ideal transition probability satisfies

$$P_j = \frac{\rho_{s_{\text{search}}}(A_j)}{\sum_{b \in C_j} \rho_{s_{\text{search}}}(A_{j,b})} \geq \frac{\rho_{s_{\text{search}}}(A_j)}{\rho_{s_{\text{search}}}(A_{j-1})}.$$

Multiplying these lower bounds gives

$$\prod_j P_j \geq \frac{\rho_{s_{\text{search}}}(A_{\text{leaf}})}{\rho_{s_{\text{search}}}(A_{\text{anchor}})} \geq \frac{\rho_{s_{\text{search}}}(\mathbf{z})}{\rho_{s_{\text{search}}}(\Lambda_h)} \geq p_B.$$

Here the leaf contains the witness, while the anchor is a subset of Λ_h . Once all $\mathbf{u}$ coordinates are fixed, the centered representative completes $\mathbf{v}$ uniquely because $B < q/2$. Hence one run of the correct branch outputs $\mathbf{z}$ with probability at least $c_2 p_B$ for an absolute $c_2 > 0$.

Repetition and verification. The probability that all T repetitions miss this fixed witness is at most $(1 - c_2 p_B)^T \leq e^{-c_2 p_B T} \leq \varepsilon_{\text{hit}}$ after choosing C_3 sufficiently large. Each coordinate of $\mathbf{v}$ lies in $(-q/2, q/2)$, so the centered representative of $\mathbf{H}_h \mathbf{u} \bmod q$ is exactly $\mathbf{v}$ and the witness passes the final verification. Every anchored run contributes at most one raw pair. Retaining both signs proves (B.20). $\square$

For a random witness, condition on the realized pair $(h, \mathbf{z})$, apply (B.19), and average. The uniform bound for each fixed witness gives the same guarantee under any distribution on such pairs.

B.3 Falcon: quality, Gaussian mass, and basis completion

This subsection gives two ways to obtain the Gaussian mass bound needed for recovery in the negacyclic ring. We first record the shared lattice duality and basis completion facts. We then

prove Theorem 12.2 from Falcon's key generation quality condition in Section B.3.2. The final part derives a separate recovery guarantee from spectral bounds on a short pair.

Let

$$n = 2^k, \quad R = \mathbb{Z}[x]/(x^n + 1), \quad R_q = R/qR,$$

where $q \equiv 1 \pmod{2n}$ is prime. We use the coefficient embedding $\text{coeff} : R \rightarrow \mathbb{Z}^n$ and its induced Euclidean norm. For $h \in R_q$, let $\mathbf{H}_h$ be its negacyclic multiplication matrix, so that $\text{coeff}(hu) = \mathbf{H}_h \text{coeff}(u) \pmod{q}$, and put

$$\Lambda_h = \text{coeff}\{(u, v) \in R^2 : hu - v = 0 \pmod{q}\} = \ker_q[\mathbf{H}_h \mid -\mathbf{I}_n]. \quad (\text{B.21})$$

Here and below coeff acts blockwise on ring pairs and sets. We identify a ring submodule with its coefficient lattice when taking intersections, orthogonal projections, or Gaussian masses.

Lemma B.4. *Write $a^* = a(x^{-1})$ for coefficient conjugation. For every $h \in R_q$, the orthogonal map*

$$\mathbf{J} \text{coeff}(u, v) = \text{coeff}(v^*, -u^*) \quad \text{satisfies} \quad q\Lambda_h^* = \mathbf{J}\Lambda_h. \quad (\text{B.22})$$

Proof. Let $\mathbf{H}$ be an integer negacyclic multiplication matrix for h and let $\mathbf{S}$ be the signed permutation matrix of conjugation. Then $\mathbf{H}^T \mathbf{S} = \mathbf{S} \mathbf{H}$, and the systematic form of (B.21) gives

$$q\Lambda_h^* = \{(\mathbf{a}, \mathbf{b}) \in \mathbb{Z}^{2n} : \mathbf{a} + \mathbf{H}^T \mathbf{b} = 0 \pmod{q}\}.$$

For $(\mathbf{u}, \mathbf{v}) \in \Lambda_h$, $\mathbf{S} \mathbf{v} - \mathbf{H}^T \mathbf{S} \mathbf{u} = \mathbf{S}(\mathbf{v} - \mathbf{H} \mathbf{u}) = 0 \pmod{q}$. Thus $\mathbf{J}\Lambda_h \subseteq q\Lambda_h^*$. Both lattices have determinant q^n , proving equality. $\square$

B.3.1 Basis completion

For the conventional row basis orientation, put

$$\Lambda_h^{\text{std}} = \text{coeff}\{(a, b) \in R^2 : a + bh = 0 \pmod{q}\}. \quad (\text{B.23})$$

It is the image of (B.21) under the signed coordinate permutation $\text{coeff}(u, v) \mapsto \text{coeff}(v, -u)$. For $u, v, F, G \in R$, define

$$\mathbf{B}(u, v, F, G) = \begin{pmatrix} v & -u \\ G & -F \end{pmatrix}. \quad (\text{B.24})$$

The equation

$$uG - vF = q \quad (\text{B.25})$$

makes its determinant equal to q . If u is invertible modulo q and $vu^{-1} = h$, the coefficient vectors of both rows of (B.24) lie in Λ_h^{std} . Their coefficient row lattice has determinant q^n , equal to the index of (B.23), so they form an NTRU basis of that lattice.

Lemma B.5. *Given q, u, v , one can decide in time polynomial in their coefficient bit length and in n whether (B.25) has an integer solution, and return a solution of polynomial bit length whenever it exists. If $u \pmod{q}$ is a unit and $vu^{-1} = h$, this solution gives a basis of Λ_h^{std} .*

Proof. Let $\mathbf{M}_u, \mathbf{M}_v$ be the integer coefficient multiplication matrices, defined by $\mathbf{M}_u \text{coeff}(a) = \text{coeff}(ua)$ and likewise for v . The equation is the integer linear system

$$[\mathbf{M}_u \mid -\mathbf{M}_v] \begin{pmatrix} \text{coeff}(G) \\ \text{coeff}(F) \end{pmatrix} = q\mathbf{e}_0.$$

Here $\mathbf{e}_0 = \text{coeff}(1)$. Polynomial-time Smith or Hermite normal form algorithms, with their transformation matrices, decide solvability and produce a solution of polynomial bit length [KB79]. The determinant and public relation argument above proves the basis claim. $\square$

Define $\text{CompleteAndCheck}_{q,B}(h, u, v)$ to perform the unit, public relation, and norm checks below, solve the integer system using Lemma B.5, and reject if a check fails or no solution exists. It accepts only after verifying

$$\begin{aligned} u \bmod q \in R_q^\times, \quad vu^{-1} &= h \pmod{q}, \quad \|(u, v)\| \leq B, \\ uG - vF &= q \text{ in } R. \end{aligned} \tag{B.26}$$

All checks use exact integer or rational arithmetic. The procedure decides the displayed conditions and returns an integer completion for every accepted pair. For $B = O(\sqrt{q})$, each candidate is processed in $T_{\text{comp}}(n, q) = \text{poly}(n, \log q)$ time.

B.3.2 Recovery from the key quality condition

For $a^* = a(x^{-1})$, define

$$\gamma_q(u, v) = \max \left\{ \|(v, -u)\|_2, \left\| \left(\frac{qu^*}{uu^* + vv^*}, \frac{qv^*}{uu^* + vv^*} \right) \right\|_2 \right\}. \tag{B.27}$$

Compute the inverse in $R \otimes \mathbb{Q}$ and set $\gamma_q(u, v) = +\infty$ if it does not exist. All norms are Euclidean coefficient norms. Fix a rational constant $C \geq 1$, independent of n , and impose the exact quality predicate $\gamma_q(u, v)^2 \leq C^2 q$. Falcon uses $C = 117/100$ in this predicate [Falcon20, Equation (3.28) and Algorithm 5]. A quality-qualified key is one admitting the algebraic and quality witness in (1.2).

Lemma B.6. *Suppose $u \bmod q$ is a unit, $vu^{-1} = h \pmod{q}$, and $uG - vF = q$ in R . Order the $2n$ coefficient rows as the n shifts of $(v, -u)$ followed by the n shifts of $(G, -F)$. They form a basis of Λ_h^{std} with maximum Gram–Schmidt norm at most $\gamma_q(u, v)$, for every integer completion (F, G) . The quality predicate is decidable exactly in polynomial bit complexity.*

Proof. The ring matrix (B.24) has determinant q , so its full coefficient matrix has determinant of absolute value q^n . Its rows lie in Λ_h^{std} , whose determinant is also q^n , and hence form a full basis. Put $d = uu^* + vv^*$, $b_1 = (v, -u)$, $b_2 = (G, -F)$, and $\mu = (Gv^* + Fu^*)/d$. The NTRU equation ensures that u, v have no common complex zero, so d is invertible. It also gives

$$b_2 - \mu b_1 = \left(\frac{qu^*}{d}, \frac{qv^*}{d} \right). \tag{B.28}$$

The right-hand side is orthogonal to every coefficient shift of b_1 , and μb_1 lies in their real span. All shifts of b_1 have norm $\|b_1\|_2$, so every Gram–Schmidt length in the first block is at most this value. After projecting away that block, the second block consists of shifts of the right-hand side of (B.28). Further orthogonalization cannot increase their common norm. This proves the quality bound for any completion.

For exact verification, form the integer multiplication matrix of d . Rational linear algebra either detects singularity or computes d^{-1} with polynomial coefficient bit length. Exact squared coefficient norms and comparison with the rational $C^2 q$ therefore take polynomial bit complexity. $\square$

Lemma B.7. *Let L be a full-rank lattice of dimension d , with basis $(b_1, \dots, b_d)$ and Gram–Schmidt lengths $r_i = \|\tilde{b}_i\|_2 > 0$. For every $s > 0$,*

$$\rho_s(L) \leq \prod_{i=1}^d \theta(s/r_i).$$

Proof. For real Gram–Schmidt coefficients μ_{ji} ,

$$\left\| \sum_{i=1}^d z_i b_i \right\|_2^2 = \sum_{i=1}^d r_i^2 \left(z_i + \sum_{j>i} \mu_{ji} z_j \right)^2.$$

Fixing $z_2, \dots, z_d$, sum first over z_1 . The shifted one-dimensional Gaussian mass is at most $\theta(s/r_1)$ by Poisson summation. Repeating for $z_2, \dots, z_d$ gives the product bound. Nonnegativity justifies the iterated sums. $\square$

Lemma B.8. *Let L be a full-rank d -dimensional lattice with $qL^* = JL$ for an orthogonal map J . If L has a basis whose Gram–Schmidt lengths are at most $C\sqrt{q}$, then for every $s > 0$,*

$$\rho_s(L) \leq \theta(Cs/\sqrt{q})^d.$$

Proof. Let $(b_1^\vee, \dots, b_d^\vee)$ be the dual basis, so that $\langle b_i, b_j^\vee \rangle = \delta_{ij}$. The reversed dual basis $(b_d^\vee, \dots, b_1^\vee)$ has Gram–Schmidt lengths $1/r_d, \dots, 1/r_1$. This follows by inverse transposition of a QR decomposition and reversal of the orthogonalization order. Hence $qJ^{-1}(b_d^\vee, \dots, b_1^\vee)$ is a basis of L with Gram–Schmidt lengths at least $\sqrt{q}/C$. Lemma B.7 and monotonicity of θ prove the claim. $\square$

Proposition B.9. *For every quality-qualified public key and every $s > 0$,*

$$\rho_s(\Lambda_h) \leq \theta(Cs/\sqrt{q})^{2n}. \quad (\text{B.29})$$

In particular, for fixed $1/2 < \beta < 1$ and the finite rational width square $s_{\text{mass}}^2 \leq q/n^\beta$, for all sufficiently large n ,

$$\begin{aligned} \rho_{s_{\text{mass}}}(\Lambda_h) &\leq \theta(Cn^{-\beta/2})^{2n} \\ &\leq \exp\left(6n \exp(-\pi n^\beta/C^2)\right) = 1 + n^{-\omega(1)}. \end{aligned} \quad (\text{B.30})$$

Thus the public mass budget $A_{\text{mass}} = 2$ works uniformly for these keys.

Proof. The orthogonal transformation $P(u, v) = (v, -u)$ maps Λ_h to Λ_h^{std} . Apply P^{-1} to the full quality basis from Lemma B.6. Its Gram–Schmidt lengths remain at most $C\sqrt{q}$. Lemmas B.4 and B.8 give (B.29) in dimension $2n$. Eventually $Cn^{-\beta/2} \leq 1$. Using $\theta(t) \leq 1 + 3e^{-\pi/t^2}$ for $0 < t \leq 1$ then proves (B.30). $\square$

Proof of Theorem 12.2. Apply Theorem 12.1 with $m = n$, fixed rational $1/2 < \beta < 1$, and $c_{\text{search}} = 1$. Use its finite schedule

$$\begin{aligned} 0 \leq \ln \ln n - D_Q &\leq 2^{-n^2}, \quad (1 - 2^{-n^2})q/n^\beta \leq s_{\text{mass}}^2 \leq q/n^\beta, \\ s_{\text{search}}^2 &= qD_Q/n. \end{aligned}$$

The congruence on q gives $q \geq 2n + 1$, and the modulus window implies $q \leq n^2$ eventually. Proposition B.9 supplies $A_{\text{mass}} = 2$, with $\ln A_{\text{mass}} = O(n/D_Q)$. For finite arithmetic, the conditions $C^2 \leq n^\beta$ and $6n \exp(-\pi n^\beta/C^2) \leq \ln 2$ can be certified by one-sided rational approximations as in Appendix A. They use only public constants and parameters. Moreover,

$$\frac{s_{\text{mass}}^2}{s_{\text{search}}^2} \geq (1 - 2^{-n^2}) \frac{n^{1-\beta}}{D_Q} \rightarrow \infty,$$

so $s_{\text{mass}}^2 \geq 8s_{\text{search}}^2$ for all sufficiently large n . Choose the public rational radius square $B_Q^2 = C^2q$. Then $1 \leq B_Q = C\sqrt{q} < q/2$ eventually.

The verifier first checks nonzeroness, modular invertibility of u , $hu = v \pmod{q}$, and $\|(u, v)\|_2^2 \leq C^2q$. It then uses Lemma B.5 to solve $uG - vF = q$ over R , rejecting when no integer solution exists, and applies the exact quality predicate $\gamma_q(u, v)^2 \leq C^2q$. All candidate coefficients have absolute value at most $C\sqrt{q}$. The completion and quality checks therefore have polynomial bit complexity and return F, G of polynomial bit length.

The witness in the hypothesis passes every check, so the common search theorem returns an accepted candidate with probability at least $1 - n^{-K}$ in time and space $2^{O(n/\ln \ln n)}$. Lemma B.6 gives the stated quality of its full coefficient basis. The quality basis gives the mass bound, and the algorithm uses the public budget $A_{\text{mass}} = 2$. Its probability bound is uniform over all qualified public keys. $\square$

B.3.3 Recovery from spectral bounds

We now derive a mass bound from the Fourier values of a short pair. These values control lengths in the pair's polynomial span and in the orthogonal projection. The resulting bound gives the recovery theorem below and its distributional corollary.

Definition B.10. Fix public constants $1/2 < \alpha < \beta < 1$, $C_{\text{sp}} > 0$, and rational $C_{\text{len}} > 0$. For a fixed public quotient $h \in R_q$, an admissible witness is a pair $(f, g) \in R^2$ for which there exist $F_N, G_N \in R$ satisfying

$$\begin{aligned} f \bmod q &\in R_q^\times, & h &= gf^{-1} \pmod{q}, & fG_N - gF_N &= q, \\ \|(f, g)\|_2 &\leq C_{\text{len}}\sqrt{q}, \end{aligned} \tag{B.31}$$

and whose spectrum satisfies

$$\frac{q}{n^\alpha} \leq G_{\min} := \min_{\zeta^n = -1} (|f(\zeta)|^2 + |g(\zeta)|^2) \leq G_{\max} := \max_{\zeta^n = -1} (|f(\zeta)|^2 + |g(\zeta)|^2) \leq C_{\text{sp}}q \ln n. \tag{B.32}$$

The extrema $G_{\min}, G_{\max}$ range over the complex roots of $x^n + 1$. The pair and its completion establish the existence of a valid target. The algorithm takes the public quotient and bounds as input.

The spectral bounds control the Gaussian mass in the span of the short pair and in its orthogonal projection. The NTRU equation makes the short pair sublattice primitive, so these two estimates together bound the mass of the full public lattice.

Lemma B.11. *If $f \bmod q$ is a unit, then $(f, g) = R$ if and only if there exist $F_N, G_N \in R$ with $fG_N - gF_N = q$. In particular, every pair satisfying (B.31) has $(f, g) = R$.*

Proof. Since f is a unit modulo q , there are $a, k \in R$ with $af + qk = 1$. Substituting $q = fG_N - gF_N$ gives

$$1 = f(a + kG_N) - g(kF_N),$$

so $1 \in (f, g)$. Conversely, if $rf + tg = 1$, take $G_N = qr$ and $F_N = -qt$. $\square$

With the constants of Definition B.10, put

$$s_{\text{mass}}^2 = \frac{q}{n^\beta}. \tag{B.33}$$

Proposition B.12. *If (B.31)–(B.32) hold, then, for every $s > 0$,*

$$1 \leq \rho_s(\Lambda_h) \leq \theta\left(\frac{s}{\sqrt{G_{\min}}}\right)^n \theta\left(\frac{s\sqrt{G_{\max}}}{q}\right)^n. \quad (\text{B.34})$$

Consequently, for every fixed $c_0 > 0$,

$$\rho_{c_0 s_{\text{mass}}}(\Lambda_h) = 1 + n^{-\omega(1)}. \quad (\text{B.35})$$

If $0 < D = D(n) \leq 1 + \ln \ln n$ and $s_{\text{search}}^2 = c_{\text{search}} q D / n$ for fixed $c_{\text{search}} > 0$, then also

$$\rho_{\sqrt{2} s_{\text{search}}}(\Lambda_h) = 1 + n^{-\omega(1)}. \quad (\text{B.36})$$

Proof. Let $K = \mathbb{Q}[x]/(x^n + 1)$, let $W = K \cdot (f, g) \subset K^2$, and define $A : K \rightarrow K^2$ by $A(a) = a(f, g)$. Write $\mathbf{A}$ for its coefficient matrix, so that $\text{coeff}(A(a)) = \mathbf{A} \text{coeff}(a)$. The negacyclic Fourier transform is unitary in coefficient norm, so the eigenvalues of $\mathbf{A}^T \mathbf{A}$ are precisely $|f(\zeta)|^2 + |g(\zeta)|^2$.

Lemma B.11 implies

$$\Lambda_h \cap W = AR. \quad (\text{B.37})$$

Indeed, $AR \subseteq \Lambda_h$. Conversely, if $a(f, g) \in R^2$ and $rf + tg = 1$ with $r, t \in R$, then $a = r(af) + t(ag) \in R$.

Lemma B.4 gives $q\Lambda_h^* = \mathbf{J}\Lambda_h$, and $\mathbf{J}W = W^\perp$. If $Q = \pi_{W^\perp} \Lambda_h$, duality and (B.37) give

$$Q^* = \Lambda_h^* \cap W^\perp = q^{-1} \mathbf{J}(AR), \quad Q \simeq q(AR)^*. \quad (\text{B.38})$$

For $a \in R$, $\|A(a)\|^2 \geq G_{\min} \|a\|^2$, and hence

$$\rho_s(AR) \leq \theta(s/\sqrt{G_{\min}})^n.$$

Writing $(AR)^* = \mathbf{A}(\mathbf{A}^T \mathbf{A})^{-1} \mathbb{Z}^n$ shows that the vector indexed by $\mathbf{z} \in \mathbb{Z}^n$ has squared norm $\mathbf{z}^T (\mathbf{A}^T \mathbf{A})^{-1} \mathbf{z} \geq \|\mathbf{z}\|^2 / G_{\max}$. Equation (B.38) therefore gives

$$\rho_s(Q) \leq \theta(s\sqrt{G_{\max}}/q)^n.$$

Decompose Λ_h into the fibres of its orthogonal projection onto Q . Every fibre is a translate of AR , and $\rho_s(AR + \mathbf{c}) \leq \rho_s(AR)$ by the Gaussian shift inequality recalled in Section 3. This proves (B.34).

For $0 < x \leq 1$,

$$\theta(x) - 1 \leq 3e^{-\pi/x^2}, \quad \theta(x)^n \leq \exp(3ne^{-\pi/x^2}).$$

At $s = c_0 s_{\text{mass}}$, the squared theta arguments in (B.34) are at most

$$c_0^2 n^{\alpha-\beta}, \quad C_{\text{sp}} c_0^2 \frac{\ln n}{n^\beta},$$

so both theta contributions equal $1 + n^{-\omega(1)}$. This proves (B.35). Finally, $2s_{\text{search}}^2/s_{\text{mass}}^2 = 2c_{\text{search}} D n^{\beta-1} = o(1)$. Monotonicity of Gaussian mass and (B.35) prove (B.36). $\square$

Theorem B.13 (Negacyclic recovery). *Fix the public constants of Definition B.10. Along any given sequence of powers of two n and primes $q \equiv 1 \pmod{2n}$ with*

$$\ln(q/n) \leq (\ln n)^{1-c}$$

for a fixed $c \in (0, 1)$, let h be any fixed public quotient admitting an admissible witness. From h , the public parameters, and the prescribed bounds alone, a randomized procedure returns $u, v, F, G \in R$ satisfying

$$\begin{aligned} u \bmod q &\in R_q^\times, & vu^{-1} &= h \pmod{q}, \\ \|(u, v)\|_2 &\leq C_{\text{len}}\sqrt{q}, & uG - vF &= q, \end{aligned}$$

and hence a certified NTRU basis of Λ_h^{std} , with probability $1 - o(1)$ over its internal randomness, uniformly over these instances. Its time and space are

$$\exp(O(n/\ln \ln n)). \quad (\text{B.39})$$

Proof. Take the finite schedule of Theorem 12.1 with $m = n$ and $B_Q = C_{\text{len}}\sqrt{q}$. The modulus window gives $q \leq n^2$ eventually, and $s_{\text{mass}}^2/s_{\text{search}}^2 \rightarrow \infty$. Proposition B.12 and monotonicity in the width supply the public mass bound $A_{\text{mass}} = 2$ at these finite parameters. Use `CompleteAndCheck` as the verifier: Lemma B.5 proves that it runs in polynomial time and accepts the admissible witness. The common theorem gives the short pair, its completion, and the stated cost. $\square$

Corollary B.14. *Under the parameter conditions of Theorem B.13, let a distribution on (h, f, g) satisfy (B.31) for every output and (B.32) except with probability ω_n . Then recovery of a short pair and certified integer NTRU basis succeeds with probability at least $1 - \omega_n - o(1)$ over the key and independent algorithmic randomness, in time and space $2^{O(n/\ln \ln n)}$.*

Proof. Apply the uniform guarantee of Theorem B.13 to each key on the spectral event and average. $\square$

More generally, if a proposal law λ is conditioned on an acceptance event $\mathcal{A}$ of positive probability, then any required event $\mathcal{E}$ satisfies

$$\Pr_\lambda[\mathcal{E}^c \mid \mathcal{A}] \leq \frac{\Pr_\lambda[\mathcal{E}^c]}{\Pr_\lambda[\mathcal{A}]}. \quad (\text{B.40})$$

Here $\mathcal{A}$ is the full acceptance event. A direct estimate under the accepted law also supplies the corollary's hypothesis.

B.4 Cyclic NTRU: saturation and recovery

We first express the Gaussian mass of the cyclic public lattice in terms of spectral bounds and a saturation index. We then give conditions under which this mass bound yields recovery. For the HPS and HRSS key distributions, the remaining parts establish the spectral and saturation bounds and combine them after key generation acceptance.

Let N be an odd prime, put

$$R_N = \mathbb{Z}[x]/(x^N - 1), \quad S_N = \mathbb{Z}[x]/(\Phi_N), \quad q = 2^{b_q},$$

and use the standard coefficient norm on $R_N \simeq \mathbb{Z}^N$. The specification gives

$$hf = 3g \pmod{(q, x^N - 1)}. \quad (\text{B.41})$$

Since 3 is a unit modulo q , the public scaling

$$\hat{h} = 3^{-1}h \pmod{q} \quad (\text{B.42})$$

turns (B.41) into $\widehat{h}f = g \pmod{q}$. Let $\mathbf{H}_h^{\text{cyc}}$ be the cyclic multiplication matrix, acting on coefficient vectors, and set

$$\Lambda_h^{\text{cyc}} = \text{coeff}\{(u, v) \in R_N^2 : \widehat{h}u - v \in qR_N\} = \ker_q[\mathbf{H}_h^{\text{cyc}} \mid -\mathbf{I}_N]. \quad (\text{B.43})$$

Write $\mathcal{T}$ for the nonzero ternary representatives of degree at most $N - 2$, $\mathcal{T}^+$ for those with nonnegative adjacent correlation, and $\mathcal{T}(d)$ for those with exactly $d/2$ coefficients of each sign. For $\mathbf{P} \in \{\text{HPS}, \text{HRSS}\}$, define $\text{Valid}_{\mathbf{P}}(h; u, v)$ by the following exact checks:

- for HPS, $u \in \mathcal{T}$, $v \in \mathcal{T}(q/8 - 2)$, and $\widehat{h}u = v \pmod{q}$.
- for HRSS, $u \in \mathcal{T}^+$ and $v = (x - 1)v_0$ as an equality between the standard integer representatives of degree less than N , for some $v_0 \in \mathcal{T}^+$, together with $\widehat{h}u = v \pmod{q}$.
- in either profile, $h(1) = 0 \pmod{q}$, and the required inverses of u in $S_N/3S_N$ and S_N/qS_N , and the public inverse of h in S_N/qS_N , exist.

The condition $h(1) = 0 \pmod{q}$ is equivalent to h lying in the image of the canonical $R_{q,0}$ public key encoding. Every test is polynomial-time and exact.

For the profile interpretation in the next lemma, inherit $\text{ord}_N(2) = \text{ord}_N(3) = N - 1$ and the respective HPS weight or HRSS correctness conditions from [NTRU20a, Sections 1.3.2–1.3.3], stated explicitly in (B.55). The geometric statements below use only the hypotheses displayed in each result.

Lemma B.15. *For an HPS witness,*

$$\|(f, g)\|_2^2 \leq (N - 1) + (q/8 - 2),$$

and for an HRSS witness, writing $g = (x - 1)g_0$,

$$\|(f, g)\|_2^2 \leq 3(N - 1).$$

Thus, along families with $q = \Theta(N)$, a public constant C_{len} gives $\|(f, g)\|_2 \leq C_{\text{len}}\sqrt{q}$, and this radius is below $q/2$ for all sufficiently large N . Any pair accepted by $\text{Valid}_{\mathbf{P}}$ yields the tuple

$$(u, f_p, h_q) = (u, \text{S3_inverse}(u), \text{Sq_inverse}(h)),$$

consisting of the secret polynomial and the inverses used by DPKE decryption and by KEM decapsulation on valid inputs. We call this tuple the short key core. Sampling seeds and KEM fallback strings are separate data.

Proof. For HPS, f is ternary and g has exactly $q/8 - 2$ nonzero coefficients. For HRSS, $f, g_0 \in \mathcal{T}^+$ and

$$\|(x - 1)g_0\|_2^2 = 2\|g_0\|_2^2 - 2 \sum_i (g_0)_i (g_0)_{i+1} \leq 2(N - 1).$$

For the output statement put $G = 3v$. In S_N/qS_N , the checked congruence gives $hu = G$, and hence $(Gu)^{-1}G^2 = h$ and $(Gu)^{-1}u^2 = h^{-1}$. The checked $R_{q,0}$ condition makes the first identity agree with the canonical public representative, so the public-key computation reproduces the same h . The remaining claims follow from the exact membership and unit tests in the specification [NTRU20a, Sections 1.2, 1.8.3–1.8.4, 1.10.1, 1.11.1–1.11.2]. $\square$

B.4.1 Saturation and Gaussian mass

The short pair sublattice can have a nontrivial saturation index. Fix an integer lift of $\hat{h}$ and put

$$k = \frac{\hat{h}f - g}{q} \in R_N, \quad K_N = \mathbb{Q}[x]/(x^N - 1). \quad (\text{B.44})$$

Define the rational map $A(a) = a(f, g)$, its real image

$$W = A(K_N \otimes_{\mathbb{Q}} \mathbb{R}) \subset \mathbb{R}^{2N},$$

and the fractional R_N -module

$$T = \{a \in K_N : af, ag, ak \in R_N\}.$$

If the integer lift of $\hat{h}$ is changed to $\hat{h} + qb$ with $b \in R_N$, then k changes to $k + bf$. Consequently both T and the ideal $fR_N + gR_N + kR_N$ are independent of the chosen lift. Put

$$\Delta_{\text{sat}} = [R_N : fR_N + gR_N + kR_N] \in \mathbb{N} \cup \{\infty\},$$

with value ∞ when this ideal is not full rank. For the N th roots of unity define the scalar spectral extrema

$$G_{\min} = \min_{\zeta^{N=1}} (|f(\zeta)|^2 + |g(\zeta)|^2), \quad G_{\max} = \max_{\zeta^{N=1}} (|f(\zeta)|^2 + |g(\zeta)|^2). \quad (\text{B.45})$$

Proposition B.16. *If $G_{\min} > 0$, then $R_N \subseteq T$ is a finite-index inclusion and*

$$\Lambda_h^{\text{cyc}} \cap W = AT, \quad \Delta_{\text{sat}} = [T : R_N] = [R_N : fR_N + gR_N + kR_N]. \quad (\text{B.46})$$

Given f, g, k , the integer Δ_{sat} is computable from the rectangular Smith form of the stacked multiplication matrices of f, g, k . For every $s > 0$,

$$1 \leq \rho_s(\Lambda_h^{\text{cyc}}) \leq \Delta_{\text{sat}} \theta\left(\frac{s}{\sqrt{G_{\min}}}\right)^N \theta\left(\frac{s\sqrt{G_{\max}}}{q}\right)^N. \quad (\text{B.47})$$

Proof. The spectral assumption makes A injective. If $A(a) = (af, ag)$ is an integer point, then $\hat{h}(af) - ag = qak$, so it belongs to Λ_h^{cyc} exactly when $a \in T$. The unique preimage of an integer point lies in K_N because A is rational and full rank on its image, proving the first equality in (B.46).

The full cyclic coefficient lattice is self-dual and multiplication matrices satisfy $\mathbf{M}_a^T = \mathbf{M}_{\bar{a}}$. Therefore

$$T^* = \bar{f}R_N + \bar{g}R_N + \bar{k}R_N.$$

Coefficient conjugation is an integral orthogonal permutation, which proves the index identity and the asserted finite-index inclusion. Equivalently, Δ_{sat} is the gcd of the full-rank minors of the rectangular matrix $[\mathbf{M}_f \ \mathbf{M}_g \ \mathbf{M}_k]$. This computation uses the witness (f, g, k) . Recovery uses the public upper bound derived below.

Cyclic conjugation is orthogonal and satisfies $\mathbf{M}_a^T = \mathbf{M}_{\bar{a}}$, so the argument of Proposition B.12 gives $q(\Lambda_h^{\text{cyc}})^* = \mathbf{J}\Lambda_h^{\text{cyc}}$ and $\mathbf{J}W = W^\perp$ for $\mathbf{J} \text{coeff}(u, v) = \text{coeff}(\bar{v}, -\bar{u})$. Thus, with $M = AT$ and $Q = \pi_{W^\perp} \Lambda_h^{\text{cyc}}$, we have $Q \simeq qM^*$ and

$$\rho_s(\Lambda_h^{\text{cyc}}) \leq \rho_s(M) \rho_s(qM^*).$$

The first factor is at most $\Delta_{\text{sat}} \rho_s(AR_N)$, whereas $M^* \subseteq (AR_N)^*$ bounds the second by $\rho_s(q(AR_N)^*)$. This gives a single factor Δ_{sat} . The unitary full cyclic Fourier transform bounds the two remaining masses by the theta factors in (B.47). $\square$

The mass estimate leads to three useful forms of the recovery condition. The first bounds the total logarithmic mass directly. The second gives sufficient bounds on the two spectral extrema and the saturation index. The third uses the bounds established by the sampling analysis below and supplies the budget for the general generator theorem.

Corollary B.17 (Recovery from a total mass bound). *Use the public finite schedule $D_N \sim \ln \ln N$, $(1 - 2^{-N^2})q/N^\beta \leq s_{\text{mass}}^2 \leq q/N^\beta$, and $s_{\text{search}}^2 = qD_N/N$, for fixed rational $1/2 < \beta < 1$. Suppose $G_{\min} > 0$ and a fixed public rational $C_{\text{mass}} \geq 0$ satisfies*

$$\ln \Delta_{\text{sat}} + N \ln \theta \left(\frac{s_{\text{mass}}}{\sqrt{G_{\min}}} \right) + N \ln \theta \left(\frac{s_{\text{mass}} \sqrt{G_{\max}}}{q} \right) \leq C_{\text{mass}} N / D_N. \quad (\text{B.48})$$

Then the public integer $A_{\text{mass}} = 4^{\lceil (C_{\text{mass}}+1)N/D_N \rceil}$ bounds $\rho_{s_{\text{mass}}}(\Lambda_h^{\text{cyc}})$ and has bit length $O(N/D_N)$. With the modulus window, witness, and radius conditions of Theorem 12.1, this supplies recovery with internal success probability $1 - N^{-K}$ and time and space $2^{O(N/\ln \ln N)}$.

Proof. Take logarithms in (B.47). The displayed budget bounds the resulting expression, and $\ln 4 > 1$ makes the public integer an upper bound. Its logarithm is $O(N/D_N)$ with a constant fixed before the instance is drawn. Moreover, $s_{\text{mass}}^2/s_{\text{search}}^2 \geq (1 - 2^{-N^2})N^{1-\beta}/D_N \rightarrow \infty$. Theorem 12.1 applies for large N . The algorithm uses this public value of A_{mass} . $\square$

Corollary B.18 (Recovery from spectral and saturation bounds). *In the schedule of Corollary B.17, the conditions*

$$G_{\min} \geq \frac{q}{N^\beta} \ln D_N, \quad G_{\max} \leq \frac{qN^\beta}{\ln D_N}, \quad \ln \Delta_{\text{sat}} \leq C_{\text{sat}} N / D_N \quad (\text{B.49})$$

with fixed public rational $C_{\text{sat}} \geq 0$ suffice, for large N , with $C_{\text{mass}} = C_{\text{sat}} + 1$ and

$$A_{\text{mass}} = 4^{\lceil (C_{\text{sat}}+2)N/D_N \rceil}, \quad \ln A_{\text{mass}} \leq 2(C_{\text{sat}} + 3)N/D_N. \quad (\text{B.50})$$

Proof. Both squared theta arguments in (B.47) are at most $1/\ln D_N$. Eventually these arguments are at most one, and $\ln \theta(t) \leq 3e^{-\pi/t^2}$ gives

$$\ln \rho_{s_{\text{mass}}}(\Lambda_h^{\text{cyc}}) \leq C_{\text{sat}} N / D_N + 6ND_N^{-\pi} \leq (C_{\text{sat}} + 1)N/D_N.$$

Corollary B.17 gives the budget and recovery claim. $\square$

Section 12.3 uses Corollary B.18 to obtain its public budget. The next corollary applies the sampling estimates directly and supplies the budget for Theorem B.21.

Corollary B.19 (Mass bound for the sampling estimates). *Fix computable rational constants $C_{\text{sp}} > 0$, $C_{\text{sat}} \geq 0$, and $2/3 < \alpha < \beta < 1$. Let $D = D(N)$ be a public positive dyadic rational of polynomial bit length with $D \leq 1 + \ln \ln N$. Put $s_{\text{ref}}^2 = q/N^\beta$, and let s_{mass}^2 be a positive dyadic rational satisfying*

$$(1 - 2^{-N^2})s_{\text{ref}}^2 \leq s_{\text{mass}}^2 \leq s_{\text{ref}}^2,$$

and suppose

$$G_{\min} \geq q/N^\alpha, \quad G_{\max} \leq C_{\text{sp}} q \ln N, \quad \ln \Delta_{\text{sat}} \leq C_{\text{sat}} N / D.$$

Then

$$\ln \rho_{s_{\text{ref}}}(\Lambda_h^{\text{cyc}}) \leq C_{\text{sat}} N / D + o(1).$$

Put $s_{\text{search}}^2 = qD/N$. For all sufficiently large N , $s_{\text{mass}}^2 \geq 8s_{\text{search}}^2$ and

$$\rho_{s_{\text{search}}}(\Lambda_h^{\text{cyc}}) \leq \rho_{s_{\text{mass}}}(\Lambda_h^{\text{cyc}}) \leq \rho_{s_{\text{ref}}}(\Lambda_h^{\text{cyc}}) \leq \bar{A}_N, \quad \bar{A}_N := 4^{\lceil (C_{\text{sat}}+1)N/D \rceil}, \quad (\text{B.51})$$

where $\bar{A}_N$ is an explicitly computable integer and $\ln \bar{A}_N = O(N/D)$.

Proof. Evaluate the upper bound in (B.47) at $s = s_{\text{ref}}$. The squared arguments of the two theta factors are then at most $N^{\alpha-\beta}$ and $C_{\text{sp}}(\ln N)/N^\beta$. The standard bound $\theta(x)^N \leq \exp(3Ne^{-\pi/x^2})$ for $0 < x \leq 1$ makes both contributions $1 + N^{-\omega(1)}$. The final statements follow from monotonicity, the dyadic lower approximation, and $s_{\text{mass}}^2/s_{\text{search}}^2 \geq (1 - 2^{-N^2})N^{1-\beta}/D \rightarrow \infty$. $\square$

B.4.2 Recovery from geometric bounds

The following theorem turns joint spectral and saturation bounds into recovery for a key generator. The next parts verify these bounds for the HPS and HRSS distributions and give explicit failure probabilities. Take a sequence of primes N and power-of-two q satisfying the profile's arithmetic constraints and the modulus window below.

Definition B.20 (Conditions for recovery). Fix a profile $\mathbf{P} \in \{\text{HPS}, \text{HRSS}\}$ and public computable rational constants

$$2/3 < \alpha < \beta < 1, \quad C_{\text{len}}, C_{\text{sp}}, C_{\text{sat}}, A_q > 0, \quad 0 < c < 1,$$

and use the public positive dyadic schedule D_N of polynomial bit length satisfying

$$0 \leq \ln \ln N - D_N \leq 2^{-N^2}.$$

A generator $\text{CycNTRUGen}_{\mathbf{P},N}$ is a distribution on (h, f, g, aux) . It satisfies the recovery conditions when every output satisfies

$$\text{Valid}_{\mathbf{P}}(h; f, g) = 1, \quad \|(f, g)\|_2 \leq C_{\text{len}}\sqrt{q},$$

and the single joint event

$$\mathcal{E}_N = \{G_{\min} \geq q/N^\alpha, \quad G_{\max} \leq C_{\text{sp}}q \ln N, \quad \ln \Delta_{\text{sat}} \leq C_{\text{sat}}N/D_N\} \quad (\text{B.52})$$

has probability at least $1 - \omega_N$, where $\omega_N = o(1)$ and Δ_{sat} is defined from the generated witness by (B.44) and the extended-index convention preceding Proposition B.16. The constants and the bound C_{sat} are fixed before the generator is sampled.

Theorem B.21 (Cyclic NTRU recovery). *Suppose*

$$q = 2^{b_q}, \quad 2N + 1 \leq q \leq N^{A_q}, \quad \ln(q/N) \leq (\ln N)^{1-c},$$

and let $\text{CycNTRUGen}_{\mathbf{P},N}$ satisfy Definition B.20. For every fixed $K > 0$ and all sufficiently large N , there is one randomized algorithm, depending only on K , the public profile, parameter sequence, and specified constants, such that on public input $(\mathbf{P}, N, q, h)$ it returns a valid short pair and the associated decryption tuple of Lemma B.15, with probability at least

$$1 - \omega_N - N^{-K}. \quad (\text{B.53})$$

Its time and space are

$$\exp\left(O\left(\frac{N}{\ln \ln N}\right)\right). \quad (\text{B.54})$$

The hidden constants depend only on K and the specified constants. More precisely, for every fixed generator output that satisfies $\text{Valid}_{\mathbf{P}}(h; f, g) = 1$, the displayed norm bound, and $\mathcal{E}_N$, the same algorithm has conditional success probability at least $1 - N^{-K}$ over independent internal coins.

Proof. On the joint event $\mathcal{E}_N$, Corollary B.19 supplies the public rational bound $A_{\text{mass}} = \bar{A}_N$ in (B.51), with $\ln A_{\text{mass}} \leq 2(C_{\text{sat}} + 2)N/D_N$ for large N . Apply Theorem 12.1 to the multiplication matrix of $\hat{h}$, with $m = N$, $c_{\text{search}} = 1$, and $B_Q = C_{\text{len}}\sqrt{q}$. Its width condition follows from $s_{\text{mass}}^2/s_{\text{search}}^2 \rightarrow \infty$. Lemma B.15 supplies the polynomial-time verifier Valid_p and the decryption inverses. The given valid pair is an accepted witness, so the pointwise success probability is at least $1 - N^{-K}$, with the stated cost. Averaging over the generator and charging $\Pr[\mathcal{E}_N^c] \leq \omega_N$ proves (B.53). $\square$

B.4.3 Arithmetic parameters and accepted sampling laws

For the distributional applications, take a supplied growing sequence of primes satisfying

$$N \geq 5, \quad \text{ord}_N(2) = \text{ord}_N(3) = N - 1, \quad q = 2^{b_q} = \Theta(N), \quad q \geq 2N + 1. \quad (\text{B.55})$$

These are the common arithmetic requirements of the two profiles in [NTRU20a, Sections 1.3.2–1.3.3]. For HPS put $w = q/8 - 2$, $m = w/2 = q/16 - 1$, and require $w \leq 2N/3$. For HRSS retain the correctness condition $q > 8\sqrt{2}(N - 1)$; the specification's choice $q = 2^{\lceil 7/2 + \log_2 N \rceil}$ satisfies it. For each sufficiently large supplied HPS degree, $q = 2^{\lceil \log_2(2N+1) \rceil}$ meets its modulus and weight conditions. The parameter sequence is part of the family definition.

Put $L = N - 1$, $a = 86/256$, and $b = 85/256$. Let $\mathbf{X} = (X_0, \dots, X_{L-1})$ have independent coordinates with

$$\Pr[X_j = 0] = a, \quad \Pr[X_j = 1] = \Pr[X_j = -1] = b, \quad P_{\mathbf{X}}(x) = \sum_{j=0}^{L-1} X_j x^j. \quad (\text{B.56})$$

This is the ternary law produced by independent uniform input bytes. Write $U_{\mathbf{X}} = \text{TernaryPlus}(\mathbf{X})$: if the adjacent correlation is negative, reverse the signs of the even-indexed coefficients, and otherwise keep the coefficients. In particular,

$$U_{\mathbf{X}}(z) = P_{\mathbf{X}}(z) \quad \text{or} \quad U_{\mathbf{X}}(z) = -P_{\mathbf{X}}(-z). \quad (\text{B.57})$$

Zero input blocks are included at this stage.

Definition B.22 (HPS and HRSS key distributions). The two raw profiles use independent blocks and are summarized as follows:

profile	f	g	profile condition
HPS	$P_{\mathbf{X}}$	uniform fixed type	$w = q/8 - 2 \leq 2N/3$
HRSS	$U_{\mathbf{X}}$	$(x - 1)U_{\mathbf{Y}}$	$q > 8\sqrt{2}(N - 1)$

For HPS, the fixed type has $m = w/2$ coefficients of each sign and $L - 2m$ zeros. The optional sorting law attaches independent uniform b_N -bit labels before sorting by label and category.

Let $\pi_N : R_N \rightarrow S_N$ be the cyclotomic projection and define

$$\mathcal{A}_N = \{\pi_N(f) \bmod 3 \in (S_N/3S_N)^\times, \quad \pi_N(f), \pi_N(g) \bmod 2 \in (S_N/2S_N)^\times\}. \quad (\text{B.58})$$

For each raw law, the accepted law is $\mu = \nu \mid \mathcal{A}_N$. Its public key is the unique representative modulo $x^N - 1, q$ with $h(1) = 0$ and cyclotomic component $3\pi_N(g)\pi_N(f)^{-1}$.

Lemma B.23. Under (B.55), the acceptance probabilities are

$$a_N^{\text{HPS}} = 1 - a^{N-1}, \quad a_N^{\text{HRSS}} = (1 - a^{N-1})^2. \quad (\text{B.59})$$

The HPS formula holds for both the ideal law and every sorting label length. Every accepted pair passes the corresponding profile verifier. For every event E defined on the raw polynomials,

$$\Pr_\mu[E^c] \leq \Pr_\nu[E^c]/a_N^P. \quad (\text{B.60})$$

Proof. For $p = 2, 3$, the order condition makes Φ_N irreducible of degree $N - 1$ over $\mathbb{F}_p$. A ternary polynomial of degree at most $N - 2$ is zero modulo p exactly when all its coefficients are zero; otherwise its cyclotomic image is a unit. TernaryPlus preserves the zero-block event. Also $x - 1$ is a unit modulo $\Phi_N, 2$, since $\Phi_N(1) = N$ is odd. Thus HPS acceptance is precisely $\mathbf{X} \neq 0$, since g has positive fixed weight, and HRSS acceptance is precisely $\mathbf{X}, \mathbf{Y} \neq 0$. Independence gives (B.59).

A cyclotomic element is a unit modulo 2^{b_q} exactly when it is a unit modulo two. Since N is odd, $x - 1$ and Φ_N are coprime modulo q , so the Chinese remainder theorem supplies the asserted public representative. As $g(1) = 0$, it satisfies $hf = 3g$ in the full cyclic ring. The sampler profiles and the accepted units are exactly the checks in Lemma B.15. Finally, conditional probability gives (B.60). $\square$

Lemma B.24. *For $L = N - 1$ and $m = q/16 - 1$, the raw HPS laws satisfy*

$$\text{TV}(\nu_{N,b_N}^{\text{HPS}}, \nu_N^{\text{HPS}}) \leq \varepsilon_{\text{FT}}(N) := \min \left\{ 1, \frac{m^2 + 2m(L - 2m)}{2^{b_N}} \right\}. \quad (\text{B.61})$$

In particular, $b_N \geq (2 + \delta) \log_2 N$, with fixed $\delta > 0$, gives $\varepsilon_{\text{FT}} = o(1)$, and $b_N = \lceil (K_{\text{tag}} + 3) \log_2 N \rceil$ gives $\varepsilon_{\text{FT}} = O(N^{-K_{\text{tag}} - 1})$.

Proof. Couple the specified ordering to one using an independent uniform ordering within each equal-label group. Exchangeability makes the latter ordering a uniform permutation, whose category word has the ideal fixed-type law. The category words can differ only if two entries of different categories share a label. There are m^2 pairs between the nonzero categories and $2m(L - 2m)$ pairs involving zero; each collision has probability 2^{-b_N} . The union bound proves the estimate and its two asymptotic specializations. $\square$

For the specification's fixed 30-bit labels, the same formula is a finite distributional error bound. For the asymptotic sorting result we use growing $b_N = O(\log N)$, so both the random-bit cost and sorting cost remain polynomial.

B.4.4 Logarithmic saturation from common cyclotomic ideals

The lemmas in this subsection hold for general odd prime N . Put $\zeta_N = e^{2\pi i/N}$, $E_N = \mathbb{Q}(\zeta_N)$, and $O_N = \mathbb{Z}[\zeta_N] \simeq S_N$; write $\pi_N(f) = f(\zeta_N)$. This notation distinguishes the cyclotomic projection from coefficient conjugation. We use the standard facts that O_N is the full ring of integers, its nonzero ideals have unique prime ideal factorization, the only ramified rational prime is N , and a prime above $p \neq N$ has residue degree $\text{ord}_N(p)$ [Was97].

Lemma B.25. *Let $f, g \in R_N$, with $g(1) = 0$, $f(1) \neq 0$, and $\pi_N(g) \neq 0$. For any public relation defining $k = (\widehat{hf} - g)/q \in R_N$, put $I = (\pi_N(f), \pi_N(g)) \subseteq O_N$. Then*

$$\Delta_{\text{sat}} = [R_N : (f, g, k)] \leq N|f(1)| \text{Norm}(I). \quad (\text{B.62})$$

Proof. Embed R_N in $A = \mathbb{Z} \oplus O_N$ by evaluation at $1, \zeta_N$. The resultant of $x - 1$ and Φ_N is $\Phi_N(1) = N$, so $[A : R_N] = N$. For $J = fR_N + gR_N$ its extension is $JA = f(1)\mathbb{Z} \oplus I$, and hence $[A : JA] = |f(1)| \text{Norm}(I)$. The well-defined map of finite abelian groups

$$(A/R_N)^2 \longrightarrow JA/J, \quad ([a_1], [a_2]) \longmapsto [fa_1 + ga_2]$$

is surjective. Thus $[JA : J] \leq N^2$. The nonzero hypotheses make JA , and therefore J , full rank. Multiplying indices gives

$$[R_N : J] = \frac{[A : JA][JA : J]}{[A : R_N]} \leq N|f(1)| \text{Norm}(I).$$

Since $(f, g, k) \supseteq J$, its index can only be smaller. $\square$

Lemma B.26. *Let $X_1, \dots, X_s$ be independent with law (B.56). For every prime p , all $c_i \in \mathbb{F}_p^\times$, and every $t \in \mathbb{F}_p$,*

$$\Pr \left[\sum_{i=1}^s c_i X_i = t \right] \leq \frac{1}{p} + \frac{2}{\sqrt{s}}. \quad (\text{B.63})$$

Proof. Let $\psi(u) = a + 2b \cos(2\pi u)$. The independent-difference characteristic function gives

$$1 - |\psi(u)|^2 \geq 4ab(1 - \cos(2\pi u)) \geq 32ab\|u\|_{\mathbb{R}/\mathbb{Z}}^2,$$

and therefore $|\psi(u)| \leq e^{-c\|u\|^2}$ with $c = 16ab > 1$. Fourier inversion and Hölder's inequality for uniform measure on $\mathbb{F}_p$ yield

$$\begin{aligned} \Pr \left[\sum c_i X_i = t \right] &\leq \frac{1}{p} \sum_v \prod_i |\psi(vc_i/p)| \\ &\leq \prod_i \left(\frac{1}{p} \sum_v |\psi(vc_i/p)|^s \right)^{1/s} = \frac{1}{p} \sum_v |\psi(v/p)|^s. \end{aligned}$$

Each nonzero c_i permutes $\mathbb{F}_p$. Centering the last sum and bounding its nonzero terms by a Gaussian integral gives $p^{-1} + \sqrt{\pi/(cs)} \leq p^{-1} + 2/\sqrt{s}$. $\square$

Lemma B.27. *Let $F = P_{\mathbf{X}}$ or $F = U_{\mathbf{X}}$. For all sufficiently large odd primes N and every nonzero prime ideal $\mathfrak{p} \subset O_N$,*

$$\Pr[\pi_N(F) \in \mathfrak{p}] \leq \epsilon_N := 32 \exp\left(-\frac{1}{4}\sqrt{\ln N}\right). \quad (\text{B.64})$$

Proof. Suppose first that $\mathfrak{p}$ lies above $p \neq N$, has residue degree d , and write $\xi = \zeta_N \bmod \mathfrak{p}$. Its order is N , so $p^d \geq N + 1$. The powers $1, \xi, \dots, \xi^{d-1}$ form an $\mathbb{F}_p$ -basis. Condition on all other coefficients of $P_{\mathbf{X}}$. The equation $P_{\mathbf{X}}(\xi) = 0$ prescribes the residues of these d coefficients uniquely. Each residue has probability at most $\eta = 170/256$, including $p = 2$, where the two nonzero signs merge. Thus the probability is at most η^d .

For a second bound, fix any nonzero linear functional $\lambda : \mathbb{F}_{p^d} \rightarrow \mathbb{F}_p$. Each block of d consecutive powers of ξ is a basis, so at least $s = \lfloor (N-1)/d \rfloor$ of the weights $\lambda(\xi^j)$ are nonzero. Apply Lemma B.26 to these coefficients, conditioning on the rest. Consequently

$$\Pr[P_{\mathbf{X}}(\xi) = 0] \leq \min\{\eta^d, p^{-1} + 2/\sqrt{\lfloor (N-1)/d \rfloor}\}.$$

For $d \geq \sqrt{\ln N}$ the first term is at most $e^{-\sqrt{\ln N}/4}$, since $-\ln \eta > 1/4$. For $d < \sqrt{\ln N}$, the second is at most $e^{-\sqrt{\ln N}} + 4(\ln N)^{1/4}/\sqrt{N}$ for large N . These explicit bounds give (B.64).

The sole prime ideal above N is $(1 - \zeta_N)$, with residue field $\mathbb{F}_N$. The membership condition becomes $P_{\mathbf{X}}(1) = 0 \bmod N$, and the same scalar lemma gives $N^{-1} + 2/\sqrt{N-1}$.

For TernaryPlus, membership implies that either $P_{\mathbf{X}}(\xi) = 0$ or $P_{\mathbf{X}}(-\xi) = 0$. The element $-\xi$ generates the same residue field and its consecutive powers are again bases. The preceding two bounds apply unchanged; at the ramified prime use weights 1 or alternating signs. A union bound costs a factor of two, already allowed by the stated constant. All comparisons above have effective thresholds. For example, the displayed estimates apply once $\ln N \geq 16$. $\square$

Lemma B.28. *Let f have either ternary law of the preceding lemma and let g be independent of f . For each fixed g with $\pi_N(g) \neq 0$,*

$$\mathbb{E}_f[\ln \text{Norm}(\pi_N(f), \pi_N(g))] \leq \epsilon_N \ln |\text{Norm}_{E_N/\mathbb{Q}}(\pi_N(g))|. \quad (\text{B.65})$$

If the cyclic coefficients of g have absolute value at most two, the right side is at most $\epsilon_N N \ln(3N)$.

Proof. For a prime ideal dividing $(\pi_N(g))$, put $e_{\mathfrak{p}} = v_{\mathfrak{p}}(\pi_N(g))$. Even when $f = 0$,

$$v_{\mathfrak{p}}(\pi_N(f), \pi_N(g)) \leq e_{\mathfrak{p}} \mathbf{1}_{\{\pi_N(f) \in \mathfrak{p}\}}.$$

Take expectations in the finite sum over these prime ideals, weighted by $\ln \text{Norm}(\mathfrak{p})$, and apply Lemma B.27. This proves (B.65), including all prime ideal powers. Finally, full cyclic Parseval and the arithmetic–geometric mean give

$$|\text{Norm}_{E_N/\mathbb{Q}}(\pi_N(g))| = \prod_{j=1}^{N-1} |g(\zeta_N^j)| \leq \left(\frac{N \|g\|_2^2}{N-1} \right)^{(N-1)/2}.$$

Using $\|g\|_2^2 \leq 4N$ bounds its logarithm by $N \ln(3N)$. $\square$

Define the raw-polynomial proxy

$$\Xi_N(f, g) = \begin{cases} N|f(1)| \text{Norm}(\pi_N(f), \pi_N(g)), & f(1) \neq 0, \pi_N(g) \neq 0, \\ \infty, & \text{otherwise.} \end{cases} \quad (\text{B.66})$$

It is defined before acceptance or public-key construction. On accepted inputs with finite proxy, Lemma B.25 gives $\Delta_{\text{sat}} \leq \Xi_N$.

Proposition B.29 (Saturation bound). *Use any of the raw laws in Definition B.22, and the public dyadic $D_N \sim \ln \ln N$. For sufficiently large N ,*

$$\begin{aligned} \Pr[\ln \Xi_N > N/D_N] &\leq r_{\text{sat}}(N), \\ r_{\text{sat}}(N) &:= 64D_N \ln(3N) e^{-\sqrt{\ln N}/4} + 12N^{-1/2} = o(1). \end{aligned} \quad (\text{B.67})$$

In particular the fixed saturation constant is $C_{\text{sat}} = 1$.

Proof. At $p = N$, Lemma B.26 bounds $\Pr[P_{\mathbf{X}}(1) = 0]$ by $N^{-1} + 2/\sqrt{N-1}$. For TernaryPlus use the two weights 1 and alternating signs and (B.57), giving at most twice this bound. In HPS, $\pi_N(g) \neq 0$ follows from its positive weight and degree less than $N-1$. In HRSS it vanishes exactly when $\mathbf{Y} = 0$, an event of probability a^{N-1} . These exceptional probabilities sum to at most $12/\sqrt{N}$ for large N .

Otherwise $|f(1)| \leq N-1$, so $\ln \Xi_N \leq 2 \ln N + \ln \text{Norm}(I)$. For large N , $N/D_N - 2 \ln N \geq N/(2D_N)$. For each nonexceptional g , Markov's inequality and Lemma B.28 bound the remaining probability by $2\epsilon_N D_N \ln(3N)$. Average over g and add the exceptional probabilities. This gives the displayed formula. The proof uses only independence, bounded coefficients, and nonzero fixed weight for HPS g , so it also covers its sorting law. $\square$

B.4.5 Spectra of the raw samplers

The following bounds hold before key generation rejection, uniformly over the stated frequencies and small-ball centres.

Lemma B.30. *Let $N \geq 3$ be odd, $\zeta = e^{2\pi i/N}$, and $P_{\mathbf{X}}(z) = \sum_{k=0}^{N-2} X_k z^k$, where $\mathbf{X} = (X_0, \dots, X_{N-2})$ has independent coordinates with probabilities $86/256, 85/256, 85/256$ on $0, 1, -1$. There are absolute constants $C, c > 0$ such that, for every $r \geq 0$,*

$$\sup_{w \in \mathbb{C}} \Pr[|P_{\mathbf{X}}(z) - w| \leq r] \leq \frac{C(r+1)^2}{N}, \quad z \in \{\zeta^j, -\zeta^j : 1 \leq j < N\}, \quad (\text{B.68})$$

$$\sup_{w \in \mathbb{R}} \Pr[|P_{\mathbf{X}}(z) - w| \leq r] \leq \frac{C(r+1)}{\sqrt{N}}, \quad z \in \{1, -1\}. \quad (\text{B.69})$$

Moreover, for every $|z| = 1$ and $t \geq 0$,

$$\Pr[|P_{\mathbf{X}}(z)| > t] \leq 4e^{-ct^2/N}.$$

Proof. We first record the elementary Fourier smoothing inequality used here. For an s -dimensional random vector $\mathbf{Z}$, $s \in \{1, 2\}$, write $\phi_{\mathbf{Z}}(\mathbf{u}) = \mathbb{E}[e^{i\langle \mathbf{u}, \mathbf{Z} \rangle}]$ and $R = r+1$. For the fixed choice $c_0 = 1/16$,

$$\Pr[\|\mathbf{Z} - \mathbf{a}\|_{\infty} \leq r] \leq C_s R^s \int_{[-c_0/R, c_0/R]^s} |\phi_{\mathbf{Z}}(\mathbf{u})| d\mathbf{u}. \quad (\text{B.70})$$

Indeed, the nonnegative function $\prod_{\ell=1}^s \text{sinc}^2(c_0(x_{\ell} - a_{\ell})/(2R))$, where $\text{sinc}(v) = \sin(v)/v$ and $\text{sinc}(0) = 1$, is bounded below by a positive constant on the indicated cube. Its Fourier transform is supported on $[-c_0/R, c_0/R]^s$ and has absolute value at most $C_s R^s$. Fourier inversion and expectation prove (B.70).

For z in the first line of the lemma, put $\mathbf{v}_k = (\text{Re } z^k, \text{Im } z^k)^T$. Since z^2 is a nontrivial N th root,

$$\sum_{k=0}^{N-1} \mathbf{v}_k \mathbf{v}_k^T = \frac{N}{2} \mathbf{I}_2, \quad \sum_{k=0}^{N-2} \mathbf{v}_k \mathbf{v}_k^T = \frac{N}{2} \mathbf{I}_2 - \mathbf{v}_{N-1} \mathbf{v}_{N-1}^T \succeq \frac{N}{6} \mathbf{I}_2.$$

This holds for both signs of z ; it does not require $z^N = 1$. The common scalar coefficient characteristic function is $\phi_{\text{coef}}(u) = (86 + 170 \cos u)/256$, and $|\phi_{\text{coef}}(u)| \leq e^{-cu^2}$ for $|u| \leq 1$. Consequently the characteristic function of $\sum_{k=0}^{N-2} X_k \mathbf{v}_k$ has absolute value at most $e^{-c'N\|\mathbf{u}\|_2^2}$ whenever $\|\mathbf{u}\|_2 \leq 1$. The integration cube in (B.70) lies in this ball. The Gaussian integral is $O(N^{-1})$, proving (B.68). At $z = \pm 1$ the same argument is one-dimensional with $\sum_{k=0}^{N-2} (z^k)^2 = N-1$, and the integral is $O(N^{-1/2})$. Finally, the real and imaginary parts are sums of independent centred bounded random variables; applying Hoeffding's inequality to both parts proves the last assertion. $\square$

Lemma B.31. *Fix $\eta > 0$. Let N be an odd prime and let m be an integer with $\eta N \leq m \leq (N-1)/2$. Let Q be uniform among polynomials of degree at most $N-2$ having m coefficients of each sign and all other coefficients zero. There are constants $C, c > 0$, depending only on η , such that for every $1 \leq j < N$ and $r \geq 0$,*

$$\sup_{w \in \mathbb{C}} \Pr[|Q(\zeta^j) - w| \leq r] \leq \frac{C(r+1)}{\sqrt{N}} + e^{-cN}. \quad (\text{B.71})$$

Also, $Q(1) = 0$ and, uniformly for $|z| = 1$ and $t \geq 0$,

$$\Pr[|Q(z)| > t] \leq 4e^{-ct^2/N}.$$

Proof. Put $d = N - 1$. Draw a uniform ordering $\pi(1), \dots, \pi(d)$ of $\{0, \dots, N - 2\}$, pair its first $2m$ entries, and assign one plus and one minus to each pair using independent fair signs σ_a . The resulting law is invariant under all permutations of the d positions and has the required counts, so it is exactly uniform. Conditional on π ,

$$Q(z) = \sum_{a=1}^m \sigma_a (z^{\pi(2a-1)} - z^{\pi(2a)}).$$

Fix $z = \zeta^j$. Let A and B consist of positions whose powers have real part at least $1/2$ and at most $-1/2$, respectively. Because N is prime, multiplication by j permutes all N residues. After deleting one position, each set therefore has size at least $N/3 - 2$, and hence at least $N/4$ for sufficiently large N . If L counts pairs joining A and B , then

$$\mathbb{E}[L] = \frac{2m|A||B|}{d(d-1)} \geq c_1 N,$$

where $c_1 > 0$ depends only on η . Transposing two entries of π changes at most two pair indicators. In the Doob martingale exposing the permutation one entry at a time, two possible next entries can be coupled by transposing their labels in the remaining completion. Its increments thus have absolute value at most 2. Azuma's inequality gives

$$\Pr[L < \mathbb{E}[L]/2] \leq \exp\left(-\frac{(\mathbb{E}[L])^2}{32d}\right) \leq e^{-c_2 N}.$$

On the complementary event, at least $c_1 N/2$ of the real coefficients $b_a = \operatorname{Re}(z^{\pi(2a-1)} - z^{\pi(2a)})$ have $|b_a| \geq 1$, while all satisfy $|b_a| \leq 2$. For $|u| \leq 1/4$ the conditional characteristic function therefore obeys

$$\left| \mathbb{E}[e^{iu \operatorname{Re} Q(z)} \mid \pi] \right| = \prod_{a=1}^m |\cos(ub_a)| \leq e^{-c_3 N u^2}.$$

Apply the one-dimensional version of (B.70) and use that a complex ball projects into a real interval of the same radius. Averaging over π proves (B.71); increasing C covers the finitely many smaller N . For the upper tail, no good-pair event is needed: conditional on every π , the summands are independent centred signs with complex coefficients of modulus at most 2. Hoeffding's inequality for each component and $m \leq N/2$ give the stated uniform estimate. The identity $Q(1) = 0$ follows from balance. $\square$

Lemma B.32. *For the input $\mathbf{X}$ of Lemma B.30, put $U = \operatorname{TernaryPlus}(\mathbf{X})$. The same small-ball estimates hold for U with at most twice the constant, and*

$$\Pr[|U(z)| > t] \leq 8e^{-ct^2/N} \quad (|z| = 1).$$

Outputs obtained from independent input blocks remain independent.

Proof. The specification's sign adjustment gives, for every input, $U(z) = P_{\mathbf{X}}(z)$ or $U(z) = -P_{\mathbf{X}}(-z)$. Thus the event $|U(z) - w| \leq r$ is contained in the union of $|P_{\mathbf{X}}(z) - w| \leq r$ and $|P_{\mathbf{X}}(-z) + w| \leq r$. Lemma B.30 covers both signs of every nontrivial frequency, and also the two real frequencies $1, -1$. The same union bound proves the upper tail. This argument does not condition on which sign adjustment was selected. Each output is a deterministic function of its own input block, which proves the final assertion. $\square$

Proposition B.33 (Spectral bound). *Suppose $q = \Theta(N)$, $q \geq 2N + 1$, and fix $2/3 < \alpha < 1$ and a positive integer K_{sp} . For the raw HRSS and ideal HPS laws of Definition B.22,*

$$\Pr[G_{\min} < q/N^\alpha \text{ or } G_{\max} > C_{\text{sp}}q \ln N] \leq r_{\text{sp}}(N) := C \left(N^{1-3\alpha/2} + N^{-\alpha/2} + N^{-K_{\text{sp}}} \right) \quad (\text{B.72})$$

where C_{sp} is a fixed public rational constant and C is effective, depending only on K_{sp} and the fixed modulus window. One may take $C_{\text{sp}} = 32(K_{\text{sp}} + 3)$.

Proof. Put $T_0 = q/N^\alpha$. Since $q = \Theta(N)$ and $\alpha < 1$, we may assume $T_0 \geq 4$ throughout the asymptotic estimates. For HPS, $m = q/16 - 1 \geq N/16$ for sufficiently large N , while $2m \leq 2N/3$ ensures the required support size. Lemmas B.30 and B.31, applied to the independent polynomials f, g with radius $\sqrt{T_0}$, give, for $1 \leq j < N$,

$$\Pr[|f(\zeta^j)|^2 + |g(\zeta^j)|^2 < T_0] \leq C \frac{T_0}{N} \left(\sqrt{\frac{T_0}{N}} + e^{-cN} \right).$$

Summing over the nontrivial frequencies gives $O(T_0^{3/2}/\sqrt{N} + T_0 e^{-cN}) = O(N^{1-3\alpha/2})$. At the trivial frequency, $g(1) = 0$, and (B.69) bounds the lower-tail probability by $O(\sqrt{T_0/N}) = O(N^{-\alpha/2})$.

For HRSS, write $f = \text{TernaryPlus}(\mathbf{X})$ and $g = (x - 1) \text{TernaryPlus}(\mathbf{Y})$, with independent coefficient vectors $\mathbf{X}, \mathbf{Y}$. If $a_j = |\zeta^j - 1|$, Lemma B.32 and $\sqrt{T_0}/a_j \geq 1$ give

$$\Pr[|f(\zeta^j)|^2 + |g(\zeta^j)|^2 < T_0] \leq C \frac{T_0}{N} \min \left\{ 1, \frac{T_0}{Na_j^2} \right\}.$$

Writing $d_j = \min(j, N - j)$, we have $a_j \geq 4d_j/N$ and hence

$$\sum_{j=1}^{N-1} \min \left\{ 1, \frac{T_0}{Na_j^2} \right\} \leq 2 \sum_{d=1}^{(N-1)/2} \min \left\{ 1, \frac{T_0 N}{16d^2} \right\} = O(\sqrt{T_0 N}).$$

The last bound follows by splitting the sum at $\sqrt{T_0 N}$ and using $\sum_{d>L} d^{-2} \leq 1/L$. The HRSS lower-tail sum is therefore also $O(T_0^{3/2}/\sqrt{N}) = O(N^{1-3\alpha/2})$. Here $g(1) = 0$ again, and the real-frequency part of Lemma B.32 gives $O(N^{-\alpha/2})$.

For both profiles the upper-tail bounds in the lemmas, together with $|\zeta^j - 1| \leq 2$ for HRSS, imply

$$\Pr[G_{\max} > C_{\text{sp}}q \ln N] \leq 16N \exp(-C_{\text{sp}}q \ln N / (32N)) \leq N^{-K_{\text{sp}}}$$

for the stated C_{sp} and sufficiently large N . Indeed the scalar Hoeffding estimates above allow the common upper tail $8 \exp(-t^2/(8N))$; for HRSS the independent-coefficient TernaryPlus tail is $8 \exp(-t^2/(4N))$, and the factor $|\zeta^j - 1| \leq 2$ gives the denominator $32N$ displayed here. This proves (B.72) without assuming independence between different frequencies. All small-ball constants are obtained from the fixed smoothing kernel, Gaussian integrals, and the displayed permutation concentration bound, and thus have effective bounds. $\square$

B.4.6 Geometry of accepted keys

We now combine the spectral and saturation bounds for the raw polynomials. Conditioning on the key generation acceptance event gives the joint geometric event used in Section 12.3. The raw failure bound is $r_{\text{sp}}(N) + r_{\text{sat}}(N)$, and the acceptance probability accounts for the denominator in the recovery guarantee.

For the HPS sorting sampler, the distribution comparison adds $\varepsilon_{\text{FT}}(N)$ to the same raw event before conditioning.

Proposition B.34. Fix rational $2/3 < \alpha < \beta < 1$ and use the finite schedule $D_N \sim \ln \ln N$. Define on the raw polynomials

$$\mathcal{E}_N^{\text{raw}} = \{G_{\min} \geq q/N^\alpha, \quad G_{\max} \leq C_{\text{sp}} q \ln N, \quad \ln \Xi_N \leq N/D_N\}. \quad (\text{B.73})$$

For $P \in \{\text{HPS}, \text{HRSS}\}$ and its ideal accepted law,

$$\Pr_{\mu_N^P}[(\mathcal{E}_N^{\text{raw}})^c] \leq \frac{r_{\text{sp}}(N) + r_{\text{sat}}(N)}{a_N^P} = o(1). \quad (\text{B.74})$$

For the accepted HPS sorting law, the same bound holds with $r_{\text{sp}} + r_{\text{sat}} + \varepsilon_{\text{FT}}$ in the numerator. On $\mathcal{E}_N^{\text{raw}}$, the actual saturation index satisfies $\ln \Delta_{\text{sat}} \leq N/D_N$.

Proof. The raw failure probability is at most $r_{\text{sp}} + r_{\text{sat}}$ by Propositions B.33 and B.29 and the union bound. Lemma B.23 transfers this event to the accepted law. For sorting, first apply Lemma B.24 once to the complete raw event, then condition using the same HPS acceptance probability. Finally, Lemma B.25 bounds the actual index by Ξ_N . $\square$

C Changing the Public Distribution

ML-KEM and ML-DSA generate their public matrices by expanding a seed. In ML-KEM, key generation derives the public matrix from the seed ρ [FIPS203, Section 5.1, Algorithm 13]. ML-DSA constructs its matrix $\mathbf{A}$ from ρ through **ExpandA** [FIPS204, Section 6.1, Algorithms 6 and 32]. Each scheme therefore specifies an induced distribution on its public operator. Since the geometric bounds in this paper are stated for specified operator distributions, we record conditions under which a bound transfers to another public law.

Proposition C.1. Fix a public space Ω_n , a schedule, and its geometric event $\mathcal{G}_n$. If a reference law μ_n satisfies $\mu_n(\mathcal{G}_n^c) \leq \omega_n$, then any law ν_n obeys

$$\|\nu_n - \mu_n\|_{\text{TV}} \leq \varepsilon_n \implies \nu_n(\mathcal{G}_n^c) \leq \omega_n + \varepsilon_n, \quad (\text{C.1})$$

$$\nu_n(x) \leq K_n \mu_n(x) \quad (x \in \Omega_n) \implies \nu_n(\mathcal{G}_n^c) \leq K_n \omega_n. \quad (\text{C.2})$$

If the new input law has energy failure at most p_E and the fixed-operator algorithm has internal failure at most ξ_n , total recovery failure is at most $p_E + \nu_n(\mathcal{G}_n^c) + \xi_n$, with unchanged resource bounds.

Proof. Apply the event bound for total variation or sum the density inequality over $\mathcal{G}_n^c$, then add the energy and internal failures. $\square$

Seed-expanded operators.

Remark C.2. If μ_n is uniform and ν_n expands an L_n -bit seed, then

$$\|\nu_n - \mu_n\|_{\text{TV}} \geq \max\{0, 1 - 2^{L_n}/|\Omega_n|\}, \quad |\Omega_n| \max_x \nu_n(x) \geq |\Omega_n|/2^{L_n}. \quad (\text{C.3})$$

Here $|\Omega_n|$ is q^{n^2} for scalar matrices, q^{kn} for rank- k square module matrices, and q^n for field multipliers. A short seed alone gives neither transfer bound. One may instead prove the geometry directly for the induced law, or use pseudorandomness covering the recovery algorithm's running time and success experiment.

D Discussion of Fine-Grained Reductions

A recovery algorithm also constrains reductions into its input regime. The question is quantitative: how much dimension growth is needed when a reduction produces a low-energy target? We compare the recovery exponent with the number of variables of a source 3-SAT instance. The comparison requires a matching recovery task and a bound on the exceptional probability across all oracle queries.

D.1 Recovery Cost and the Reduction Model

Let n count the secret coefficients of a linear recovery instance. For fixed module rank, this is the rank times the ring degree. Use the normalized candidate radius U_n from (1.4) and put

$$q = n^{\kappa+o(1)}, \quad \kappa > 1/2, \quad Q_n = \ln(q/\sqrt{n}), \quad D_n = \ln \frac{Q_n}{1 + \ln U_n}. \quad (\text{D.1})$$

In the recovery window $D_n \rightarrow \infty$, Theorem 5.1 and the finite implementation in Appendix A give the cost scale $2^{O(n/D_n)}$. The timeout construction there supplies a computable worst-case time bound for every public input, while retaining constant success probability on each good instance. This suffices for composition.

Fix a target law μ_n specifying the public operator, labels, observation count, energy bounds, and a uniquely specified correct recovery output. Let $\mathcal{G}_n$ be the set on which both the energy and geometric conditions hold, with

$$\mu_n(\mathcal{G}_n^c) \leq \delta_n.$$

On every instance in $\mathcal{G}_n$, the time-bounded recovery procedure returns the specified output with probability at least $2/3$. The integer secret task of the framework for linear relations supplies this uniquely specified output.

Consider a classical randomized reduction from 3-SAT on N variables. We use formulas without repeated clauses, whose length is polynomial in N ; arbitrary formulas can first be normalized in polynomial input time. Assume the following conditions, uniformly over source formulas:

1. Apart from its oracle calls, the reduction takes polynomial time and makes at most $a(N) = \text{poly}(N)$ queries. Each query has dimension $n(N) \rightarrow \infty$, with $n(N) \leq \text{poly}(N)$, and a polynomial-length encoding.
2. With the specified correct oracle, the reduction decides the source instance with probability at least $5/6$.
3. Conditional on any positive-probability query history in this correct-oracle execution, the law of the next query has total variation distance at most ζ_N from $\mu_{n(N)}$.
4. The accumulated exceptional probability satisfies

$$a(N)(\delta_{n(N)} + \zeta_N) = o(1). \quad (\text{D.2})$$

The distribution-transfer principle in Appendix C then bounds the bad-instance probability of each query by $\delta_{n(N)} + \zeta_N$. It is important to specify the full target task: a short-error law alone does not fix the secret distribution, number of observations, or geometric guarantee.

D.2 Composition and Randomized ETH

Proposition D.1. *Under the reduction conditions above, 3-SAT has a bounded-error randomized algorithm with running time*

$$2^{O(n(N)/D_{n(N)})} \text{poly}(N). \quad (\text{D.3})$$

Proof. Answer a query by the most frequent output of $c \ln((a(N) + 2)(N + 2))$ independent runs of the time-bounded recovery procedure, with deterministic tie breaking. For a sufficiently large constant c , a Chernoff bound makes its error on every good instance at most

$$\eta_N = \frac{1}{100(a(N) + 1)(N + 2)^2}.$$

The specified output has per-run probability at least $2/3$, so a strict majority of correct outputs implies that this rule is correct.

Use the correct-oracle execution as a reference transcript, and attach fresh independent recovery randomness to each query on that transcript. Its conditional query-law assumption bounds the chance that a query is bad by $\delta_{n(N)} + \zeta_N$. On a good query the internal error is at most η_N . A union bound over the reference transcript gives

$$a(N)(\delta_{n(N)} + \zeta_N + \eta_N) = o(1).$$

Couple the actual simulation to this transcript with the same reduction randomness. The transcripts agree until the first incorrect simulated answer, so the same bound controls the probability of any divergence. Adding the correct-oracle reduction's error $1/6$ leaves total error below $1/3$ for all sufficiently large N . The number of queries and repetitions contributes only a polynomial factor to the time bound. $\square$

We use randomized ETH, denoted **rETH**, for the assertion that 3-SAT on N variables has no bounded-error randomized algorithm in time $2^{o(N)} \text{poly}(|\phi|)$, where $|\phi|$ is the formula length. This is the randomized counterpart of the exponential-time hypothesis [IP01, IPZ01]. Randomization is essential to the formulation because the composed recovery algorithm is randomized.

Corollary D.2. *Under **rETH**, no reduction satisfying the conditions above has*

$$n(N)/D_{n(N)} = o(N).$$

Proof. Otherwise Proposition D.1, together with formula normalization, would give the algorithm excluded by **rETH**. $\square$

Repeating recovery improves its internal success on a good instance. It does not reduce the probability that a query itself violates the energy or geometry assumptions. This is why the accumulated budget (D.2) is separate from amplification.

D.3 Dimension and Hidden Energy

The simplest consequence concerns linear dimension growth. If $n(N) = O(N)$ and $D_{n(N)} \rightarrow \infty$, then

$$\frac{n(N)}{D_{n(N)}} = o(N).$$

Thus **rETH** excludes such a matching linear-dimension reduction throughout the recovery window.

For polylogarithmic U_n and polynomial modulus, $D_n \sim \ln \ln n$. In this common window, the corollary also excludes

$$n(N) = o(N \ln \ln N). \tag{D.4}$$

Indeed, if $n(N) \leq \sqrt{N}$, then $n(N)/D_{n(N)} = o(N)$ directly. For the remaining dimensions, $\sqrt{N} < n(N) \leq \text{poly}(N)$ gives $\ln \ln n(N) = \ln \ln N + O(1)$, and (D.4) again implies the excluded exponent scale. These conclusions exclude the stated asymptotic growth windows.

More generally, fixing the target dimension and modulus while increasing U_n decreases D_n and raises the recovery cost. For example,

$$U_n = \exp(\ln n / L_n), \quad 1 \ll L_n \leq \ln n \implies D_n = \ln L_n + O(1).$$

The composed exponent is then $O(n / \ln L_n)$. Lower target energy therefore requires more dimension growth to avoid the excluded windows. Dimension and energy must be compared together.

Known reductions provide background for this matching question, including Regev’s LWE reduction [Reg05], small-parameter SIS and LWE reductions [MP13], and short-distribution module-LWE reductions [BJRW23]. Applying the composition above to any such route also requires a dimension-controlled reduction from 3-SAT and matching its actual target law, output task, observation count, and error budget. A quantum reduction would require a separate quantum-oracle simulation.

E Finite-Parameter Substitutions

For each listed parameter set, we substitute its dimension and modulus, together with the deterministic joint energy bound, into the depth expression. The resulting formal leading term values are shown in Table 3. From (1.4),

$$R^2 = nV + 3B^2, \quad U = \sqrt{V + 3B^2/n},$$

$$D = \ln \frac{\ln(q/\sqrt{n})}{1 + \ln U}, \quad b_D = \frac{n}{2D}, \quad b_\infty = \frac{n}{2 \ln \ln n}.$$

Here nV bounds the joint secret and error energy, B bounds each secret coefficient, and all logarithms are natural. The quantity b_D is the finite formal substitution into the leading exponent $n/(2D)$, while b_∞ is the asymptotic comparison obtained by replacing D with $\ln \ln n$. The displayed nonintegral values of V are rounded only for presentation. All calculations use the exact fractions in the diagnostic script.

For ML-KEM, public key generation uses $\text{CBD}(\eta_1)$ for both secret and error, giving $V = 2\eta_1^2$ and $B = \eta_1$ [Kyber21, FIPS203]. For FrodoKEM, the support bounds $H = 12, 10, 6$ give $V = 2H^2$ and $B = H$. Here n is the dimension of one secret column, and all eight columns share the Gaussian list [Frodo20]. For SABER, $B = \mu/2$ and the rounding step $q/p = 8$ gives $V = B^2 + 16$, including the rounding error [SaberSpec]. For ML-KEM and SABER, n is the module rank times the ring degree. For NTRU LPrime, $n = p$ and a ternary secret of exact weight w gives $V = 1 + w/n$ and $B = 1$ [NTRU20b]. This fraction is evaluated exactly before computing D .

For ML-DSA, $n = \ell m$, $B = \eta$, and $V = \eta^2 + (\eta + 2^{d-1})^2$, as in (11.4). The public key rounding error has coefficient magnitude at most 2^{d-1} and is included together with $\mathbf{s}_2$ in the lifted error. The ML-DSA parameter sets use $m = 256$, $d = 13$, and $\eta = 2, 4, 2$, respectively [FIPS204].

The modulus and energy jointly determine the available depth. The quantity b_D records this finite comparison, while b_∞ records the common asymptotic scale.

Remark E.1. The ML-DSA family in Definition 11.1 extends the standard compressed public key relation by increasing the ring degree and modulus at fixed module ranks, with the secret coefficient bound η and rounding step Δ bounded or polylogarithmic. In particular, η and the number d of discarded bits may stay fixed. This choice separates dimension growth from the coefficient scales contributing to the joint energy.

The depth can be written exactly as

$$D = \ln \ln n + \ln \left(\frac{\ln(q/\sqrt{n})}{\ln n} \right) - \ln(1 + \ln U).$$

Table 3: Parameter values, joint energy bounds, and formal leading term values for five linear public families.

Parameter set	n	q	V	b_D	b_∞
ML-KEM-512	512	3329	18	359.1	139.8
ML-KEM-768	768	3329	8	450.2	202.8
ML-KEM-1024	1024	3329	8	622.5	264.5
FrodoKEM-640	640	32768	288	511.3	171.5
FrodoKEM-976	976	65536	200	659.6	253.0
FrodoKEM-1344	1344	65536	72	772.8	340.3
LightSaber	512	8192	41	354.0	139.8
Saber	768	8192	32	524.0	202.8
FireSaber	1024	8192	25	679.4	264.5
ntrulpr653	653	4621	1.3859	218.3	174.7
ntrulpr761	761	4591	1.3285	254.0	201.1
ntrulpr857	857	5167	1.3279	283.8	224.3
ntrulpr953	953	6343	1.3620	311.9	247.4
ntrulpr1013	1013	7177	1.3870	329.5	261.8
ntrulpr1277	1277	7879	1.3359	412.0	324.5
ML-DSA-44	1024	8380417	16793608	1754.6	264.5
ML-DSA-65	1280	8380417	16810016	2263.3	325.2
ML-DSA-87	1792	8380417	16793608	3329.4	444.9

The modulus bounds $q \geq 2m + 1$ and $q \leq n^{C_q}$, with $n = \ell m$ and fixed ℓ , give $\ln(q/\sqrt{n}) = \Theta(\ln n)$. Hence the middle term is $O(1)$, while $U = (\ln n)^{O(1)}$ makes the last term $O(\ln \ln \ln n)$. Thus $D \sim \ln \ln n$. At the listed ML-DSA dimensions, the rounding contribution $2^{d-1} = 4096$ dominates the deterministic energy bound. For ML-DSA-44, $U \approx 4098$ gives $D \approx 0.292$, compared with $\ln \ln n \approx 1.936$. Consequently, $b_D/b_\infty = \ln \ln n/D \approx 6.63$. The finite correction accounts for the separation between the two formal exponents in the ML-DSA rows.